Vincent Liu
WiseMotif.ai

WiseMotif Monograph of Mathematical Intelligence
Series 02 · Mathematics and Artificial Intelligence

COMPLETE FIVE-VOLUME MONOGRAPH · VOLUMES I–V

Topos-Native Intelligence

A Sheaf-Theoretic Architecture for Verifiable Machine Intelligence

RESEARCH ARCHITECTURE EDITION 1

I Diagnosis and the Source Mathematics
II The Architecture and Its Mixture of Experts
III Learning, Systems, Validation, and Position
IV Provenance and Adjudication
V Transferable Cores and the Replacement Case
A Apparatus · A0–A16

Original five-volume topos design
English · 2026

GM-Omega-Infinity: Topos-Native Intelligence

Complete Five-Volume Monograph

GM-Omega-Infinity: Topos-Native Intelligence

A Sheaf-Theoretic Architecture for Verifiable Machine Intelligence

Vincent Liu

Publication and Edition Record

WiseMotif.ai
WiseMotif Monograph of Mathematical Intelligence
Series 02 · Mathematics and Artificial Intelligence
Research Architecture Edition 1
Version 1.0.0
English
Copyright 2026 Vincent Liu
No named person or institution has reviewed, authorized, sponsored, or endorsed this publication unless such involvement is explicitly documented.

Licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (https://creativecommons.org/licenses/by-nc-nd/4.0/). Code listings and machine-readable schemas printed in this work are additionally available under the Apache License 2.0. Text and data mining rights are reserved under Article 4(3) of Directive (EU) 2019/790.

Series Note

GM-Omega-Infinity: Topos-Native Intelligence is the second work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme, whose central part is *The Grothendieckian Machine*, the first work of the series. That work set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. The present work addresses the question its predecessor leaves open: what an intelligence must be, internally, if it is to reason inside such worlds. What makes an inference local, what obliges local inferences to agree, and what may be concluded when they do not, are its subject.

Its answer is that reasoning should be native to a topos. A context is treated as a site, the evidence it supports as a sheaf over that site, and a conclusion as a global section, which exists only where the local data glue. Coherence is then a condition the architecture must discharge, where an ordinary model can only imitate it, and incoherence becomes an obstruction that can be computed and located, where an ordinary model registers an unexplained error. On this footing the work builds a reasoning kernel of nine layers over a single base, a mixture of experts in which each expert is an open sub-topos and routing is the construction of a sufficient cover, a learning theory that proves which objectives preserve this structure and which destroy it, and a reinforcement scheme in which reward is verified by descent rather than predicted.

The mathematics is drawn from seven lineages: Grothendieck, Deligne, Faltings, Voevodsky, Scholze, Lurie and Mochizuki. Each borrowing is admitted on a single condition, that removing it would break a named definition, theorem or algorithm of this book. What survives that test is restated without its source vocabulary and made into algorithms with stated costs, degeneration conditions and refutation gates; what does not survive is recorded, with the reason, among the negative results.

Within the Grothendieckian Programme this work is the meta-reasoning kernel. It does not propose a replacement for the Transformer; it states the structure that any machine reasoning over local evidence must answer to, precisely enough that existing systems can be measured against it and later systems built upon it. Every claim carries its grade where it is made. Theorems are proved, or cited at the level of their statements; design decisions are never used as premises. The experiments that will test the theory belong to the research it makes possible.

Vincent Liu
WiseMotif.ai

Dedication

For the mathematicians whose programmes are quoted in these pages, and who owe this book nothing; and for the reader who checks.

Preface

The question of the local

This book began with a narrow question: what does the word *local* mean for a language model? A model reads a context and makes claims that the context is supposed to support. Ask which parts of the context may support a claim, and which collections of such parts determine support on their union, and the standard architecture is silent. It has positions, masks and learned attention weights, and these certainly relate one position to another. What they are not known to provide is a notion of covering. Without one, the sheaf condition, that data agreeing on overlaps glue to data on the whole, constrains nothing, because there are no overlaps on which it can act. Such a model may still behave coherently. It is not obliged to, and the distance between exhibiting a property and being obliged to have it is the subject of this book.

The distance matters because the failures that are hardest to repair in present systems live inside it. A model asserts what its context does not support, and nothing in the computation separates support from a good fit of form. It composes two steps of reasoning reliably and four steps unreliably, and there is no account of where the composition breaks. It is adapted to a narrow domain and is quietly damaged elsewhere, and nothing it computes records what was preserved. Its capacity grows with scale, and nothing shows that the quantity which grows is the quantity that composition requires. Chapter 1.1 reads five such failures as one structural deficit: locality without gluing, composition without compositionality, confidence without justification, adaptation without preservation, and capacity without structure.

The reading is a diagnostic convention, and it is graded as one. That the five failures share a single cause is stated as a conjecture, with the condition under which it would be false. What the reading supplies is a direction. If the deficit is structural, a larger system of the same kind will not remove it, any more than more points will make a set into a space. A different kind of structure is required, and the question becomes which structure, built how, and obliged to what.

The Grothendieckian turn

Grothendieck's way with a hard problem was seldom to attack it where it stood. He built the setting in which the problem became natural, and let the solution come to it as, in his own image, a rising sea slowly submerges what it surrounds. Schemes, sites and topoi are the most famous of those settings. Each replaced a space by the system of its local data and of the rules by which local data agree, and each turned questions about points into questions about the gluing of sections. The relative point of view made every construction a construction over a base, and base change a first-class operation. The functor of points made an object known by everything that maps into it. Descent made the passage from local to global a matter of data and of the compatibilities those data must satisfy, with obstruction classes measuring exactly how they fail.

This book applies that turn to reasoning itself. A context is treated as a site: its admissible pieces of evidence, the ways in which one piece refines another, and the families of pieces that jointly cover a claim. What the context supports is a sheaf on that site, an assignment of local information that restricts consistently from larger pieces to smaller ones. A conclusion is a global section, and it exists only where the local information glues. Where it does not glue, the failure is not noise. It is a cohomology class: an obstruction with a definite location and a definite size, which a system can compute, report and act upon. Inference becomes native to a topos instead of being borrowed from one, and that is the sense of the title.

Three consequences follow, and the whole architecture is organised around them. Agreement is a condition to be discharged, so every layer must say whether it preserves restriction and, if it does not, what it measures in its place. Disagreement is information, so the obstruction to gluing is an output of the system and not an error to be suppressed. Change of context is a morphism, so adaptation, transfer and retrieval are expressed as base change along a map of sites, and the question of what survives a change becomes the question of what the map preserves. A fourth consequence is quieter and perhaps deeper. A topos carries an internal logic, and in that logic truth is local: a proposition holds on the part of the site where it is forced, and the subobject classifier records where that part lies. A reasoning system built in this way returns, with every verdict, the part of the site on which it holds.

The place of this work in the Grothendieckian Programme

The Grothendieckian Programme of which this series is part has a centre and a kernel. Its centre is *The Grothendieckian Machine*, the first work of the series, which described how mathematical worlds are built, compared and transported, and how a machine may hold them as typed, versioned and verifiable objects: worlds with their ambient categories and logics, bridges with their preservation profiles, obligations with their evidence. That work specified the worlds and the traffic between them. It did not specify the reasoning that must take place inside a world when evidence arrives piecemeal, overlaps imperfectly and must be reconciled before anything can be concluded.

The present work supplies that kernel. It is the theory of local inference, gluing, obstruction, internal logic and change of base on which every reasoning component of the programme depends. Where the first work asks what a mathematical world is and how two worlds may be compared, this one asks what it is to reason within one: when a body of local evidence licenses a conclusion, how the licence is computed, what is reported when it is withheld, and how the licence behaves when the context changes. The two works meet at the site. A world of the first work presents its evidence as a site; the kernel of this work reasons over it.

It is therefore a work of foundations. Its object is the meta-reasoning core of a machine intelligence, the part that decides what a body of evidence licenses and what it does not, specified with enough precision that it can be implemented, tested and, if it is wrong, refuted. It is not a proposal for one more model. The structures it describes are those to which any system reasoning over local evidence must answer, whether or not it was designed with them in view, and the book is written so that existing architectures can be measured against them.

Worlds, sites and the interface with the first work

The two works are written to meet, and the place where they meet can be stated exactly. *The Grothendieckian Machine* holds a mathematical world as a record: its ambient category and logic, its admitted objects and maps, its invariants, its supported fragments and the obligations it has not yet discharged. A bridge between two worlds records its direction and

what it preserves, reflects, forgets or leaves unknown. An obligation records what must still be shown and what would count as showing it. None of these records says how a body of evidence inside a world becomes a conclusion. That is the gap this book fills, and it fills it by giving every such world a site.

A world presents its evidence as a site: the admissible pieces of evidence, the refinement order among them, and the families that cover a claim. The reasoning kernel of this book takes that site, places a sheaf of local information over it, and computes what glues. A bridge of the first work becomes, in this book, a morphism of sites, and the question that the first work asks of a bridge, what it preserves, becomes a question this book can answer by computing what the morphism carries across and what it loses. An obligation of the first work becomes an open cell of a contract here, printed as open until it is discharged. The correspondence is not decorative. It is what allows the programme to be built as one system rather than two: the worlds are specified once, and every reasoning component of the programme reasons over them in the same way, under the same discipline of gluing and with the same account of failure.

This also fixes the order of dependence. The present book does not depend on the full apparatus of the first; it needs only that evidence be presented as a site and that changes of context be presented as morphisms. The first book, in turn, needs from this one an account of reasoning that respects the boundaries it draws between theorem and design, between established and conjectural, between a result and its proxy. The grading discipline of this book is that account in its sharpest form.

The kernel in seven operations

Stripped to its operations, the kernel does seven things, and each of them is a piece of mathematics before it is a piece of architecture.

It **covers**. From a pool of evidence it constructs a site: atoms of evidence, the order in which they refine one another, and covering families that say when a collection of pieces is enough. Covering is posed as construction, with a stated sufficiency criterion, and not as the selection of the highest-scoring items.

It **assigns**. Over the site it places a sheaf, with stalks whose dimensions may differ from piece to piece and restriction maps that say how information on a larger piece determines information on a smaller one.

It **transports**. Attention is rebuilt as descent: values are carried along restriction maps into the shared stalk of an overlap and compared there, so that what is mixed has first been made comparable.

It **glues**. A solver reconciles local sections into a global one where that is possible, by a named normative program with an exact reverse, and reports the residual where it is not.

It **obstructs**. What fails to glue is measured as a class, located on the overlaps where it lives, and returned as an output; a head that cannot obtain independent overlap data abstains rather than inventing it.

It **decides**. An internal-logic head evaluates propositions by Kripke–Joyal forcing, so that every verdict is returned together with the part of the site on which it is forced.

It **changes base**. Transfer to a new context is a morphism of sites, and the layer that implements it states which structure the morphism preserves and which it forgets.

Two further invariants run alongside these operations. Holonomy measures the non-abelian capacity of the sheaf to carry consistent information around loops, and the book proves that on orthogonal cellular sheaves this capacity is either zero or bounded away from zero by a constant that depends only on the stalk dimension. Coherence across experts is

measured by descent and, at the next order, by a gerbe, whose class can be non-trivial only on configurations rich enough to carry it.

The architecture of the work

The work is in five volumes, bound as one and divided by argument rather than by length. Each volume ends where a reader may stop with something complete in hand.

Volume I, *Diagnosis and the Source Mathematics*, states what is wrong and supplies the mathematics with which to say it. Part I names the five deficits, states the central thesis together with the claims on which it rests, and fixes the method by which mathematics may be carried into an architecture. Part II is the source layer: sites and the ladder of Grothendieck topologies; sheaves, nuclei and internal logic; dependent type theory and univalent transport; infinity-topoi; cohomology, descent and the obstruction to gluing; Deligne weights and strictness; almost mathematics; higher descent and monadic reconstruction; Kan extensions and coends, and what attention actually computes; enriched categories and profunctor composition; non-abelian holonomy; fundamental groups and Galois categories; the monodromy filtration; separation complexity; the six operations; motives; and anabelian reconstruction. Every statement in Part II is a theorem about mathematics, proved or cited with its hypotheses, and none is a statement about a neural network. Volume I is the only place where the grade of theorem is licensed for the source mathematics, and nothing later in the work may raise a grade that it did not license.

Volume II, *The Architecture and Its Mixture of Experts*, builds the object. Part III spends the mathematics of Volume I on nine layers over a single base, from the context site to the transfer layer, together with the engines attached to them: an engine that grades representations by weight, an almost-descent engine that separates a categorical defect from a small norm, and the units that carry certificates from one layer to the next. Part IV turns the same structure into a mixture of experts in which an expert is an open sub-topos, routing is the construction of a sufficient cover, the dispatch and combine of experts are read through an essential adjoint triple, and disagreement among experts is measured by descent and by the gerbe it defines. Every architectural choice in this volume is a design decision, named as one and tied to the source result it answers to. A reader who wants the machine and not its provenance may stop here with a specification in hand, its open contracts printed beside it.

Volume III, *Learning, Systems, Validation, and Position*, carries the architecture into learning and into systems. Part V asks which objectives may be optimised without destroying the structure they were meant to enforce, and proves that some may not; it treats pretraining as the construction of coverage and descent, continual learning as base change along morphisms of sites, and data as the material from which sites are compiled. Part VI develops reinforcement learning in which reward is verified by descent, shows that the group-relative advantage of current methods is a Hodge-theoretic quantity on the nerve of the sampling group, derives an algorithm from that identity, and constrains self-evolution by a geometric restriction with calibrated gates. Parts VII and VIII specify the execution model, kernels, distributed training, serving, the interface and a reference implementation with typed claim records. Parts IX and X give the instruments by which the whole may be checked, the rigour ledger in which every principal claim is graded, and an account of position and risk.

Volume IV, *Provenance and Adjudication*, turns to the sources of the design. Part XI sets out Grothendieck's programme and what of it transfers, the functor of points, motives and the motivic fundamental group, and inter-universal Teichmüller theory, and then rules, import by import, on what may be carried into the architecture. It develops the indeterminacy calculus that binds every reported number in the book to the gauge, basepoint and refinement choices

on which it could depend. It is deliberately the most sparingly graded volume in the book, although it contains some of its deepest mathematics, because its business is adjudication. Its one theorem of its own is the composition law for typed transfers. Inter-universal Teichmüller theory is treated at length and kept in quarantine: what is taken from it is structural, and nothing in the architecture depends on whether the theory is correct.

Volume V, *Transferable Cores, Their Algorithms, and the Replacement Case*, reduces seven lineages to the cores that survive restatement without their vocabulary. Part XII isolates each core and records where it fails; Part XIII turns the surviving cores into algorithms with costs, degeneration conditions and refutation gates, proves an exact one-way degeneration under a declared numerical contract, an overhead theorem and a conditional separation, and specifies a machine-checking kernel with statement-level encodings in Lean 4. Part XIV assesses the replacement case in one place and states what is proved and what remains a research target.

The **apparatus** belongs to the whole work rather than to any volume: notation, collected and expanded proofs, a glossary, a fully worked example with its code, the algorithm compendium, normative pseudocode, the experimental protocol and its pre-registered statistical plan, deterministic task generators, the formalisation sources, the attribution register, the record of negative results and the bibliography. It is consulted from everywhere, and it is where the obligations of the work are collected rather than summarised.

What each volume asks, and what it delivers

Each volume is organised around one question and ends with something a reader can take away and use.

Volume I asks whether the deficits of present models are structural, and whether there is mathematics exact enough to name them. It delivers the diagnosis, the discipline of transfer, and a source layer in which every theorem the rest of the book may rely on is stated with its hypotheses. A mathematician may read Part II as a self-contained account of the sheaf-theoretic, cohomological and higher-categorical mathematics the architecture needs, organised around what each piece makes computable.

Volume II asks what an architecture looks like when that mathematics is spent on it. It delivers a specification of nine layers and of a mixture of experts, with every construction graded, every invariant stated, every degeneration to the standard stack written down, and every open contract printed. Three disciplines govern the layers, and they are proved to be independent of one another: a tower discipline, under which a layer takes the level of the context tower as an explicit argument and preserves both the level and the base at that level; a weight discipline, under which a layer's value map respects the filtration by weight and publishes, for each of its linear maps, whether it is strict; and an almost discipline, which separates a categorical defect from a small numerical error. A type discipline runs across all three. Structure-carrying data, the site, the cover, the nerve and the restriction maps, determine what a value is a value of; value-carrying data, the sections, activations, weights and rewards, live in the fibres that structure determines; and any map that lets values change structure must be declared as a bridge. Much of what the book forbids follows from that one rule.

Volume III asks how such an architecture is trained, run and checked. It delivers objectives with proofs of admissibility, a reinforcement scheme with verified reward, an execution model, a distributed and serving design, an interface with typed claim records and a reference implementation, and the instruments, metrics and ledger by which every claim of the first three volumes can be tested.

Volume IV asks where the design comes from and what of its sources it is entitled to keep. It delivers an adjudication, import by import, of Grothendieck's programme, the functor of points, motives and the motivic fundamental group, and inter-universal Teichmüller theory, together with the indeterminacy calculus and the law by which typed transfers compose.

Volume V asks what survives when each source is restated without its vocabulary, and whether what survives is enough to build on. It delivers seven transferable cores, their algorithms, a unified stack in which they interoperate, a machine-checking kernel, and a replacement assessment that states, in one place and without qualification beyond what the proofs support, what has been shown and what has not.

The disciplines of a layer

A layer in this architecture is not defined until five things are stated about it: what it computes, over which base, what it preserves, what it strips, and what it reports when it cannot do what it is asked. The first two are familiar. The other three are where the theory does its work.

What a layer preserves is stated against the three disciplines and the type rule, and against the sheaf condition itself: of the nine layers, three do not preserve restriction, two of them by design and one by a defect that the naturality residual measures. Each claim of preservation is a theorem or is marked as a design decision. What a layer strips is recorded in a dévissage ledger: the architecture is a filtration, from the full block program down to the token skeleton of the standard stack, and each successive quotient names the structure removed at that step and what the removal forces. The global degeneration table of Chapter 3.1 collects these, so that for every layer a reader can see under which hypotheses it reduces to its standard counterpart and what is lost when it does. What a layer reports when it cannot proceed is part of its type. A solver that cannot certify its result returns a residual and a status; a head that lacks the data it would need abstains; an instrument that cannot decide returns undecided. None of them returns a number in place of an answer it does not have.

These disciplines cost freedom, and the cost is stated. A designer may not reuse a value-carrying quantity to reshape the site without declaring a bridge; may not report an observable without stating the transformations under which it is invariant; may not claim a degeneration without naming the hypotheses under which it holds. The return is a specification in which every component can be checked against its own statement, and in which a failure has an address.

Seven lineages, seven repairs

The mathematics behind the architecture comes from seven lineages, and each contributes a different repair to the typing of a reasoning system rather than a different metaphor for it. Grothendieck's programme makes the base and the ambient category variable, and then asks which finite presentations are tame enough to compute. Deligne's theory of weights adds filtrations and refuses to infer strictness from the mere preservation of a filtration. Faltings's almost mathematics changes equality by an idempotent ideal and separates a categorical defect from a small norm. Voevodsky's constructions separate locality, declared invariance, finite correspondence, effective layers and identity by equivalence. Scholze's mathematics repairs the coefficient category, completion, quotient and the problem of reversible transfer. Lurie's higher-categorical work replaces pairwise compatibility by coherent descent at all levels. Mochizuki's transferable residue keeps independently interpreted structures apart until a typed link states what crosses between them.

Whether these repairs can inhabit one specification is a question the book answers rather than assumes, and the answer is conditional. They interoperate only after three common errors are prohibited: a numerical tolerance is not an almost ideal; a finite pairwise consistency check is not higher descent; and equivalence within one typed universe is not permission to identify objects that belong to different species of specification. The interoperability object of Chapter 12.8 carries refusal states as part of its structure, so that an incompatible pair is guarded rather than coerced into one category.

Six principles drawn from these lineages are written into the architecture as obligations. Every construction is stated over a base, and base change is an operation in its own right. Value-carrying and structure-carrying data are never interchanged. Experts share no ambient universe, so every comparison between them names its link. No quantity is reported without a proof of its invariance or a stated budget for its indeterminacy. The specification is invariant under a declared class of changes of implementation. And structure is removed one layer at a time, each degeneration naming what it removed. Deleting the part of the book that supplies these principles would break definitions, theorems and algorithms elsewhere, which is the test by which their place is earned.

Numbers that know what they depend on

A reasoning system reports numbers: confidences, defects, capacities, costs. A number is worth reporting only if it would not change under choices that should not matter, and the book makes this a rule. Every reported quantity carries either a proof that it is invariant under a declared class of transformations or a stated budget for how far it may move under them. The rule is enforced by an indeterminacy calculus, developed in Volume IV, which treats three kinds of ambiguity as first-class: the choice of gauge in a stalk, the choice of basepoint from which a loop is read, and the choice of cover at which a quantity is computed. Each kind has its own budget, and the budgets compose.

The consequence reaches every part of the book. An attention score that depends on an arbitrary rotation of a stalk is not reported as a score. A holonomy measurement that depends on where a loop starts is reported with that dependence or not at all. A defect computed on one cover and compared with a defect computed on another is compared only through the refinement that relates them. The calculus is where three indeterminacies that inter-universal Teichmüller theory names in its own setting meet three ambiguities that this book carries independently and proves things about; the correspondence is used structurally, and nothing in the calculus depends on whether that theory is correct.

Experts as local topoi

A mixture of experts is the part of present practice that comes closest to the structure this book requires, and the part where the difference is easiest to see. In current systems an expert is a slice of a weight matrix and a router chooses the highest-scoring few. Nothing says what an expert is competent about, what happens where two competences overlap, or whether the selected experts jointly cover the input.

Here an expert is an open sub-topos: the topos of sheaves over an open part of the context site, with its own local logic and its own sections. Routing becomes the construction of a cover by such parts, sufficient for the claim at hand under a stated criterion, rather than the selection of a fixed number of them. The dispatch of an input to an expert and the combination of their outputs are read through an essential adjoint triple, and the implementation that gathers and scatters tokens is designed with that adjunction in view while remaining distinct from it.

Where experts overlap, their sections must agree on the overlap, and the extent to which they fail to agree is measured by the cohomology of the cover. At the next order, the consistency of their pairwise reconciliations is measured by a gerbe, and here the book records one of its own corrections: a first construction of that gerbe was withdrawn after its class was shown to vanish by construction, two no-go theorems explain why every repair of the same shape must fail, and the replacement is built to survive both. Second-order disagreement is, moreover, vacuous on a graph and on a contractible simplex; the smallest configuration on which it can be non-trivial is four experts on the boundary of a tetrahedron.

Experts in this architecture share no ambient universe. Each lives over its own part of the site, and every comparison between two of them names the link along which it is made. That rule, taken from the inter-universal setting and restated without its vocabulary, is what keeps the mixture well typed: no expert's output is read in another's terms without a declared translation.

Mathematics admitted, mathematics refused

A book that draws on this much mathematics owes its reader a rule for what is allowed in, and the rule must be one the author cannot set aside when it is inconvenient. The rule here is **typed transfer**. A construction from another field earns its place only if deleting it would leave a named definition, theorem, algorithm or gate of this book ill-typed: a definition losing a term, a theorem losing a premise, an algorithm losing an input, a gate losing a check. Resemblance is not enough, and neither is a fruitful analogy. Chapter 1.3 makes the rule exact in two stages. A type check asks what in the source is being matched with what in this book, and which statement witnesses the match; a typing without witnesses is not a weak transfer but no transfer at all. An evaporation test then removes the imported vocabulary and asks what remains. What remains is the transferable core. What evaporates is cut.

The rule removes one failure in particular, a failure common enough to have become a genre: a theorem of arithmetic geometry restated in the language of neural networks and thereby made to look as if it had been proved about them. Here a cited result is used at the level of its statement, labelled as such, and never placed as a premise in a proof about the architecture. The checking of hypotheses is a separate obligation that comes second and often fails, and when it fails the book says so. Structure that is stated and never used has no place in the argument. The negative-results appendix records every construction cut and every claim demoted under the rule, with its reason, so that the reader can see what the discipline cost and judge whether it was applied evenly.

The rule is not a test of truth. A construction can pass the evaporation test and still be a poor idea, and the test can show that a transfer is empty without ever showing that it is deep. Its virtue is narrower and more useful: it makes the dependence of every architectural decision on its mathematics visible, so that a reader who doubts a decision knows exactly which source result to examine.

Grades, and the status of every sentence

Claims are graded where they are made. A **theorem** is proved here, or cited with its hypotheses stated and checked. A **measurable** quantity comes with its definition and an algorithm that computes it, even where its interpretation remains open. A **design** decision is named as a decision and is never used as a premise. A **metaphor** is marked as illustration and may justify nothing. Obligations not yet discharged carry their own tokens, so that an open contract is visibly open, an untrainable component is visibly untrainable, and an instrument that

cannot decide a question abstains instead of returning a number. The grade belongs to the sentence and not to the prose around it, and no overview, figure or table may state a claim more strongly than the chapter that establishes it.

This apparatus of grades is not caution for its own sake. A theory of reasoning that could not say, of each of its own statements, what licenses it, would be refuted by its own first page. The book holds itself to the standard it sets for the systems it describes: every conclusion is to be a section that glues from the evidence actually presented, and every place where the evidence does not glue is to be printed as an obstruction rather than smoothed into prose.

What is established

The mathematics of Volume I is established at the grade of theorem, and several results about the architecture's own constructions are proved. The non-abelian capacity of an orthogonal cellular sheaf takes the value zero or a value bounded below by a constant depending only on the stalk dimension, so that it is a binary certificate rather than a continuous score (Chapter 2.11). Generic weighted attention does not factor through the coend that is often invoked to explain it. For every k at least two there is a matched-budget instance on which top-k routing is strictly redundant. Second-order disagreement among experts is vacuous on a graph and on a contractible simplex, so that four experts arranged on the boundary of a tetrahedron are the smallest configuration in which it can be non-trivial. The gluing loss that sheaf neural networks have used as a training term has a minimiser independent of the data, which explains their reported collapse to the identity sheaf. The advantages used by group-relative policy optimisation are the degree-zero Laplacian of the rollout nerve applied to the reward, under two different normalisations (Chapter 6.2). Typed transfers compose according to an explicit law. And, under a declared numerical contract, a parameter injection realises the Transformer's forward map inside this architecture bitwise, with equal gradients and an equal training trajectory (Chapter 13.11). That last equality runs in one direction only. It is an equality of programs, not of architectures, and it is not a claim of replacement.

Several results are negative, and they carry weight. A pairwise difference of per-vertex values is a coboundary by construction, which removed the first form of the obstruction head and forced the form that survives. There is no positive-margin ping-pong certificate between spherical caps for isometries of Euclidean space, so the holonomy dichotomy stands while its certification must admit a third outcome, undecided. The argument against collapse is partial, and the book prints the part it does not cover.

Learning under the obligation to glue

An architecture that is obliged to glue can still be trained out of its obligation, and much of the learning theory in Volume III is about preventing exactly that. The most instructive case is a negative one. Sheaf neural networks were proposed to learn restriction maps along with everything else, and they were observed, again and again, to learn the identity sheaf and become ordinary graph networks. The book shows why. The gluing loss they minimised has a minimiser that does not depend on the data, the zero restriction map, so that an optimiser can satisfy the loss by destroying the structure the loss was meant to enforce. A loss of that kind may never be a primary training term. The consequence is general: an objective is admissible only if the structure it rewards cannot be reached by collapsing the site, and the book proves, objective by objective, which ones meet that condition and which do not.

Pretraining is then read as the construction of coverage and descent rather than as the prediction of the next token alone; continual learning becomes base change along a morphism

of sites, so that what fine-tuning may and may not damage is a statement about what that morphism preserves; and the data from which a site is compiled become an object of the theory, with the attacks that would trivialise a site stated alongside the defences against them. Reinforcement learning is brought under the same discipline. A reward that is verified by descent can be checked against the evidence it claims to summarise, and a self-evolving system is allowed to change only along a declared geometric restriction, with gates whose calibration is itself specified. The identity between group-relative advantage and the degree-zero Laplacian of the rollout nerve is more than a reformulation. It shows that the common practice of subtracting a group mean is a particular choice of cover, and it opens the choice of other covers, weighted by the structure of the sampled group, as a design variable with consequences that can be stated in advance.

Validation as part of the theory

A theory of reasoning that cannot be refuted is not yet a theory, and the book treats its own falsification as part of its content. Every module is required to name a measurement whose outcome would force it to be abandoned. Part IX collects these measurements into instruments and an experimental programme, with pre-registered effect sizes, strong rather than convenient baselines, and an explicit statement of what must be given up in each case of failure. The statistical plan is fixed before any data exist; the task generators are deterministic and carry negative controls designed to expose contamination; and a separating task family of three grammars with one hard core is constructed so that the advantage the theory predicts, if it exists, has somewhere to show itself, and the controls that would refute it are built in alongside.

The book also specifies how its own mathematics is to be checked by machine. A kernel of statement-level encodings in Lean 4, with finite fixtures and a complete trust ledger, separates what has been formalised from what has only been stated, and the replacement ledger records, for every result on which a replacement argument would rest, the grade at which it is held and the machine check, if any, that supports it. Part X then gathers the principal claims into one ledger and states, beside each, what the book does not claim. Read together, these parts turn the programme into a set of decisions that can be taken by experiment and by proof, one at a time, without any of them depending on the author's confidence.

The implementable subset is named precisely. It is a hybrid with a frozen site, token-level top-k routing and a sequence-level solver of fixed form, and it has no measured result yet. That is where the experiments begin.

From this kernel onward

Three lines of work follow from this book, and each is already specified in its pages. The first is experimental: to build the implementable subset, run the pre-registered programme, and let the outcomes decide which modules survive. The second is mathematical: to close the obligations printed as open, of which the most consequential are the certification of holonomy beyond its present third outcome, a composition of the local specialisations into a single forward map, if one exists, and an argument against collapse that also excludes the functional collapse of transmission. The third belongs to the Grothendieckian Programme as a whole: to place this kernel inside the worlds of *The Grothendieckian Machine*, so that a machine which can hold a mathematical world can also reason within it, with every conclusion licensed by the evidence that world presents and every failure to conclude recorded as the obstruction it is.

Limits

Three limits are stated here so that no reader has to look for them. The book does not claim to replace the Transformer: several local specialisations are proved, their hypotheses differ from one another, and no theorem here composes them into a single forward map; the replacement assessment says exactly this and no more. The book reports no experiment: there is no trained model in these pages, and every empirical statement is a protocol with pre-registered endpoints awaiting its run. And it claims no priority over the work on which it builds. Cellular sheaf theory, neural sheaf diffusion, categorical deep learning and equivariant position encoding all exist and are cited. The result in the literature most useful to this book was a negative one, that sheaf neural networks frequently learn the identity sheaf and collapse to ordinary graph networks, and a substantial part of Volume III exists to explain why that happens rather than to hope it will not happen here.

These limits are the shape of a foundational work at its first statement. The mathematics is complete where it is claimed to be; the architecture is specified, with its open contracts printed; the experiments are designed and not yet run. A theory that is right will survive them, and a theory that is wrong should be easy to refute. I have tried to make this one exact enough that its failure would be unambiguous, and to record every argument I could not complete as an open obligation rather than disguise it.

An invitation to the research community

A book of this kind is finished only when other people have tried to break it, and it is written to make that easy. The mathematics of Part II can be checked by anyone who knows it, and the collected and expanded proofs are given in full where the book claims them. The architecture is specified to the level of typed contracts and normative pseudocode, so that an implementation can be compared with the specification clause by clause. The experimental programme is pre-registered: its endpoints, estimators, controls and refuting observations are fixed in advance, and the task generators are deterministic, so that a result obtained by one group can be reproduced by another and a negative result means what it says.

Three kinds of contribution would matter most. The first is refutation: a counterexample to a theorem, a failure of a stated degeneration, or an experiment in which a module fails the gate it names. Each would remove something from the book, and each is welcome. The second is completion: the obligations printed as open are stated precisely enough to be taken up, and several of them are mathematical problems of independent interest. The third is construction: an implementation of the frozen-site subset, run against the pre-registered programme, is the first experiment the theory needs and the one most likely to decide its future. The attribution register and the bibliography record the work on which the book builds; work that builds on it should be able to do the same.

Reading the work

The volumes may be read in order, and that is the order the argument was built for. A mathematician may begin with Part II and the collected proofs, and come to the architecture afterwards. An engineer may begin with the architecture overview of Chapter 3.1 and the compliance matrix of Chapter 1.3, which shows for every module which contracts are discharged and which remain open. A researcher in learning may go from Part I to Parts V and VI and then to the experimental programme of Part IX. A sceptical reader should begin with the rigour ledger of Part X and the replacement assessment of Part XIV, where the book

sets out plainly what it has and has not shown. *How to read this book*, which follows, gives these routes in full and explains how to read a grade.

What this work offers

What the book offers is a theory of the reasoning kernel and a programme for testing it: a stack of constructions meaningful over a non-trivial site, a mixture of experts in which an expert is a local topos, a learning theory that knows which objectives destroy the structure they are meant to enforce, a reinforcement scheme whose reward is checked rather than guessed, and a falsification programme that names, module by module, the measurement whose outcome would end it. Within the Grothendieckian Programme it is the part on which the rest must reason. Whether it is right will be decided by the work that follows, and I have tried to leave that work nothing to guess.

Vincent Liu
WiseMotif.ai
2026

Acknowledgements

This book takes its method from seven mathematical programmes, and the debt is specific enough to be itemised.

From **Grothendieck** I take the relative point of view — that a construction should be stated over a base and that base change should be a first-class operation — together with dévissage, the discipline of stripping structure one layer at a time and naming what was removed at each step. Both run through the book as design law rather than as decoration; the second is what makes the degeneration arguments of Part XIII auditable at all.

From **Deligne** I take weights and strictness: the insight that a filtration can be forced to be canonical rather than chosen, and that morphisms which respect it are thereby constrained. That is the pattern behind the weight-graded layer engine and behind the refusal to let a filtration be a tunable hyperparameter.

From **Faltings** I take almost mathematics: the practice of declaring, in advance, an ideal of differences that are to count as negligible, and then proving statements *up to* that ideal rather than approximately. The book's numerical tolerance discipline is written in that shape, and the difference between "almost equal by declaration" and "equal to within a constant nobody wrote down" is the whole of the borrowing.

From **Voevodsky** I take the insistence that a transfer is a *datum* carried by a construction and not a property it happens to enjoy, and with it a plain accounting of what invariance costs: an invariance assumption buys theorems and forbids distinctions, and both sides must be declared.

From **Scholze** I take tilting as a method: a reversible change of setting in which a hard question becomes a tractable one, with the reversibility proved rather than assumed. The demand that a change of representation come with its inverse is used throughout Part XIII.

From **Lurie** I take higher descent: that coherence data is itself data, that it must be carried rather than assumed, and that the obstruction to gluing lives in a tower rather than at a single level.

From **Mochizuki** I take species and mutations — implementation-invariant specification, in which what is specified is invariant under a declared class of implementation change — and the indeterminacy calculus, in which no quantity is reported without either a proof of invariance or a stated budget for its ambiguity.

Those are debts of method, and I must be equally specific about what they are not. **None of the mathematicians named above has endorsed this work, reviewed it, or is in any way responsible for it.** No theorem of theirs is asserted here of a neural network. Where a result of theirs is used at all it is used at the level of its *statement*, under the grade discipline of Table FM.1, and every such use is registered by result and by source in Appendix A9, which caps the grade at which the import may be consumed. Inter-universal Teichmüller theory is treated at length and quarantined: Scholze and Stix published a specific objection, the author of the theory disputes it, there is no consensus, and nothing in this book depends on the outcome — the dependency quarantine is checked rather than promised.

I also owe a debt to the authors of the negative results this book leans on, several of whom reported that an idea close to one of mine does not work. That literature was worth more to me than the literature that reported success, and Part V is largely an attempt to deserve it.

Errors, overstatements and gaps that remain are mine.

Vincent Liu · WiseMotif.ai

On the imported mathematics, and how it is attributed

A book that borrows this heavily owes its reader a mechanism, not a promise. The mechanism has three parts, and all three are checkable.

The register. Appendix A9 is the source boundary. Its unit is not a work but a *result actually consumed*: a theorem, comparison, classification, construction or status statement whose truth or historical identity matters to a numbered declaration or to a binding policy of this book. Repeated uses of one external result occupy one row, with every internal alias collected, so that citing the same premise in six chapters cannot make one imported fact look like six verified ones. Each row names a controlling source and says whether it is primary — the publication in which the result was first proved — or secondary, in which case the row must explain why a later reformulation is the thing actually being used.

The cap. No declaration receives grade **T** merely because a cited authority proves a load-bearing step. If this book does not print a complete proof of every load-bearing step, the declaration is capped at grade **M** for release purposes, however impeccable its source. A chapter that prints **T** while its proof says *standard*, or *by* a named theorem, or *the cited result gives*, is a grade-reconciliation defect and is recorded as one. This rule is the reason the book contains long proofs of things a specialist would consider routine: the alternative is a grade the reader cannot audit.

The tiers. Every imported item sits in exactly one of three tiers. **Tier G** is transferred mathematics whose hypotheses are verified here and which is used as a theorem. **Tier S** is a pattern of reasoning imported as a design principle, with no claim that any theorem transfers. **Tier A** is analogy, which may motivate and may never justify. A **Tier S** or **Tier A** item appearing as a premise in a grade-**T** argument is an error, and it is an error that can be found mechanically rather than by taste.

The practical consequence for a reader is simple. To learn what is this book's own mathematics, read the register and the proofs; to learn what is somebody else's, read the register and the sources. If a chapter appears to claim more than the register allows, the register governs and the chapter is the defect.

How to read this book

The work is in five volumes bound as one. The volumes are an internal division of a single book, not a series of separate ones: no volume is issued, cited or paginated on its own, a citation to this work is a citation to the bound object and gives the part, chapter and declaration number, and the appendices belong to the whole rather than to any one volume. The division is by argument rather than by length. Each volume ends where a reader may stop with something complete in hand, not at a fixed page span.

The five volumes.

Volume	Title	Parts
I	*Diagnosis and the Source Mathematics*	Preface + I–II
II	*The Architecture and Its Mixture of Experts*	III–IV
III	*Learning, Systems, Validation, and Position*	V–X
IV	*Provenance and Adjudication: The Grothendieck Programme and Inter-Universal Structure*	XI
V	*Transferable Cores, Their Algorithms, and the Replacement Case*	XII–XIV
—	*Apparatus* (appendices, shared by the whole work)	A0–A16

Volume I — *Diagnosis and the Source Mathematics*. Preface and Parts I–II. This volume states what is wrong and supplies the mathematics with which to say it. Part I names five structural deficits and fixes the transplantation discipline: what is carried across is a method of solution, never a source theorem. Part II is the source layer, and the only one: every theorem here is stated at grade **T** and proved or cited, and no claim is made about neural networks. Nothing later in the work may raise a grade this volume did not license.

Volume II — *The Architecture and Its Mixture of Experts*. Parts III–IV. This volume specifies the object with its **OPEN** cells and contracts visible. Part III spends the mathematics of Volume I on a concrete architecture; Part IV specialises it into a mixture of experts whose gating is a site of local topoi. Every construction here is grade **D** — a design decision — and each one names the source result it answers to. The reader who wants the machine and not its provenance may stop at the end of this volume with an architectural specification in hand, and with the currently implementable subset and the open contracts explicitly identified as such. It is not a warrant that the full architecture is implementable, trainable end to end, or production ready: the ledger of Part X names the implementable subset as the frozen-site, token-top-k, sequence-level six-slot-PCG hybrid, and that subset has no measured result either.

Volume III — *Learning, Systems, Validation, and Position*. Parts V–X. This volume specifies how to train, run and measure the object, then states where the programme stands. Objectives, reinforcement learning, execution, distribution and the reference interface are specified to their current contract status, with **OPEN** cells and contracts identified; validation and falsification follow, then the rigour ledger and the risk register. No experiment has been run anywhere in this work: every empirical statement here is a protocol or endpoint that remains unsigned and unrun, or an explicit **UNDETERMINED**; no threshold has been signed.

Volume IV — *Provenance and Adjudication: The Grothendieck Programme and Inter-Universal Structure*. Part XI. This volume is placed here on purpose. About half of it states the deep mathematics — sites, weights, almost mathematics, motives, perfectoid and condensed methods, higher topoi, and inter-universal Teichmüller theory — and the rest adjudicates what may be carried into the architecture of Volumes II and III. It is not a second volume of principles. Its verdicts are grade **D**, **M** or **X**, never **T**, except for the composition law it proves for itself. Read after the earlier volumes, so that its claims can be checked against mathematics already graded.

Volume V — *Transferable Cores, Their Algorithms, and the Replacement Case*. Parts XII–XIV. This volume takes the seven lineages one at a time: it restates each source, isolates the core that survives the evaporation test, records the hypotheses that fail, and turns only the survivors into algorithms with costs, degeneration conditions and falsification gates. It also repairs the earlier volumes where the deep sources demand it. It closes with the replacement case, which is assessed and not asserted, and with the ledger of what would refute it.

***Apparatus*.** Appendices A0–A16. The apparatus serves the whole work and is attached to no volume: the notation, the collected proofs, the glossary, the bibliography, the worked example and its code, the algorithm compendium, the experimental protocol, the attribution register, the pre-registered statistical plan, the expanded proofs and the negative-results appendix. It is consulted from everywhere and is where the book's obligations are collected rather than summarised.

Read across the volumes, the grade profile is the shape of the argument. Volume I is the only place where grade **T** is licensed about the source mathematics, and everything downstream that carries **T** either proves its own statement or descends from a Volume I licence. Volume II is grade **D** for every architectural decision; its **T** statements prove properties of its own constructions, never of the source mathematics. Volume III mixes **D** with the licence tokens, none of which exceeds **S0**. Volume IV is deliberately the weakest-graded volume in the book despite containing some of its deepest mathematics, for the reason its title states. Volume V carries **T** on statements this book proves about its own constructions.

Why the division falls where it does

The obvious alternative division would place Part XI beside Part II, on the ground that both expound mathematics. That reading is half right and is rejected for four reasons, each of which is a measurement rather than a preference.

First, the two parts do different work. Part XI is roughly half exposition of source mathematics and roughly two fifths adjudication — extraction audits, transferability verdicts and stated limits — carried at grades **D**, **M** and **X**. Part II is source mathematics at grade **T** throughout. Binding them under one title that says *principles* would give audit material the cover of a theorem, which is the confusion this book exists to remove; one of the audit chapters says so of itself, recording that it is an extraction audit and not a claim that a neural network carries the structures of arithmetic geometry.

Second, Part II is the single foundation and cannot be divided. It receives 1,744 inbound cross-references, more than any other part, of which 512 come from Part III and 334 from Part IV. A foundation that everything stands on belongs at the front, alone.

Third, Part XI is downstream of Part II in the dependency graph, not parallel to it. Part XI cites Part II 405 times while Part II cites Part XI 63 times, a ratio of about 6.4 to 1. An adjudication layer built on a foundation is not a second foundation.

Fourth, the placement is a named authorial decision rather than a consequence of pagination: the opening of the Grothendieck programme states that it is set late so that everything it

says can be checked against a substantial body of mathematics that has already been graded. The measured alternative is also worse mechanically — merging the two parts drives the imbalance across volumes from about eleven per cent to about one hundred and fifty-six per cent and increases the number of backward references — but the grading argument is the one that decides it.

The apparatus is not a sixth volume for the same kind of reason. Its material is consulted rather than read, it is addressed from every part, and three of its members — the attribution register, the expanded proofs and the negative-results appendix — are the places where the book records what it owes and what it failed to establish. Binding them as a terminal volume would suggest that they close the argument, when their function is to remain open to the whole of it. They are therefore apparatus in the technical sense: the equipment with which the text is checked, not a stage of the text.

A reader who stops at a volume boundary should know what is in hand and what is not. After Volume I: a diagnosis, a transplantation discipline, and a body of mathematics that is true independently of whether the architecture is a good idea. After Volume II: a specification with discharged cells distinguished from **OPEN** cells and contracts, not a claim of complete buildability. After Volume III: the object's training, execution, instrumentation and competitive position specified, with the empirical column still open. After Volume IV: an account of where the mathematics came from and a verdict on how much of it may be carried, which is the volume most likely to change the reader's estimate of the programme in either direction. After Volume V: the algorithms, a machine-checking contract set with **OPEN** cells marked, and the replacement case assessed, not asserted. No boundary is a claim that the material before it has been validated.

How to read a grade

Every substantive claim carries a token, and two different scales are in use. They are not interchangeable and confusing them is the most common way to misread this book.

Token	Scale	What it means	What a reader may write on its strength
T	claim grade	Proved here in full, or a correctly cited external result used at the level of its statement only	The statement, with its hypotheses; nothing about a neural network unless the statement is about one
D	claim grade	A design decision: motivated, possibly by deep mathematics, but not a transplanted theorem	That the book made this choice, and why; never as a premise in a **T** argument
S0	licence	The formal statement has been written down, and where a proof is printed, that proof establishes it	An assertion about a formal statement
S1	licence	A declared finite fixture has been checked	An assertion about that fixture and no larger object
S2	licence	A length-uniform or configuration-uniform check has been performed	An assertion quantified over the declared family

The claim grade is printed in parentheses on the head of a declaration and is fixed by Table FM.1, which also defines the two weaker grades **M** and **X**. The licence token is fixed by Definition 1.2.1 and is assigned by the strongest check *actually executed*, never by the strength of the claim; it appears in ledger columns and in the *Deliberate demotions* section that closes each chapter. Proposition 1.2.3 records the consequence for Chapter 1.2: no claim row in Table 1.2.2 exceeds **S0**; no row carries **S1** or **S2**. The licence token **S2** belongs solely to Part XIII. Identically

named labels in Table 3.8.13's site curriculum and Table 7.4.1's serving determinism classes are unrelated.

Three paths

The book is not meant to be read linearly, and the five paths through the technical parts are set out in the section that follows. Three coarse routes are enough to start.

For a mathematician. Read Chapters 11.1, 11.2 and 11.3 first, for the relative viewpoint, the functor of points and the value/structure split, because Part II is not self-contained and takes its formulations from them. Then Part II in order, then Appendix A1 for the collected proofs and Appendix A10 for the audit-grade versions of the four principal results of Part XIII. The result I would most like checked is the holonomy dichotomy of Chapter 2.11; its *certification* is a separate matter and is currently **UNDETERMINED** for the deployed procedure.

For an artificial-intelligence researcher. Read Part I for the diagnosis and the method, then Part V for the learning theory — the collapse theorem there is the most practically consequential result in the book, because it says which popular objective has a data-independent minimiser — then Part VI for policy optimisation and self-evolution, then Part IX for the falsification programme and Chapter 10.2 for the positioning against state-space models, diffusion language models, test-time training and modern mixtures of experts.

For an engineer. Read Chapter 1.3 first, for the module compliance matrix, and do not skip it: it states for every module which contract cells are discharged and which are **OPEN** or **UNTRAINABLE**, and reading a module without it invites the assumption that an unstated backward map exists. Then Chapter 3.8 for the declared reference configuration and its closed arithmetic cells, Chapter 8.3 for the typed contracts, Part III and Part IV for the architecture, and Part VII for execution, kernels, distributed training and serving. The cheapest thing here to implement is the compositionality defect of Part I, which requires no retraining.

The reader's guarantees

Five editorial commitments follow. Their checks must be read at their recorded status; an **OPEN** check is not enforcement.

Every claim carries its grade at the point of use. No overview, appendix, figure, table or API entry may state a claim more strongly than the chapter that establishes it; where two places disagree, the typed claim registry of Chapter 8.3 governs and the discrepancy is a defect to be reported rather than a matter of interpretation.

Every gap is recorded as a gap. Where an argument could not be completed, the book prints a gap record naming the missing step, what would close it, and the grade to which the enclosing statement falls while it is open. Concealing a gap is treated as a more serious failure than having one, and the appendices that collect them are placed where they cannot be missed.

No experiment is claimed. There is no trained model behind any sentence in this book. Where a plot would ordinarily display data it displays a labelled schematic or it does not appear, and every empirical gate is unsigned and unrun, with that status printed in the compliance matrix rather than in a footnote. No empirical threshold has been signed.

Every number states its evidentiary status. Chapter 3.8 declares the reference configuration, formulae and partial cost cells that are presently closed. Complete totals, ratios, atomisation and publication-level regenerated-view verification remain **OPEN**; no signed publication guarantee is asserted.

Every import is attributed and capped. Imported mathematics is registered by result and by source, used at the level of its statement, and prevented by the release rule from lifting a declaration to a grade the book has not earned.

Two things are *not* promised. The book does not promise that the programme works; it promises only that closed contracts carry stated failure conditions, while **OPEN** contracts do not. And it does not promise completeness of the mathematics it borrows: the register records what was consumed, not what could have been.

Abstract

GM-Omega-Infinity: Topos-Native Intelligence, by **Vincent Liu** (WiseMotif.ai), is a monograph in five volumes bound as one. It sets out the meta-reasoning kernel of the Grothendieckian Programme: a theory in which reasoning over evidence is native to a Grothendieck topos, together with the architecture, learning theory and validation programme that follow from it.

Contemporary language models are strong interpolators and unreliable reasoners. The usual explanations are scale, data quality and the absence of explicit search. This work locates a large part of the deficit in structure. A model's context, as ordinarily presented, carries no non-trivial notion of covering, so the condition that local evidence must glue is satisfied vacuously; whatever coherence such a model shows is coherence it has learned to imitate, not coherence it is obliged to possess. The work makes that obligation part of the architecture. A context becomes a site, the evidence it carries a sheaf, and a conclusion a global section that exists only where local evidence glues; the failure to glue becomes an obstruction that can be computed and located. The argument proceeds in three movements.

The first is **diagnostic**. Five recurrent failure modes, locality without gluing, composition without compositionality, confidence without justification, adaptation without preservation and capacity without structure, are read as consequences of the missing covering structure. The reading is a diagnostic convention, graded as a design criterion; positional encodings, masks and learned attention do impose relations among positions, and the claim is only that those relations are not known to satisfy the axioms of a covering. That the five failures share one cause is stated as a conjecture, with the condition under which it would be false.

The second is **constructive**. Nine layers, L0 to L8, rebuild the reasoning stack over a single base: a **context site** built from evidence atoms; **sheaf embeddings** with stalks of heterogeneous dimension; **descent attention**, for which it is proved that the indiscrete identity coend has the unweighted codiagonal quotient while generic weighted attention does not factor through it; **nucleus activations**; a **descent solver** given one normative program with an exact reverse through its realised arithmetic; an **obstruction head** that consumes independently supplied overlap data and abstains without it; an **internal-logic head** implementing Kripke–Joyal forcing; a **holonomy layer** measuring non-abelian capacity, which on an orthogonal cellular sheaf is either zero or bounded below by a constant depending only on the stalk dimension; and a **transfer layer** built from morphisms of sites. Several modules carry a proved one-way specialisation to their standard counterparts under an explicit set of hypotheses. Those hypotheses differ from module to module, and no theorem in this work composes them into a single forward map.

The third movement extends the kernel to experts and to learning. In **Topos-MoE** an expert is an open sub-topos, routing is the construction of a sufficient cover rather than a top-k selection, and the essential adjoint triple $\Sigma_U \dashv U^* \dashv \Pi_U$ is the formal counterpart of dispatch and combine; the production gather–scatter implementation is motivated by that adjunction and is not identified with it. For every $k \geq 2$ there is a matched-budget instance on which top-k routing is strictly redundant, and second-order disagreement among experts vanishes identically on a graph and on a contractible simplex, so that four experts on $\partial\Delta^3$

form the smallest configuration in which it can be non-trivial. In **learning**, the gluing loss used by sheaf neural networks is shown to have a minimiser independent of the data, which explains their reported collapse to the identity sheaf and forbids that loss as a primary training term. In **reinforcement learning and self-evolution**, reward is verified by descent rather than predicted, the advantages of group-relative policy optimisation are identified with the degree-zero Laplacian of the rollout nerve, and a policy-optimisation algorithm is derived for a decoder that is parallel and revisable; it is presented as a biased trust heuristic with its bias stated.

Behind the architecture stand seven mathematical lineages, Grothendieck, Deligne, Faltings, Voevodsky, Scholze, Lurie and Mochizuki. Each import is admitted only if deleting it would leave a named definition, theorem, algorithm or gate of the work ill-typed, and only the core that survives restatement without its source vocabulary is carried across. Those cores become algorithms with stated costs, degeneration conditions and refutation gates. The work proves that typed transfers compose according to an explicit law; that, under a declared numerical contract, a parameter injection realises the Transformer's forward map inside the architecture bitwise, with equal gradients and an equal training trajectory, an equality of programs in one direction only; an overhead theorem; and a conditional separation by monodromy. Inter-universal Teichmüller theory is treated at length and quarantined: nothing in the architecture depends on its correctness.

The status of the whole is stated once and is not softened anywhere:

The status of the work. GM-$\Omega\infty$ is a research architecture and structural augmentation for Transformer and mixture-of-experts systems. It has several proved local specialisations and no proved global subsumption or replacement result.

Accordingly no part of the work describes the architecture as containing, strictly subsuming or replacing the Transformer, or as supplying a performance floor. Replacement appears only as a research target with a stated discharge condition. The reference configuration still contains grouped-query Transformer attention, so the object exhibited is a hybrid. Analytic cost subtotals for the declared configurations are reported in Table 3.8.9; complete totals and ratios remain open, and no comparison of quality exists.

Negative results are treated as findings. A pairwise difference of per-vertex values is a coboundary by construction, which invalidated the first form of the obstruction head. There is no positive-margin ping-pong certificate between spherical caps for isometries of $\mathbb{R}^d$, so the holonomy dichotomy survives while its certification must admit a third outcome, undecided. The anti-collapse argument is partial: zero and literal-identity collapse of the parameters are excluded, functional collapse of transmission is not.

Every substantive claim carries one of four grades, **T** theorem, **M** measurable, **D** design criterion and **X** metaphor, and every unfinished obligation one of four status tokens, **OPEN**, **UNTRAINABLE**, **UNDETERMINED** and **WITHDRAWN**. A principal-claim ledger in Part X lists, with each claim, what the work explicitly does not claim. Every module names a measurement whose outcome would force it to be abandoned, and Part IX collects these into an experimental programme with pre-registered effect sizes, strong baselines and a statement of what must be given up in each case of failure. No experiment has yet been run. The theory is offered in a form exact enough for those experiments to decide it.

Contents

Series Note iv

Dedication v

Preface vi
- The question of the local . . . vi
- The Grothendieckian turn . . . vi
- The place of this work in the Grothendieckian Programme . . . vii
- Worlds, sites and the interface with the first work . . . vii
- The kernel in seven operations . . . viii
- The architecture of the work . . . ix
- What each volume asks, and what it delivers . . . x
- The disciplines of a layer . . . xi
- Seven lineages, seven repairs . . . xi
- Numbers that know what they depend on . . . xii
- Experts as local topoi . . . xii
- Mathematics admitted, mathematics refused . . . xiii
- Grades, and the status of every sentence . . . xiii
- What is established . . . xiv
- Learning under the obligation to glue . . . xiv
- Validation as part of the theory . . . xv
- From this kernel onward . . . xv
- Limits . . . xvi
- An invitation to the research community . . . xvi
- Reading the work . . . xvi
- What this work offers . . . xvii

Acknowledgements xviii

On the imported mathematics, and how it is attributed xx

How to read this book xxi
- Why the division falls where it does . . . xxii
- How to read a grade . . . xxiii
- Three paths . . . xxiv

The reader's guarantees xxv

Abstract xxvi

The claim vocabulary ci

Where the generative principles are used ciii

Five paths through the book cv
Conventions . . . cvii
What this book is not . . . cvii

Part I — Diagnosis and Method 1

Chapter 1.1 — The five structural deficits 2
1.1.1 The three standard explanations, and where each stops . . . 2
1.1.2 What this chapter claims, and at what grade . . . 3
1.1.3 Locality without gluing . . . 4
1.1.4 Composition without compositionality . . . 6
1.1.5 Confidence without justification . . . 7
1.1.6 Adaptation without preservation . . . 8
1.1.7 Capacity without structure . . . 9
1.1.8 The common origin, stated as a conjecture . . . 10
1.1.9 The observables this chapter licenses . . . 10
1.1.10 Which declaration repairs which deficit, and which part is not repaired . . . 11
1.1.11 What this diagnosis is not competing with . . . 13
1.1.12 The negative results the diagnosis must respect . . . 14
1.1.13 What the diagnosis costs . . . 15
1.1.14 What would make this diagnosis wrong . . . 16
1.1.15 Deliberate demotions . . . 16

Chapter 1.2 — The central thesis and the claims it rests on 18
1.2.1 The thesis . . . 18
1.2.2 What a local specialisation does and does not mean . . . 19
1.2.3 The claim ledger . . . 19
1.2.4 The five contributions we would defend . . . 21
1.2.5 The five things we are least sure of . . . 22
1.2.6 The claim ledger read a second time: licence grade and falsification condition 22
1.2.7 The three parts of the thesis, and their logical independence . . . 26
1.2.9 Deliberate demotions . . . 27

Chapter 1.3 — The method of transplantation 28
1.3.1 The failure mode, stated precisely . . . 28
1.3.2 Six status words, and what each licenses . . . 29
1.3.3 The three moves . . . 29
1.3.4 The module contract . . . 31
1.3.5 The compliance matrix . . . 32
1.3.6 The regression obligation . . . 36
1.3.7 Where the generative principles is used . . . 37
1.3.8 What is done with the deep theorems that cannot be used . . . 38
1.3.9 The transplantation type check . . . 38
1.3.10 The evaporation test . . . 41
1.3.11 The canonical locus, the two names, and the named variants . . . 44
1.3.12 The document's own falsifiability . . . 46

1.3.13 Deliberate demotions 46

Part II — Mathematical Principles **48**

Chapter 2.1 — Topos foundations: sites, sheaves, nuclei and the internal logic **49**
2.1.1 The problem with the token sequence 49
2.1.2 Categories of contexts and presheaves 50
2.1.3 Sieves and Grothendieck topologies 51
2.1.4 The Context Site, and the hypothesis the baseline omits 51
2.1.5 Sheaves, separated presheaves, and sheafification 58
2.1.6 The subobject classifier and the Heyting structure of truth 59
2.1.7 The idempotence–differentiability trade-off 60
2.1.8 Frames, locales, nuclei and sublocales 61
2.1.9 Theorem , fully corrected: ReLU is the closed nucleus at zero 61
2.1.10 Which topos, exactly — and the accurate gap 62
2.1.11 Which activations are nuclei, and which are not 63
2.1.12 The LayerNorm and max-pool correction 65
2.1.13 The modality tower 66
2.1.14 Kripke–Joyal forcing 67
2.1.15 Geometric morphisms, the vacuity objection, and what actually survives 68
2.1.16 Points, stalks, and how much of Deligne's theorem the architecture may use 74
2.1.17 Where the architecture sits on the topology ladder 77
2.1.18 Degeneration condition 80
2.1.19 Summary of grades for this chapter 80

Chapter 2.2 — The Grothendieck-topology ladder **83**
2.2.1 The ladder as an architectural design space 83
2.2.2 Voevodsky cd-structures and finite Nisnevich descent 89
2.2.3 Condensed and solid coefficients 94
2.2.4 Motivic descent over condensed coefficients 101
2.2.5 Architectural consequences and falsifiable predictions 105

Chapter 2.3 — Internal logic, dependent type theory and univalent transport **112**
2.3.1 Forcing at general and generic stages 112
2.3.2 Martin-Löf type theory as internal language 118
2.3.3 Univalence as a transport mechanism 123
2.3.4 Higher inductive types for architectural gluing 128
2.3.5 Lean 4 formalization boundary 132

Chapter 2.4 — Infinity-topos foundations: coherence, transport, shape and approximation **138**
2.4.1 Quasicategories and the coherence that the architecture was missing . . . 138
2.4.2 Hypercompletion, Postnikov convergence and the boundary of pointwise tests 144
2.4.3 Giraud's infinity-topos axioms and the two categories that must not be confused 151
2.4.4 Straightening, unstraightening and the unresolved 12.6.3 square 155
2.4.5 Shape as a context invariant 159

2.4.6 Ambidexterity, norm maps and the pooling invariant . . . 163
2.4.7 Goodwillie calculus as a controlled interaction hierarchy . . . 166
2.4.8 Architectural consequences, costs and falsifiable predictions . . . 171
2.4.9 Rigour ledger . . . 176

Chapter 2.5 — Cohomology, descent and the obstruction to gluing **177**
2.5.1 Čech complexes on a site . . . 177
2.5.2 Cellular sheaves on graphs . . . 180
2.5.3 The constant sheaf and oversmoothing . . . 181
2.5.4 The relation between H^1, coker δ^0 and ker δ^1 . . . 182
2.5.5 Well-posedness of the architecture . . . 182
2.5.6 Numerics: spectral gap, conjugate gradients, and exact costs . . . 184
2.5.7 Descent data, stacks and the cocycle condition . . . 185
2.5.8 Gerbes, bands, and the classifying class . . . 187
2.5.9 The fatal dimension objection . . . 188
2.5.10 Cocycle failure is not a cohomology class . . . 189
2.5.11 Situating the computable theory inside the general one . . . 191
2.5.12 Exact sequences and their architectural use . . . 194
2.5.13 A complete worked computation . . . 195
2.5.14 Numerical rank, and why dim H^0 should not be reported as an integer . . . 196
2.5.15 The normalised sheaf Laplacian and preconditioning . . . 196
2.5.16 From the Context Site to a cellular sheaf . . . 197
2.5.17 Functoriality: refinement, and why the obstruction is a genuine invariant 199
2.5.18 Module template: the cohomology analyser . . . 203
2.5.19 Summary of grades for this chapter . . . 203

Chapter 2.6 — Deligne weights, strictness and canonical degeneration **206**
2.6.1 Scope, grading and source boundary . . . 206
2.6.2 Frobenius, purity and the two Weil theorems . . . 206
2.6.3 Mixed Hodge structures . . . 208
2.6.4 Strict filtered morphisms . . . 210
2.6.5 The monodromy weight filtration . . . 214
2.6.6 Neutral Tannakian reconstruction . . . 216
2.6.7 Weight-filtered representations in GM-$\Omega\infty$ . . . 219
2.6.8 Degenerating contexts and canonical monodromy weights . . . 225
2.6.9 Compatibility with the inherited topos foundations . . . 228
2.6.10 Negative and conditional results . . . 231
2.6.11 Rigour ledger . . . 234
2.6.12 Deliberate demotions . . . 236

Chapter 2.7 — Almost mathematics: dense valuations, negligible error and scale architecture **237**
2.7.1 Scope, evidence grades and the archimedean firewall . . . 237
2.7.2 Valuation-theoretic setup and idempotence . . . 238
2.7.3 Almost-zero modules and the Serre subcategory . . . 239
2.7.4 The Gabriel quotient and its universal property . . . 241
2.7.5 Quantitative almost isomorphisms and exact composition constants . . . 245
2.7.6 Almost purity in Faltings' and Scholze's forms . . . 247
2.7.7 Heights: the Arakelov decomposition and Northcott finiteness . . . 250

2.7.8 Architectural fusion: dense scales versus finite ladders 254
2.7.9 Layerwise almost equivalence and accumulated constants 258
2.7.10 Falsifiable predictions, limitations and open interfaces 260
2.7.11 Ledger of results and dependency firewall 264

Chapter 2.8 — Higher descent, monadic reconstruction and ambidextrous aggregation 270
2.8.1 Scope, interfaces and the globally open square 270
2.8.2 Hypercovers and hyperdescent 271
2.8.3 The descent spectral sequence 274
2.8.4 Infinity-categorical Beck monadicity 279
2.8.5 The norm transformation 282
2.8.6 Ambidexterity and its failure 286
2.8.7 Architectural fusion 290
2.8.8 Negative results, conditional interfaces and audit ledger 294
2.8.9 Chapter conclusions 296

Chapter 2.9 — Kan extensions, coends, and what attention actually computes 298
2.9.1 Ends and coends 298
2.9.2 The enriched setting 299
2.9.3 What attention is: setting up 300
2.9.4 The compositionality defect 302
2.9.5 Softmax does not supply missing coend data 303
2.9.6 The T-Block: a transport-and-descent design 304
2.9.7 Further structure: profunctors, limits, and the residual stream 306
2.9.8 The exact weighted-coend theorem—and its failure boundary 309
2.9.9 Summary of grades for this chapter 315

Chapter 2.10 — Enriched categories, weighted colimits, and coherent profunctor composition 318
2.10.1 The enriching base and its closure 318
2.10.2 Enriched categories, functors, and transformations 319
2.10.3 Ends, coends, and their existence 320
2.10.4 Weighted limits and colimits 322
2.10.5 Enriched Yoneda and density 323
2.10.6 The bicategory of enriched profunctors 325
2.10.7 Associativity, units, and coherence 326
2.10.8 Concrete specialisations and audit cases 328
2.10.9 Representables inside **Prof** and finite weighted sums 331
2.10.10 What this machinery does—and does not—license for attention 334
2.10.11 Summary of grades 335
2.10.12 Deliberate demotions 336

Chapter 2.11 — Non-abelian holonomy and the capacity–consistency trade-off 338
2.11.1 The orthogonal sheaf and its transport 338
2.11.2 Holonomy determines the space of global sections 342
2.11.3 The capacity–consistency trade-off 344
2.11.4 Two invariants of the holonomy group 349
2.11.5 The holonomy dichotomy 350

2.11.6 Certifying the free regime: the ping-pong lemma, and why it is subtle here 352
2.11.7 Why fixed-tolerance counting has zero asymptotic slope 355
2.11.8 Instrument N6: the falsification protocol 361
2.11.9 Holonomy as a representation, and the construction of the intermediate regime 363
2.11.10 Audit boundary: negative and conditional conclusions 365
2.11.11 What is theorem and what is wager 366
2.11.12 Summary of grades for this chapter 367

Chapter 2.12 — Fundamental groups, Galois categories, and the one-type boundary 369
2.12.1 Scope, conventions, and certification boundary 369
2.12.2 Grothendieck's Galois categories: the six axioms 370
2.12.3 The fundamental theorem: the fundamental group is the automorphism group of the fibre functor 373
2.12.4 Base points, path torsors, and outer automorphisms 381
2.12.5 The pro-étale fundamental group 385
2.12.6 The shape of an infinity-topos and the deliberate one-type downgrade . . 389
2.12.7 Architectural payload: a parallel Galois branch for L7 394

Chapter 2.13 — Monodromy filtration, lower-central weights, and the graded holonomy scan 408
2.13.1 Scope, dependencies, and the release source boundary 408
2.13.2 Nilpotence, unipotence, and finite polynomial calculus 409
2.13.3 The monodromy filtration: statement and cited existence 409
2.13.4 The monodromy filtration: full uniqueness proof 410
2.13.5 What the graded isomorphisms buy: the rigidity criterion 413
2.13.6 Small matrices: two valid filtrations and two failures 414
2.13.7 Intrinsic formula, rank profile, and an executable rigidity audit 415
2.13.8 The lower-central weight filtration on a fundamental group 419
2.13.9 Pronilpotent and Lie-algebra forms 422
2.13.10 Two distinct zero mechanisms: compact tolerance and abelianisation . . 425
2.13.11 The corrected algorithm: `GradedScan` on weight -2 428
2.13.12 Complexity against the actual loops 437
2.13.13 How the two filtrations meet, and why they remain different 438
2.13.14 Worked example I: the integer Heisenberg group 439
2.13.15 Worked example II: torsion in orthogonal holonomy 441
2.13.16 Combining monodromy rigidity with the graded scan 443
2.13.17 What the corrected scan changes in the architecture 445
2.13.18 Mandatory conformance and adversarial tests 445
2.13.19 Failure boundaries and open obligations 447
2.13.20 Rigour ledger 448
2.13.21 Final conclusion 449

Chapter 2.14 — Separation complexity: uniform circuits, width-five programs, and the accurate scan boundary 451
2.14.1 Scope, source boundary, and the one convention that must not drift . . . 451
2.14.2 Circuit families and three non-interchangeable uniformity notions 452
2.14.3 The class relation and its exact epistemic status 454
2.14.4 Why PARITY lies in TC^0 454

2.14.5 Random restrictions and the imported switching lemma 455
2.14.6 The switching-lemma proof that PARITY is not in AC^0 455
2.14.7 Why TC^0 is contained in NC^1 457
2.14.8 Branching programs: the apparently weak model 458
2.14.9 The five-state group fact 459
2.14.10 Why a non-solvable group is required 460
2.14.11 Barrington's induction in full 461
2.14.12 From NC^1 circuits to polynomial-length width-five programs 463
2.14.13 The converse: bounded width is in NC^1 463
2.14.14 The S_5 word problem is NC^1-complete 464
2.14.15 Merrill–Sabharwal: what the transformer upper bound actually says . . 466
2.14.16 Hypothesis one: logarithmic attention and arithmetic precision 466
2.14.17 Hypothesis two: one effective length-uniform transformer family 467
2.14.18 Hypothesis three: depth is fixed with respect to sequence length 468
2.14.19 The exact licensed conclusion 469
2.14.20 Associative scan: the exact work and depth 469
2.14.21 Intake from Chapters 2.11, 2.12, and 2.13 471
2.14.22 Exact S_5 scan as an NC^1-complete value-path primitive 471
2.14.23 Macro-layers are not circuit depth 472
2.14.24 Worked construction: one scan carrier, two complexity levels 473
2.14.25 The conditional architectural separation 474
2.14.26 Conditions under which the architectural claim fails 475
2.14.27 Exactness, robustness, and approximation are separate questions 476
2.14.28 What the scan buys architecturally 476
2.14.29 Uniformity audit across the whole argument 477
2.14.30 Proof anatomy of the transformer simulation 478
2.14.31 A worked width-five compilation 480
2.14.32 The exact scan implementation contract 481
2.14.33 Finite windows and asymptotic claims 483

Chapter 2.15 — The six operations, Tannakian reconstruction, and the anabelian surrogate **485**
2.15.1 The six operations 485
2.15.2 Beck–Chevalley squares 490
2.15.3 The Beck–Chevalley deviation as an order-sensitivity probe 502
2.15.4 Inverse demand: $f^!$ and reverse retrieval 503
2.15.5 Tannakian reconstruction, and why it does not apply 506
2.15.6 A note on motives proper 507
2.15.7 The anabelian programme 507
2.15.8 The finite surrogate: torsors, and a theorem that rescues the intuition . . 508
2.15.9 Degeneration and failure modes for this chapter's instruments 515
2.15.10 Open obligations and release boundary 515
2.15.11 Summary of grades for this chapter 516

Chapter 2.16 — Motives, transfers, and Tannakian reconstruction **519**
2.16.1 Scope: the Grothendieck mainline, with the conjectural joints exposed . . 519
2.16.2 Pure motives begin with cycles, not cohomology 520
2.16.3 The standard conjectures: exact statements and conditional consequences 522

2.16.4 Finite correspondences: the primitive this book actually needs 526
2.16.5 The architectural decision: use transfers before functions 530
2.16.6 Presheaves with transfers 532
2.16.7 Nisnevich descent: local agreement with residue-field control 533
2.16.8 $\mathbb{A}^1$-localisation: algebraic homotopy as an imposed equivalence 535
2.16.9 A fixed construction of Voevodsky's DM 538
2.16.10 Motivic cohomology: representability inside DM 542
2.16.11 Milnor K-theory and the norm residue theorem 543
2.16.12 From triangulated motives to an abelian heart: the open bridge 548
2.16.13 Neutral Tannakian reconstruction, recalled once and then applied conditionally 548
2.16.14 Pure motives against the Tannakian hypotheses 550
2.16.15 Mixed motives against the Tannakian hypotheses 551
2.16.16 Binding alignment with Chapter 11.3: all five failures remain in force . . 552
2.16.17 A complete finite-transfer worked example 552
2.16.18 Transfer admission and failure semantics 554
2.16.19 Two correspondence calculi, and why neither may replace the other . . . 555
2.16.20 Low-degree readings of the norm residue theorem 555
2.16.21 Result-by-result rigour ledger 556
2.16.22 Open obligations 557
2.16.23 Final dependency boundary and release contract 558

Chapter 2.17 — Anabelian depth: reconstruction, kernels, and finite shadows 559
2.17.1 Scope, source boundary, and the question this chapter answers 559
2.17.2 Grothendieck proper: the exact anabelian assertion 560
2.17.3 The restatement test at the Grothendieck stage 563
2.17.4 Faltings: the abelian shadow is a reconstruction theorem with a quotient 564
2.17.5 Deligne: two weight axes and the first information beyond Tate 568
2.17.6 Bhatt–Scholze pro-étale theory and Scholze/Fargues–Fontaine untilting . 571
2.17.7 Lurie: shape, one-type truncation, and a deliberate downgrade 574
2.17.8 Kernel calculus and the refused Voevodsky transfer 575
2.17.9 Mochizuki: structural prerequisites, mono-anabelian direction, and the binding cut 577
2.17.10 The complete finite-proxy theorem 581
2.17.11 Six explicit kernel audits 585
2.17.12 Restatement test, deletion decisions, and lineage fusion 588
2.17.13 Load-bearing lineage map 592
2.17.14 Grade and bibliography-gap ledger 593
2.17.15 Declaration-consumer ledger 594
2.17.16 Final reconstruction boundary 595
2.17.17 Release conclusion and unresolved obligations 597

Part III — The GM-Ω∞ Architecture 599

Chapter 3.1 — Architecture overview: nine layers, one dataflow, one degeneration obligation 600
3.1.1 The tower of fibrations 602
3.1.2 The nine layers 606

3.1.3 Why these nine, and what each augments . . . 607
3.1.4 The type discipline: value-carrying and structure-carrying data . . . 608
3.1.5 Dataflow . . . 615
3.1.6 The invariants each layer preserves . . . 617
3.1.7 Dévissage: what each layer strips, and what that forces . . . 618
3.1.8 The global degeneration table . . . 620
3.1.9 The end-to-end degeneration theorem . . . 621
3.1.10 A hypothetical representational floor would not be an optimisation guarantee 621
3.1.11 The invariance class of every observable . . . 622
3.1.12 Reading order . . . 629
3.1.13 Deliberate demotions . . . 630

Chapter 3.2 — L0 — the context site **632**
3.2.1 What is being replaced, and what a replacement must supply . . . 632
3.2.2 Evidence atoms . . . 633
3.2.3 The context frame . . . 634
3.2.4 The three coverage axioms, verified . . . 641
3.2.5 Covering as sufficiency: the minimal sufficient cover . . . 643
3.2.6 Why sufficiency rather than similarity: the logical motivation . . . 646
3.2.7 Reverse retrieval: coverage as a demand signal . . . 647
3.2.8 The site induced by a token stream, and what the standard objectives measure 647
3.2.9 Degeneration . . . 648
3.2.10 L0 as the base of the tower, and the architecture's only bridge . . . 648
3.2.11 Observables and falsification . . . 649
3.2.12 Deliberate demotions . . . 651

Chapter 3.3 — L1 and L2 — the sheaf embedding and descent attention **653**
3.3.1 L1 — the sheaf embedding . . . 653
3.3.2 L2 — descent attention . . . 662
3.3.3 Deliberate demotions . . . 678

Chapter 3.4 — L3 and L4 — nucleus activation and the descent solver **680**
3.4.1 L3 — nucleus activation . . . 680
3.4.2 L4 — the descent solver . . . 701
3.4.3 Deliberate demotions . . . 724

Chapter 3.5 — L5 and L6 — the obstruction head and the internal logic head **726**
3.5.1 L5 — the obstruction head . . . 726
3.5.2 L6 — the internal logic head . . . 743
3.5.3 The joint certificate . . . 753
3.5.4 Deliberate demotions . . . 755

Chapter 3.6 — L7 and L8 — holonomy and the transfer layer **758**
3.6.1 L7 — holonomy and non-abelian capacity . . . 758
3.6.2 L8 — the transfer layer . . . 778
3.6.3 The falsification register . . . 786
3.6.4 Deliberate demotions . . . 789

Chapter 3.7 — Alien Engine and T-Unit **792**

3.7.1 Position in the relative L0–L8 tower . . . 793
3.7.2 The corrected T-Unit . . . 794
3.7.3 The corrected Alien Engine . . . 803
3.7.4 Invariance and budget register . . . 810
3.7.5 Worked transport . . . 812
3.7.6 Falsification gates . . . 812
3.7.7 Executable API boundary and conformance records . . . 813
3.7.8 Dévissage and final accounting . . . 815
3.7.9 A complete T-Unit execution . . . 816
3.7.10 T-Unit conformance interface . . . 817
3.7.11 Learning the links without learning an identification . . . 819
3.7.12 Link families for the architecture . . . 819
3.7.13 Indeterminacy estimation protocol . . . 820
3.7.14 Security and provenance . . . 821
3.7.15 Additional rejection tests . . . 821
3.7.16 Experimental protocol for the restored organs . . . 821
3.7.17 Deliberate demotions . . . 823

Chapter 3.8 — Model configuration and schedules **825**
3.8.1 Configuration as a relative, typed object . . . 825
3.8.2 One normative configuration and two fixtures . . . 829
3.8.3 Parameter accounting . . . 831
3.8.4 The invalid stalk schedule, diagnosed . . . 832
3.8.5 Fixed-shape rank continuation . . . 832
3.8.6 Shape-stable active-rank schedule and the unavailable frame theorem . . . 834
3.8.7 End-to-end training schedule . . . 835
3.8.8 Safe structural edits . . . 838
3.8.9 Withdrawn global degeneration and local conformance . . . 838
3.8.10 Cost budgets . . . 839
3.8.11 Failure modes . . . 841
3.8.12 Falsification programme . . . 842
3.8.13 Reproducible manifest . . . 842
3.8.14 Final schedule ledger . . . 843
3.8.15 Research configuration bill of materials . . . 843
3.8.16 Site-construction schedule . . . 845
3.8.17 Data mixture and evidence construction . . . 849
3.8.18 Loss schedule and Iron Rule 2 . . . 850
3.8.19 Learning-rate and gate synchronisation . . . 850
3.8.20 Checkpoint and replay cadence . . . 851
3.8.21 RL and self-evolution entry schedule . . . 851
3.8.22 Evaluation cadence . . . 852
3.8.23 A worked transition . . . 853
3.8.24 Configuration review checklist . . . 853
3.8.25 Deliberate demotions . . . 854

Chapter 3.9 — The weight-graded layer engine **856**
3.9.1 Position in the relative L0–L8 tower . . . 857
3.9.2 Static schemas and the admission boundary . . . 859

3.9.3 `StrictnessCheck` . . . 859
3.9.4 `MonodromyFiltration` . . . 862
3.9.5 `GradedScan` . . . 865
3.9.6 Zero runtime, with the accounting made explicit . . . 867
3.9.7 Invariance and budget register . . . 869
3.9.8 Worked transport . . . 870
3.9.9 Falsification gates . . . 872
3.9.10 Executable API boundary and conformance records . . . 873
3.9.11 Dévissage and final accounting . . . 874
3.9.12 A complete execution . . . 875
3.9.13 conformance interface . . . 885
3.9.14 Deliberate demotions . . . 890

Chapter 3.10 — The almost-descent engine **893**
3.10.1 Position in the relative L0–L8 tower . . . 893
3.10.2 The almost-descent engine . . . 895
3.10.3 Invariance and budget register . . . 907
3.10.4 Worked transport . . . 908
3.10.5 Falsification gates . . . 911
3.10.6 Executable API boundary and conformance records . . . 912
3.10.7 Dévissage and final accounting . . . 913
3.10.8 A complete execution . . . 914
3.10.9 conformance interface . . . 927
3.10.10 Deliberate demotions . . . 930

Part IV — Topos Mixture-of-Experts **935**

Chapter 4.1 — The expert as a local topos **936**
4.1.1 Setting, notation, and the ambient semantic topos . . . 937
4.1.2 Candidate (a): the sub-topos cut out by a Lawvere–Tierney nucleus . . . 938
4.1.3 Candidate (b): the open sub-topos, equivalently the slice . . . 940
4.1.4 Candidate (c): sheaves on a sub-site . . . 945
4.1.5 The recommendation . . . 946
4.1.6 Localic families versus two-categorical ones . . . 947
4.1.7 What is imposed and what is discovered . . . 947
4.1.8 Experts that share no ambient universe: the link discipline . . . 949
4.1.9 The Topos-MoE layer: summary definition . . . 955
4.1.10 The routing objective carries the ambidexterity defect . . . 955
4.1.11 Module template for the representation layer . . . 961
4.1.12 Deliberate demotions . . . 964

Chapter 4.2 — Routing as coverage construction **967**
4.2.1 Typed cover families and the detached coverage shadow . . . 968
4.2.2 The atom frame . . . 971
4.2.3 Full MSC and the relative maximum-coverage shadow . . . 972
4.2.4 The differentiable relaxation, and the objective it is *not* . . . 974
4.2.5 The routing objective, and why coverage alone is empty . . . 976
4.2.6 A fully worked example . . . 977

4.2.7 Two negative results for top-k routing . . . 978
4.2.8 Inverse demand: the dual programme . . . 980
4.2.9 Invariance classes and budgets for every routed quantity . . . 980
4.2.10 Routing decisions are structure-carrying . . . 982
4.2.11 Routing relative to a base . . . 983
4.2.12 Module template for the routing layer . . . 983
4.2.13 Refinement of the coverage family, and the verdict a refinement audit returns 988
4.2.14 Exact reference search at bounded height . . . 992
4.2.15 Deliberate demotions . . . 995

Chapter 4.3 — Descent, gluing, and the gerbe of experts **996**
4.3.1 Local sections and the sheaf on the nerve . . . 997
4.3.2 The Čech complex and the sheaf Laplacian . . . 998
4.3.3 Computing H^0 and H^1 . . . 999
4.3.4 One-sidedness: what a detected conflict does and does not mean . . . 1000
4.3.5 The gerbe of experts: a withdrawn construction and the reason it failed . . . 1001
4.3.6 Executable diagnostic types and statuses . . . 1002
4.3.7 Five diagnostic capabilities as five computable tests . . . 1005
4.3.8 Higher coherence status . . . 1007
4.3.9 Module template for the descent layer . . . 1011
4.3.10 Deliberate demotions . . . 1012

Chapter 4.4 — Degeneration, and reconciliation with modern practice **1014**
4.4.1 The degeneration theorem . . . 1015
4.4.2 Local conformance, not an optimisation or performance floor . . . 1017
4.4.3 The degeneration ladder . . . 1017
4.4.4 Rung conformance and the terminal-residual diagnostic . . . 1018
4.4.5 Reconciliation with modern practice . . . 1022
4.4.6 No converse and no global degeneration theorem . . . 1030
4.4.7 Module template for this chapter . . . 1030
4.4.8 Deliberate demotions . . . 1031
4.4.9 References cited in this chapter . . . 1033

Chapter 4.5 — Systems, failure modes, and the restriction-map problem **1034**
4.5.1 Making overlaps cheap . . . 1034
4.5.2 Cost analysis . . . 1035
4.5.3 Failure modes and their mechanisms . . . 1038
4.5.4 The restriction-map problem . . . 1044
4.5.5 Resolving the conflict between two formulations . . . 1046
4.5.6 Implementation: the fused layer . . . 1047
4.5.7 Monitoring . . . 1048
4.5.8 Training schedule . . . 1048
4.5.9 Falsifiable claims . . . 1049
4.5.10 Module template for the systems layer . . . 1050
4.5.11 Provenance of the imported ideas . . . 1051
4.5.12 Deliberate demotions . . . 1052
4.5.13 Additional references cited in this chapter . . . 1053

Chapter 4.6 — The solid context engine **1054**

4.6.1 Position in the relative L0–L8 tower . . . 1055
4.6.2 The solid context record . . . 1056
4.6.3 `TowerExtend`: one legitimate level at a time . . . 1058
4.6.4 `TailBound`: a theorem, not a measurement . . . 1060
4.6.5 `TiltStructure`: the declared structure, and no more . . . 1065
4.6.6 `UntiltCompare`: comparison with the surviving choice exposed . . . 1066
4.6.7 Invariance and budget register . . . 1069
4.6.8 Worked transport . . . 1071
4.6.9 Falsification gates . . . 1073
4.6.10 Executable API boundary and conformance records . . . 1077
4.6.11 Dévissage and final accounting . . . 1086
4.6.12 A complete execution . . . 1088
4.6.13 conformance interface . . . 1089
4.6.14 Deliberate demotions . . . 1090

Chapter 4.7 — The Motivic Justification Unit **1092**
4.7.1 Position in the relative L0–L8 tower . . . 1093
4.7.2 Canonical claims before comparison . . . 1096
4.7.3 Proof terms and machine-enforced grade . . . 1099
4.7.4 Transporting a justification along an equivalence . . . 1103
4.7.5 Invariance is part of the measurement type . . . 1106
4.7.6 Auditing the cd-structure behind descent . . . 1109
4.7.7 Invariance and budget register . . . 1113
4.7.8 Worked transport . . . 1115
4.7.9 Falsification gates . . . 1117
4.7.10 Executable API boundary and conformance records . . . 1119
4.7.11 Dévissage and final accounting . . . 1132
4.7.12 A complete execution . . . 1136
4.7.13 conformance interface . . . 1139
4.7.14 Deliberate demotions . . . 1143

Chapter 4.8 — The coherent routing engine **1155**
4.8.1 Position in the relative L0–L8 tower . . . 1156
4.8.2 Route diagrams and certificate scope . . . 1158
4.8.3 `CoherenceWitness`: agreement on overlaps and among agreements . . . 1159
4.8.4 `NormMap`: the comparison between pooling and compatibility . . . 1160
4.8.5 `AmbiDefect`: a route objective, not a dashboard metric . . . 1161
4.8.6 `GoodwillieTower`: interaction order for ablation studies . . . 1163
4.8.7 `BCdevCheck`: no square by wishful straightening . . . 1165
4.8.8 Invariance and budget register . . . 1166
4.8.9 Worked transport . . . 1169
4.8.10 Falsification gates . . . 1171
4.8.11 Executable API boundary and conformance records . . . 1173
4.8.12 Dévissage and final accounting . . . 1175
4.8.13 A complete execution . . . 1176
4.8.14 conformance interface . . . 1197
4.8.15 Deliberate demotions . . . 1200

Part V — Learning: Objectives, Pretraining and Continual Adaptation **1204**

Chapter 5.1 — Objective design: what may and may not be optimised **1205**
5.1.1 The object under study: cellular sheaves and their consistency functional . 1206
5.1.2 The learning problem over a base site 1207
5.1.3 The collapse theorem 1208
5.1.4 Invariance classes, and the criterion that decides an objective 1211
5.1.5 The repair taxonomy, and its exhaustiveness 1214
5.1.6 Fix one: heterogeneous stalks 1216
5.1.7 Fix two: the Stiefel constraint 1217
5.1.8 The residue is on the activation side 1219
5.1.9 Fix three: the overlap-informativeness dual 1221
5.1.10 Fix four: read-out-only consistency 1224
5.1.11 The abstention read-out 1226
5.1.12 The same pathology, five times over, in the published literature 1227
5.1.13 The optimiser must respect the type of a parameter 1228
5.1.14 The backward pass is part of the specification 1232
5.1.15 The admissible objective, assembled 1236
5.1.16 Reconciliation: structural certificates and the one admitted paired defect . 1240
5.1.17 Deliberate demotions 1249

Chapter 5.2 — Pretraining objectives: coverage, descent and the evidence lower bound **1251**
5.2.1 Next-token prediction: the coverage defect 1252
5.2.2 Masked language modelling: no descent content on a discrete site 1254
5.2.3 Diffusion: an evidence lower bound with a gauge symmetry 1255
5.2.4 The pretraining recipe 1258
5.2.5 Initialisation, and why the natural choice is the dangerous one 1261
5.2.6 Optimiser, learning rate, and schedules 1263
5.2.7 Telemetry that detects collapse early 1265
5.2.8 Declared $\mathbf{A}^1$-invariance: a group action and a test 1269
5.2.9 Deliberate demotions 1278

Chapter 5.3 — Continual learning and transfer: morphisms of sites **1280**
5.3.1 Tasks as sites, adaptation as a morphism of sites 1281
5.3.2 A site relocation is additional certified data, not a consequence of fine-tuning 1282
5.3.3 Openness, made concrete and checkable 1283
5.3.4 The forgetting asymmetry 1284
5.3.5 Openness as a design target 1287
5.3.6 Transport along equivalences, weight shifts and transfer as a primitive . . 1293
5.3.7 Deliberate demotions 1303

Chapter 5.4 — Data and site construction: where the site comes from **1305**
5.4.1 Evidence atoms and their extraction 1306
5.4.2 From entailment to a partial order 1307
5.4.3 Why down-sets, and not the obvious alternative 1308
5.4.4 Learned coverage 1309
5.4.5 Cost at web scale 1311
5.4.6 Quality controls 1312

5.4.7 The trivialisation attack . . . 1314
5.4.8 What is hand-designed, and what a learned site would require . . . 1317
5.4.9 Degeneration, observables and the falsification gate . . . 1318
5.4.10 Finite cd-structure verification data . . . 1318
5.4.11 A finite-realised pro-étale covering protocol . . . 1321
5.4.12 Deliberate demotions . . . 1329

Part VI — Reinforcement Learning and Self-Evolution 1331

Chapter 6.1 — The landscape of policy optimisation and why the choice is narrowed 1332
6.1.1 Common notation and the five design slots . . . 1333
6.1.2 Proximal policy optimisation . . . 1335
6.1.3 Group relative policy optimisation . . . 1335
6.1.4 Dr. GRPO and the two biases . . . 1336
6.1.5 DAPO . . . 1336
6.1.6 GSPO and the sequence-level ratio . . . 1337
6.1.7 GMPO . . . 1338
6.1.8 RLOO, REINFORCE++, Lite PPO and CISPO . . . 1339
6.1.9 Off-policy training, staleness and the training–inference gap . . . 1340
6.1.10 The comparison table . . . 1341
6.1.11 Constraint one: mixture of experts . . . 1343
6.1.12 Constraint two: the token-level ratio has no canonical definition . . . 1344
6.1.13 Constraint three: the reward is not scalar . . . 1348
6.1.14 The forcing argument . . . 1349
6.1.15 The ratio module . . . 1350
6.1.16 Chapter references . . . 1352
6.1.17 Lineage-qualified algorithm comparison . . . 1352
6.1.18 Evidence workflow for a landscape claim . . . 1353
6.1.19 Consequences for the surveyed choices . . . 1354
6.1.20 Failure modes . . . 1355
6.1.21 Worked adjudications . . . 1355
6.1.22 Deliberate demotions . . . 1357

Chapter 6.2 — The Hodge theory of group-relative advantage 1358
6.2.1 The Čech complex of a sampling group . . . 1360
6.2.2 The identity . . . 1360
6.2.3 A worked example . . . 1361
6.2.4 From a group to a cover . . . 1362
6.2.5 Unbiasedness under a genuine cover . . . 1363
6.2.6 Optimal baselines in closed form . . . 1364
6.2.7 The corrected placement of the advantage . . . 1366
6.2.8 What the identity buys, and what it does not . . . 1367
6.2.9 Where H^1 actually lives . . . 1368
6.2.10 A worked example with a genuine cover . . . 1369
6.2.11 The effective sample size of a cover, and an accurate cost . . . 1370
6.2.12 Why the standard-deviation division destroys monotone improvement . . . 1371
6.2.13 The invariance class of the advantage, and a classification inside one estimator family . . . 1372

6.2.14 The dimension of each complex, and why the gerbe is still not claimed . . 1377
6.2.15 The grouped advantage module 1378
6.2.16 Summary of the chapter 1381
6.2.17 Weight-filtered Hodge advantage 1382
6.2.18 What fails without a compatible metric 1384
6.2.19 Filtered grouped advantage and unbiasedness 1386
6.2.20 The filtered spectral sequence and the location of failure 1387
6.2.21 Morphisms, preservation and non-preservation 1388
6.2.22 A computable filtered-Hodge audit 1389
6.2.23 Worked filtered example 1391
6.2.24 Reporting contract for the fused object 1392
6.2.25 Failure modes and exact degenerations 1392
6.2.26 Optimisation consequences of the filtration 1393
6.2.27 Quantitative defect when block preservation is approximate 1394
6.2.28 Estimating grade-dependent baseline weights 1395
6.2.29 Falsification programme for the fusion 1395
6.2.30 Equitable links and exact advantage transport 1396
6.2.31 Gauge action on filtered coefficients 1397
6.2.32 Versioned coefficient extraction 1398
6.2.33 Distributed grade aggregation 1399
6.2.34 Final reporting boundary 1400
6.2.35 Deliberate demotions 1400

Chapter 6.3 — Descent-Grouped Sequence Policy Optimisation **1402**
6.3.1 The Ω-valued reward 1403
6.3.2 Disintegration: how an Ω-valued reward becomes an objective 1404
6.3.3 What the disintegration costs 1405
6.3.4 The descent gate 1406
6.3.5 The indeterminacy trust region 1409
6.3.6 The coricity side-constraint, and why it must never be a loss term 1410
6.3.7 Entropy control 1410
6.3.8 The DGSPO objective 1411
6.3.9 Implementation 1414
6.3.10 Distributed form, memory and communication 1417
6.3.11 Hyper-parameters 1418
6.3.12 Degeneration conditions 1419
6.3.13 Verifier design, and why chart diversity is structural 1420
6.3.14 Learning to abstain 1421
6.3.15 Type discipline: value-carrying rewards, structure-carrying gates 1423
6.3.16 Observables 1424
6.3.17 Falsification gate 1424
6.3.18 Weight-lineage consumer contract 1425
6.3.19 Filtered-update conformance audit 1427
6.3.20 Interaction with the type rule 1428
6.3.21 Measurement obligations 1428
6.3.22 Failure modes 1428
6.3.23 Selection bias made explicit 1429
6.3.24 Fallback policy as a typed branch 1430

6.3.25 Deliberate demotions 1431

Chapter 6.4 — Self-evolution under a geometric restriction 1432
6.4.1 What a self-edit is 1433
6.4.2 Undecidability, and why the edit language is a fixed list 1435
6.4.3 Curriculum as coverage refinement 1437
6.4.4 Is a new expert needed? 1438
6.4.5 Memory as a stack 1439
6.4.6 The evolution loop 1439
6.4.7 Two worked edits 1443
6.4.8 Population, archive and acceptance 1444
6.4.9 Multi-agent coherence, and why pairwise consistency fails 1445
6.4.10 Does the loop converge? 1445
6.4.11 Relation to existing work, and what is new 1447
6.4.12 Placing the loop in the training pipeline 1448
6.4.13 Complexity and hardware 1449
6.4.14 Degeneration condition 1450
6.4.15 The specification as a species, and a self-edit as a mutation 1450
6.4.16 Observables 1453
6.4.17 Falsification gate 1453
6.4.18 Executable Northcott search 1454
6.4.19 Height certificates and mutation composition 1456
6.4.20 Northcott examples, non-examples and migration 1457
6.4.21 Deliberate demotions 1458

Chapter 6.5 — Reward hacking, structural attacks and what remains unsolved 1460
6.5.1 Weakness one: the exogenous site measure, and the trivial-site attack . . . 1461
6.5.2 Weakness two: descent verification catches only local hacks 1464
6.5.3 Weakness three: the schedule-replay bias on negative advantages 1465
6.5.4 Weakness four: the variance claim rests on an unmeasured correlation . . 1466
6.5.5 Weakness five: scalarisation is provably incomplete and destroys modularity 1467
6.5.6 Weakness six: the geometric restriction constrains logic, not behaviour . . 1467
6.5.7 Weakness seven: no convergence, and calibration is an assumption 1468
6.5.8 Secondary failure modes 1468
6.5.9 How much does the exogeneity defence actually buy? 1470
6.5.10 The attacks in the standard Goodhart taxonomy 1470
6.5.11 A red-team protocol 1471
6.5.12 Complexity of the defence stack 1472
6.5.13 Degeneration condition 1472
6.5.14 Observables specific to this chapter 1472
6.5.15 Falsification gate 1473
6.5.16 The measurement problem 1473
6.5.17 Open problems blocking deployment 1475
6.5.18 Decision margin as a typed attack surface 1477
6.5.19 Failure modes of DecisionMargin 1479
6.5.20 The HEIGHT-BOUNDED marker as a threat-model object 1480
6.5.21 Calibration and red-team tests 1480
6.5.22 Worked margin attacks 1481

6.5.23 Marker misuse scenarios . . . 1482
6.5.24 Deliberate demotions . . . 1483

Chapter 6.6 — Self-evolution gate calibration **1484**
6.6.1 Threat model and safety target . . . 1484
6.6.2 Mutation classification . . . 1485
6.6.3 Canonical corpus and environment-fixed structure . . . 1485
6.6.4 Observable vector . . . 1486
6.6.5 Performance-indexed floor . . . 1486
6.6.6 Parent-relative and population-relative tests . . . 1487
6.6.7 Structure certification . . . 1488
6.6.8 Calibration procedure . . . 1488
6.6.9 Online decision rule . . . 1489
6.6.10 Error rates . . . 1490
6.6.11 What happens on abstention . . . 1491
6.6.12 Cumulative small-edit detector . . . 1491
6.6.13 Multiple testing and optional stopping . . . 1492
6.6.14 Base change, links and budgets . . . 1492
6.6.15 Calibration drift and refresh . . . 1492
6.6.16 Worked decision records . . . 1493
6.6.17 Complexity and hardware cost . . . 1493
6.6.18 Degeneration . . . 1493
6.6.19 Failure modes of the calibration itself . . . 1494
6.6.20 Falsification gates . . . 1494
6.6.21 Section 63-GC — the three system-level conditions of the gate (normative) 1495
6.6.22 Integration with the evolution loop . . . 1496
6.6.23 Dévissage and final safety statement . . . 1498
6.6.24 Sample size and power . . . 1499
6.6.25 Hierarchical error allocation . . . 1499
6.6.26 Human review contract . . . 1500
6.6.27 Gate implementation isolation . . . 1500
6.6.28 Mutation-specific test selection . . . 1500
6.6.29 Gate state machine . . . 1501
6.6.30 Additional worked attack: denominator substitution . . . 1501
6.6.31 Calibration report . . . 1501
6.6.32 Typed decision margin . . . 1502
6.6.33 The HEIGHT-BOUNDED certificate . . . 1504
6.6.34 Joint use in the online decision . . . 1506
6.6.35 Failure-mode matrix . . . 1506
6.6.36 Worked certificate calculation . . . 1507
6.6.37 Calibration maintenance for the two types . . . 1508
6.6.38 Simultaneous calibration of the margin . . . 1508
6.6.39 Calibrating the normalising units . . . 1509
6.6.40 Completeness and auditability . . . 1510
6.6.41 Reason-code disclosure policy . . . 1510
6.6.42 Deliberate demotions . . . 1511

Chapter 6.7 — Reinforcement-learning infrastructure **1513**

6.7.1 System boundary and data flow . . . 1514
6.7.2 Revision rounds as MDP actions . . . 1514
6.7.3 DGSPO likelihoods and grouped baselines . . . 1517
6.7.4 Rollout–training mismatch . . . 1518
6.7.5 Frozen and replayed MoE routing . . . 1521
6.7.6 Externally committed sites . . . 1523
6.7.7 Dynamic sampling without curriculum collapse . . . 1524
6.7.8 Reference-policy resets . . . 1524
6.7.9 Abstention calibration . . . 1524
6.7.10 Staged falsification before scaling . . . 1525
6.7.11 Cost and capacity planning . . . 1526
6.7.12 The learner update as an auditable transaction . . . 1527
6.7.13 Sequence storage and revisable-state replay . . . 1528
6.7.14 Control-plane telemetry . . . 1528
6.7.15 Throughput optimisation after correctness . . . 1529
6.7.16 Deployment modes . . . 1529
6.7.17 Operational acceptance tests and incident runbooks . . . 1530
6.7.18 Complexity, degeneration and failure containment . . . 1532
6.7.19 Lineage-typed infrastructure evidence . . . 1532
6.7.20 Replay grades and atomic publication . . . 1534
6.7.21 Cost evidence as an incremental ledger . . . 1535
6.7.22 Release matrix for typed evidence . . . 1536
6.7.23 Failure modes . . . 1536
6.7.24 Worked instrument packets . . . 1537
6.7.25 Distributed incident accounting . . . 1538
6.7.26 Deliberate demotions . . . 1538

Part VII — Systems and Implementation **1540**

Chapter 7.1 — Execution model **1541**
7.1.1 The compiled site as a build artefact . . . 1542
7.1.2 The section object and its invariants . . . 1543
7.1.3 The forward pass as a descent transaction . . . 1545
7.1.4 The task graph and external suspension . . . 1547
7.1.5 The descent solve . . . 1548
7.1.6 The backward pass of the descent solve . . . 1549
7.1.7 The trainability budget . . . 1551
7.1.8 Boundary conditions and pinned evidence . . . 1553
7.1.9 Determinism and replay . . . 1553
7.1.10 The route is part of the record . . . 1556
7.1.11 Local specialisation condition . . . 1557
7.1.12 Two-component Arakelov execution cost . . . 1557
7.1.13 Observables and the falsification gate . . . 1564
7.1.14 Deliberate demotions . . . 1565

Chapter 7.2 — Kernels and hardware **1567**
7.2.1 Where the arithmetic actually goes . . . 1567
7.2.2 The fused restriction–transport–reduce kernel . . . 1569

7.2.3 The ragged problem . . . 1570
7.2.4 Roofline of the descent solve . . . 1571
7.2.5 Numerical precision . . . 1572
7.2.6 The map catalogue and the residency budget . . . 1573
7.2.7 Fixed-rank truncation and the three-witness rule . . . 1574
7.2.8 Differentiating through the solve . . . 1583
7.2.9 Second-order audit availability . . . 1583
7.2.10 Local kernel specialisation . . . 1584
7.2.11 Observables and the falsification gate . . . 1584
7.2.12 Measurement manifest . . . 1585
7.2.13 Deliberate demotions . . . 1585

Chapter 7.3 — Distributed training **1587**
7.3.1 Two placement problems, not one . . . 1587
7.3.2 Scaling without assumed locality . . . 1588
7.3.3 The all-to-all and edge-exchange schedules . . . 1589
7.3.4 Overlap without changing the deployed operator . . . 1591
7.3.5 Checkpoints, manifests and the site version . . . 1592
7.3.6 Local communication specialisation . . . 1594
7.3.7 Distributed evidence closure and component cost . . . 1594
7.3.8 Failure handling without changing the program . . . 1602
7.3.9 Distributed observables and budgets . . . 1603
7.3.10 Conformance gates . . . 1604
7.3.11 Trainability status . . . 1605
7.3.12 Deliberate demotions . . . 1605

Chapter 7.4 — Inference and serving **1607**
7.4.1 Normative serving transaction . . . 1608
7.4.2 The router is part of the action . . . 1608
7.4.3 Fixed deployed solve and certificate-only continuation . . . 1610
7.4.4 Versioned caches . . . 1611
7.4.5 Speculation and revision . . . 1612
7.4.6 Batching and scheduling . . . 1612
7.4.7 Prefill, chunking and prefix sharing . . . 1613
7.4.8 Machine-independent cost account . . . 1613
7.4.9 Service tiers and commitment . . . 1615
7.4.10 Local serving specialisation . . . 1616
7.4.11 Topology-aware envelopes, solid tails and accountable serving cost . . . 1616
7.4.12 Counterfactual serving verdicts . . . 1625
7.4.13 Observability, replay and production gates . . . 1626
7.4.14 Trainability boundary . . . 1627
7.4.15 Deliberate demotions . . . 1627

Part VIII — API and Reference Implementation **1629**

Chapter 8.1 — API specification **1631**
8.1.1 Scope, conformance levels, and the two design principles . . . 1631
8.1.2 Kinds: value-carrying and structure-carrying data . . . 1634

8.1.3 The site, morphisms of sites, and base change . . . 1635
8.1.4 Stalks, restriction maps, and heterogeneous dimensions in the type system 1640
8.1.5 The sheaf and its operators . . . 1642
8.1.6 The Section, and what it must carry . . . 1643
8.1.7 The descent solve . . . 1644
8.1.8 Routing and the minimal sufficient cover . . . 1648
8.1.9 Measured quantities and their budgets . . . 1652
8.1.10 Read-outs, and two constructions that had to be withdrawn . . . 1653
8.1.11 Error taxonomy . . . 1656
8.1.12 Degeneration condition . . . 1658
8.1.13 Lineage-facing adapters and closed conformance transactions . . . 1659
8.1.14 Deliberate demotions . . . 1667

Chapter 8.2 — Reference implementation 1668
8.2.1 Scope: one permitted mutation of the specification . . . 1668
8.2.2 The fixture: a complex on which the second-degree tests are not vacuous . 1670
8.2.3 The coboundary operators . . . 1672
8.2.4 The adjoint and the sheaf Laplacian . . . 1672
8.2.5 The descent solve . . . 1673
8.2.6 The minimal sufficient cover . . . 1677
8.2.7 The compositionality defect . . . 1677
8.2.8 The degree-one obstruction, and a cochain that cannot carry one . . . 1678
8.2.9 The gerbe class, rebuilt . . . 1679
8.2.10 The degenerate fixture and the conformance comparison . . . 1680
8.2.11 The test suite . . . 1682
8.2.12 Training configuration and parameter groups . . . 1684
8.2.13 End-to-end executable harness and shape trace . . . 1685
8.2.14 The RC-mini integration path . . . 1687
8.2.15 What is not implemented, and why . . . 1690
8.2.16 Reference slices for the five Stage-3 engines . . . 1691
8.2.17 Deliberate demotions . . . 1708

Chapter 8.3 — Typed claim record and implementation contracts 1710
8.3.1 The claim is the unit of accountable output . . . 1710
8.3.2 Grade, dependency and falsification spine . . . 1712
8.3.3 Decision rule . . . 1716
8.3.4 Realisation contract . . . 1716
8.3.5 The species criterion . . . 1717
8.3.6 Corrected Topos-MoE tensor contract . . . 1717
8.3.7 Fused post-routing execution core . . . 1718
8.3.8 MoE failure modes . . . 1720
8.3.9 Integrated claim/MoE path . . . 1720
8.3.10 Storage and communication . . . 1721
8.3.11 Base change and mutation . . . 1721
8.3.12 Falsification gates . . . 1721
8.3.13 Dévissage . . . 1722
8.3.14 Canonical serialisation . . . 1722
8.3.15 Proposition type system . . . 1723

8.3.16 Claim-kind obligations . . . 1724
8.3.17 Module compliance records . . . 1724
8.3.18 The instantiated registry . . . 1725
8.3.19 Adjudicated conflicts . . . 1730
8.3.20 Corrections and withdrawals . . . 1733
8.3.21 Fully worked claim record . . . 1734
8.3.22 Claim batching and partial failure . . . 1735
8.3.23 Counter-context record . . . 1735
8.3.24 Worked two-expert tensor execution . . . 1736
8.3.25 Backward contract . . . 1736
8.3.26 Error taxonomy . . . 1739
8.3.27 Conformance corpus . . . 1739
8.3.28 Operational falsification . . . 1740
8.3.29 API integration . . . 1740
8.3.30 Privacy and selective disclosure . . . 1741
8.3.31 Version negotiation . . . 1741
8.3.32 Additional complexity bounds . . . 1742
8.3.33 Deliberate refusal to infer claims from confidence . . . 1742
8.3.34 Proof terms, checked identity transport and bounded height . . . 1742
8.3.35 Deliberate demotions . . . 1754

Part IX — Validation and Falsification **1755**

Chapter 9.1 — Validation instruments **1756**
9.1.1 Instruments before architecture . . . 1756
9.1.2 The common experimental contract . . . 1758
9.1.3 N1 — Matched non-trivial-site test . . . 1761
9.1.4 N2 — Five-line nucleus check . . . 1763
9.1.5 N3 — Holonomy and capability against GRAPE-M . . . 1763
9.1.6 N4 — Adaptive depth and Postnikov language . . . 1765
9.1.7 N5 — Obstruction prediction and gerbe non-vacuity (withdrawn) . . . 1766
9.1.8 N6 — Identity Sheaf Network reproduction . . . 1767
9.1.9 N7 — Coverage routing against modern top-k MoE . . . 1768
9.1.10 N8 — Restriction-map ρ ablation . . . 1769
9.1.11 N9 — Nerve non-vacuity . . . 1770
9.1.12 N10 — Coverage-grouped advantage against DAPO and GSPO . . . 1771
9.1.13 N11 — Beck–Chevalley deviation probe . . . 1772
9.1.14 N12 — Species conformance across independent implementations . . . 1773
9.1.15 N13 — Functional transmission and task-bypass audit . . . 1774
9.1.16 Holonomy dichotomy test as a theorem audit . . . 1775
9.1.17 Gate ordering, pre-registration and promotion . . . 1776
9.1.18 Cross-instrument controls . . . 1778
9.1.19 Power, minimum useful effects and robustness . . . 1779
9.1.20 Decisive pilot configurations . . . 1780
9.1.21 Failure semantics, abandonment and the permissible conclusion . . . 1781
9.1.22 Theorem-assumption binding register . . . 1782
9.1.23 Cross-instrument consistency checks . . . 1787

9.1.24 Deliberate demotions . . . 1789

Chapter 9.2 — Metrics, calibration and structural observability 1790
9.2.1 Predictive and compositional performance . . . 1796
9.2.2 Structural coherence and obstruction risk . . . 1798
9.2.3 Scope, evidence and transfer . . . 1803
9.2.4 Streamed structural telemetry . . . 1805
9.2.5 Metric gaming and adversarial controls . . . 1809
9.2.6 Calibration and estimator assurance . . . 1812
9.2.7 Alert, incident and promotion policy . . . 1814
9.2.8 Evaluation report schema . . . 1815
9.2.9 Signed metric cards under the A12 contract . . . 1816
9.2.10 Decision decomposition for theorem-facing metrics . . . 1821
9.2.11 Deliberate demotions . . . 1822

Part X — Positioning, Rigour and Risk 1823

Chapter 10.1 — The rigour ledger 1824
10.1.1 The four grades . . . 1825
10.1.2 The four-axis claim record . . . 1825
10.1.3 Principal-claim ledger and adjudicated failures . . . 1827
10.1.4 Twelve things this specification does not claim . . . 1847
10.1.5 Corrections ledger . . . 1849
10.1.6 Outstanding proof and evidence obligations . . . 1851
10.1.7 Five retained claim families after demotion . . . 1854
10.1.8 T/D declarations and S0/S1/S2 sentence licences . . . 1855
10.1.9 Book-wide principal-claim distribution . . . 1857
10.1.10 Deliberate demotions . . . 1859

Chapter 10.2 — Positioning, risks and the strangler-fig roadmap 1861
10.2.1 Relation to current frontiers . . . 1862
10.2.2 Risk register . . . 1866
10.2.3 Cross-frontier interactions . . . 1871
10.2.4 Risk ownership and escalation . . . 1872
10.2.5 Systems and deployment gates . . . 1873
10.2.6 Strangler-fig roadmap . . . 1874
10.2.7 Recommended sequence . . . 1877
10.2.8 What the roadmap may legitimately establish . . . 1878
10.2.9 Priority boundaries and trigger-complete risk contracts . . . 1879
10.2.10 Trigger-complete risk register . . . 1881
10.2.11 Deliberate demotions . . . 1884

Part XI — The Grothendieck Programme and Inter-Universal Structure 1886

Chapter 11.1 — Grothendieck's programme and what transfers 1887
11.1.1 The three-tier discipline and the test that enforces it . . . 1888
11.1.2 Relative geometry: work over a base, not with absolute objects . . . 1889

11.1.3 The functor of points . . . 1891
11.1.4 Grothendieck topologies: covering is structure, not a property . . . 1891
11.1.5 Topoi as generalised spaces, and as classifying objects . . . 1893
11.1.6 Descent: global objects are local objects plus glue . . . 1895
11.1.7 The six operations . . . 1896
11.1.8 Motives . . . 1897
11.1.9 Anabelian geometry . . . 1897
11.1.10 Dévissage: reduce by stripping structure . . . 1897
11.1.11 Why Grothendieck's method is the right template for this document . . . 1900
11.1.12 What the programme does not contain, and why that is the binding constraint . . . 1902
11.1.13 The relative point of view, made usable . . . 1903
11.1.14 Dévissage as the document's method, and the hypothesis it hides . . . 1905
11.1.15 The retrofit mandate: principle, chapter, and what must change . . . 1906
11.1.16 Summary of tiers and grades for this chapter . . . 1912

Chapter 11.2 — The functor of points and what a representation is 1916
11.2.1 Grothendieck's move . . . 1916
11.2.2 Yoneda: precisely what it gives . . . 1917
11.2.3 Yoneda for an infinity-category: mapping spaces and coherence . . . 1918
11.2.4 The probe category, defined . . . 1923
11.2.5 A pro-étale probe: actions beyond finite completion . . . 1925
11.2.6 The linear probe functor, computed exactly . . . 1932
11.2.7 The two support-level vacuity mechanisms . . . 1934
11.2.8 Approximate probing is filtered, not a fixed-tolerance functor . . . 1935
11.2.9 Isomorphism, invariance and the similarity-index literature . . . 1936
11.2.10 Representability, and why the model is not one of its own probes . . . 1938
11.2.11 What the functorial view predicts, and what it merely renames . . . 1940
11.2.12 The negative result, stated as a proposition . . . 1941
11.2.13 Beyond linear readouts: what survives, and at what cost . . . 1942
11.2.14 The consequence for GM-Ω∞: the test family is declared . . . 1942
11.2.15 What may be reported from a probe suite, and under what invariance . . . 1944
11.2.16 The probing contract and implementation species . . . 1947
11.2.17 Degeneration, failure modes, and grades . . . 1948

Chapter 11.3 — Motives, universality and the limits of the analogy 1951
11.3.1 The problem that motives were invented to solve . . . 1951
11.3.2 Correspondences and the category of pure motives . . . 1952
11.3.3 The Standard Conjectures, and the precise sense in which the theory is conditional . . . 1953
11.3.4 The transfer question, stated so that it can be answered . . . 1956
11.3.5 The closest object that actually exists . . . 1958
11.3.6 The five failures, stated precisely . . . 1960
11.3.7 Reconciliation with Chapter 2.16: five falsifiable non-overturns . . . 1962
11.3.8 The type discipline the failures force . . . 1968
11.3.9 Why the chapter is nevertheless here . . . 1970
11.3.10 The open problem, stated without sentiment . . . 1977
11.3.11 Summary of tiers and grades . . . 1978

Chapter 11.4 — Motivic fundamental groups, mixed Tate periods, and the Galois action **1982**
11.4.1 Scope, ownership, and grade boundary . . . 1982
11.4.2 From a based variety to a motivic fundamental group . . . 1983
11.4.3 Deligne weights on non-abelian paths . . . 1986
11.4.4 The mixed Tate corridor inside DM . . . 1987
11.4.5 The mixed Tate Galois group and the unramified category over $\mathbb{Z}$ . . . 1990
11.4.6 Periods as coordinates on a comparison torsor . . . 1991
11.4.7 Multiple zeta values as path periods . . . 1994
11.4.8 Transfer audit for GM-Ω^{∞} . . . 1996
11.4.9 Graded transfer verdict . . . 2000
11.4.10 The deletion test: why Deligne and Voevodsky are load-bearing here . . . 2000
11.4.11 Construction audit: bar complexes, free paths, and one mixed Tate extension 2002
11.4.12 Four false positives and the experiment that rejects them . . . 2006
11.4.13 The motivic Galois action on the fundamental group itself . . . 2007
11.4.14 Weight dimensions and the $\{2, 3\}$ recurrence . . . 2009
11.4.15 Result and source ledger . . . 2010
11.4.16 Open obligations and final synthesis . . . 2013

Chapter 11.5 — What inter-universal Teichmüller theory actually is **2015**
11.5.1 How to read this chapter . . . 2016
11.5.2 Provenance of this account, and what each source is used for . . . 2016
11.5.3 The shape of the argument, in outline . . . 2018
11.5.4 The Diophantine target . . . 2019
11.5.5 Anabelian geometry, and why hyperbolic curves . . . 2021
11.5.6 Why "inter-universal" . . . 2024
11.5.7 Frobenioids, and the Frobenius-like versus étale-like distinction . . . 2026
11.5.8 Hodge theatres . . . 2029
11.5.9 Why "Teichmüller" . . . 2030
11.5.10 The log-theta lattice . . . 2031
11.5.11 Multiradiality . . . 2033
11.5.12 The three indeterminacies of the multiradial representation . . . 2034
11.5.13 How a Diophantine inequality is supposed to emerge . . . 2036
11.5.14 Common misdescriptions, corrected . . . 2037
11.5.15 Effectivity, and the subsequent literature . . . 2038
11.5.16 A cut-aware correspondence instrument . . . 2039
11.5.17 Which strata of this chapter become programme statements . . . 2043
11.5.18 Every item of this chapter, with its tier and its grade . . . 2044

Chapter 11.6 — Coherence and weights at the theta-link **2047**
11.6.1 Scope, sources and grade boundary . . . 2047
11.6.2 Thesis 1 — Hodge-theatre gluing as a coherence problem . . . 2048
11.6.3 Thesis 2 — a weight mismatch as a conditional obstruction to ring structure 2056
11.6.4 The fusion — coherent coarse gluing with an obstructed ringed lift . . . 2061
11.6.5 What this fusion does not settle . . . 2064
11.6.6 Falsification protocols . . . 2064
11.6.7 Application status and research programme . . . 2067
11.6.8 Preservation of the Chapter 11.7 cuts . . . 2070

11.6.9 Import deletion test . . . 2070
11.6.10 Declaration-consumer ledger . . . 2073
11.6.11 Rigour, bibliography and unresolved ledgers . . . 2074
11.6.12 Final synthesis . . . 2077

Chapter 11.7 — The transferable core **2079**
11.7.1 The discipline . . . 2079
11.7.2 Interface contract for every lineage kernel . . . 2080
11.7.3 What was considered and rejected . . . 2085
11.7.4 Transfer 1 — Working without a common ambient, and the consequence for expert links . . . 2086
11.7.5 Transfer 2 — Value-carrying and structure-carrying data . . . 2092
11.7.6 Transfer 3 — Specification by formulas rather than by objects . . . 2099
11.7.7 Transfer 4 — Ambiguity admitted, bounded and carried . . . 2102
11.7.8 The retrofit register . . . 2108
11.7.9 The four transfers, graded . . . 2118

Chapter 11.8 — Limits **2123**
11.8.1 The non-canonical untilt: locating the disputed arrow . . . 2123
11.8.2 What inter-universal Teichmüller theory does not give us . . . 2133
11.8.3 Two audits, and the difference between them . . . 2135
11.8.4 The danger, and the guards . . . 2145
11.8.5 The Scholze–Stix objection . . . 2147
11.8.6 Mochizuki's response . . . 2148
11.8.7 Why we do not adjudicate, and what the reader should do . . . 2149
11.8.8 What would change our mind . . . 2150
11.8.9 What survives if the reader deletes the framing . . . 2151
11.8.10 The Tier A material that is retained, and why . . . 2152
11.8.11 Formulations that are not supported . . . 2153
11.8.12 What is claimed . . . 2154

Chapter 11.9 — The indeterminacy calculus **2157**
11.9.1 Presentations, indeterminacies and budgets . . . 2158
11.9.2 The first indeterminacy: presentation gauge . . . 2160
11.9.3 The second indeterminacy: basepoint . . . 2165
11.9.4 The third indeterminacy: cover refinement . . . 2167
11.9.5 Composition: how budgets propagate through a layer . . . 2172
11.9.6 The constants for L0–L8, and where orthogonal-frame sensitivity lives . . . 2175
11.9.7 The invariance audit . . . 2177
11.9.8 Degeneration: the calculus must say nothing about a standard Transformer 2181
11.9.9 Budgets over a base, and the calculus as an obligation on the whole document 2183
11.9.10 Algebraic completion: when a budget becomes a composable defect . . . 2187
11.9.11 The consumption schedule . . . 2192
11.9.12 The calculus restated without borrowed vocabulary . . . 2193
11.9.13 What the chapter has and has not established . . . 2195

Chapter 11.10 — Faltings completion of the indeterminacy calculus **2198**
11.10.1 Scope, notation and the missing datum . . . 2199
11.10.2 Ideal closure and its tensor strengthening . . . 2202

11.10.3 Realising an indeterminacy by an almost defect . . . 2209
11.10.4 Why a free linearisation does not close the gap automatically . . . 2218
11.10.5 The categorical defect and the metric budget are different ledgers . . . 2220
11.10.6 Boundaries that the completion does not cross . . . 2223
11.10.7 The exact strength of the thesis . . . 2224
11.10.8 Import, bibliography and status registers . . . 2226
11.10.9 Coherence obligations checked against Chapter 11.17 . . . 2228

Chapter 11.11 — Indeterminacy in practice **2235**
11.11.1 Estimating presentation-gauge and frame-perturbation records . . . 2235
11.11.2 Estimating the basepoint budget . . . 2240
11.11.3 Estimating the refinement budget . . . 2242
11.11.4 Budgets in the type system . . . 2244
11.11.5 Reporting discipline . . . 2249
11.11.6 Falsification gates tied to budgets . . . 2250
11.11.7 A fully worked illustration . . . 2251
11.11.8 Cost . . . 2253
11.11.9 Engineering the budgets away, and what it costs . . . 2254
11.11.10 Estimating the link budget, and the schedule the calculus imposes . . . 2255
11.11.11 Open problems . . . 2259
11.11.12 Predeclared allocation budgets and certificate adapters . . . 2260

Chapter 11.12 — Deligne's transferable core: strict boundaries, declared grade effects, and canonicality **2279**
11.12.1 The discipline . . . 2280
11.12.2 What was considered and rejected . . . 2281
11.12.3 Transfer I: strictness is consumer authority, not descriptive metadata . . . 2285
11.12.4 Transfer II: every operation must declare its grade effect . . . 2289
11.12.5 Transfer III: existence of a decomposition is not canonicity . . . 2294
11.12.6 Transfer IV: exact single-grade claims are hypotheses to certify . . . 2298
11.12.7 The obligations are independent and must be conjoined . . . 2302
11.12.8 What does not transfer . . . 2304
11.12.9 Import and deletion audit . . . 2306
11.12.10 Retrofit register . . . 2306
11.12.11 Graded transfer table . . . 2311
11.12.12 T-proof audit and declaration inventory . . . 2313
11.12.13 Source-request table . . . 2315
11.12.14 Worked conformance bundle . . . 2316
11.12.15 Limits . . . 2320
11.12.16 Final architectural conclusion . . . 2320

Chapter 11.13 — Faltings's transferable core: structural invisibility requires proved closure **2322**
11.13.1 The restatement discipline . . . 2322
11.13.2 Candidate 1: numerical smallness is not structural invisibility . . . 2324
11.13.3 Candidate 2: a fixed discard class survives the test . . . 2325
11.13.4 Candidate 3: closure must be checked operation by operation . . . 2327
11.13.5 Candidate 4: an error ideal is useful, but idempotence is not automatic . . . 2330
11.13.6 Candidate 5: closure can be certified on a finite operation algebra . . . 2333

11.13.7 Candidate 6: local tolerance does not control an unbounded composite . 2334
11.13.8 Candidate 7: quantisation error is not automatically an ideal 2336
11.13.9 Candidate 8: floating-point rounding is not an almost-isomorphism . . . 2338
11.13.10 Candidate 9: dropout does not make arbitrary models structurally equivalent 2339
11.13.11 Candidate 10: pruning is not an arithmetic purity theorem 2339
11.13.12 The existing module calculus and the `ALMOST-CERT` boundary . . . 2340
11.13.13 Candidate 11: the arithmetic theorem has no neural-network consumer 2343
11.13.14 Candidate 12: there is no hidden valuation, completion, or étale site in a network 2343
11.13.15 Candidate 13: approved observers can determine the largest safe discard subspace 2344
11.13.16 A small operation library for common neural primitives 2347
11.13.17 The closure hierarchy is strict 2349
11.13.18 Candidate 14: representation changes must transport the discard class 2350
11.13.19 Import and deletion audit 2352
11.13.20 Retrofit register 2352
11.13.21 Graded transfer table 2356
11.13.22 Worked end-to-end closure transaction 2358
11.13.23 Degeneration and falsification conditions 2361
11.13.24 T-proof audit 2362
11.13.25 Source-request table 2363
11.13.26 Final extraction 2366

Chapter 11.14 — Scholze's transferable core: reversible change of ambient, choice-sensitive return, and adequate sites 2367
11.14.1 The discipline 2368
11.14.2 What was considered and rejected 2369
11.14.3 Transfer 1 — Move across an equivalence, solve, and return 2371
11.14.4 Transfer 2 — A non-canonical return creates an invariance obligation . . 2374
11.14.5 Transfer 3 — Enlarge the site only to represent a named missing object . 2378
11.14.6 Transfer 4 — Choose a topology to state the descent you need, and print the price 2381
11.14.7 Compatibility of the four retained disciplines 2385
11.14.8 Further deductions needed by implementations 2386
11.14.9 Four worked audit traces 2389
11.14.10 The retrofit register 2394
11.14.11 Graded transfer table 2395
11.14.12 Falsification programme 2398
11.14.13 Finite holonomy as a counterexample to pairwise return agreement . . 2398
11.14.14 Source-request table 2402
11.14.15 Final accounting 2402

Chapter 11.15 — Voevodsky's transferable core: correspondences, declared localisation, minimal descent, and checkable proof 2404
11.15.1 The discipline 2405
11.15.2 What was considered and rejected 2405
11.15.3 Transfer 1 — Correspondences are a primitive, not a decorated map . . . 2408

11.15.4 Transfer 2 — Declare invisible deformations before measuring invariance 2413
11.15.5 Transfer 3 — Use the least topology whose named gluing equations hold 2417
11.15.6 Transfer 4 — Proof-assistant acceptance is a falsifiable certificate claim . 2421
11.15.7 The three motivic relations are independently typed 2426
11.15.8 Further mathematics of the correspondence primitive 2427
11.15.9 Localisation as an explicit reconstruction quotient 2429
11.15.10 The coarsest-topology claim, stated without vacuity 2429
11.15.11 Certificate dependency and grade propagation 2430
11.15.12 Four worked admission traces 2431
11.15.13 What would be required to reopen the Chapter 2.17 branch 2433
11.15.14 Normalisation cannot replace transfer multiplicity 2434
11.15.15 The retrofit register 2436
11.15.16 Graded transfer table 2438
11.15.17 Retrofit deletion and consumer graph 2440
11.15.18 Falsification programme 2441
11.15.19 Source-request table 2442
11.15.20 Final accounting 2443

Chapter 11.16 — Lurie's transferable core: obligations that survive restatement 2444
11.16.1 The discipline 2444
11.16.2 What was considered and rejected 2445
11.16.3 Transfer I: coherence is an obligation with a degree 2448
11.16.4 Transfer II: a varying expert family must include transport 2454
11.16.5 Transfer III: sheaf semantics can carry more than a one-type export . . . 2458
11.16.6 Transfer IV: agreement of adjoints is a theorem, not a convenience 2460
11.16.7 Independence and joint use of the four transfers 2464
11.16.8 Falsification fixtures and evidence packets 2465
11.16.9 Import and deletion audit 2467
11.16.10 Retrofit register 2468
11.16.11 Graded transfer table 2475
11.16.12 Limits and bounded claims 2478
11.16.13 Source-request audit 2479

Chapter 11.17 — The composition law for typed transfers 2480
11.17.1 The exact obligation inherited from the current Chapter 11.10 2480
11.17.2 Strict defect equipment 2481
11.17.3 The central composition theorem 2483
11.17.4 The category and its order enrichment 2486
11.17.5 Iterated composition and sharpness 2487
11.17.6 Exact module defects and the extension envelope 2488
11.17.7 Faithful module realisation of an indeterminacy 2491
11.17.8 Idempotent ideals and closure of the negligible class 2492
11.17.9 The complete almost-module transfer theorem 2494
11.17.10 Exact base change and explicit non-exactness charges 2495
11.17.11 The instance-verification protocol 2497
11.17.12 Grothendieck: filtered probing and unavoidable amplification 2498
11.17.13 Deligne: filtration degree and strictness 2500
11.17.14 Faltings: almost defects and the fixed ideal 2502

11.17.15 Scholze: canonical tilting does not compose the inverse-choice defect . 2502
11.17.16 Voevodsky: a non-zero finite-transfer skeleton 2503
11.17.17 Lurie: local mate pasting is not a global defect equipment 2505
11.17.18 Mochizuki: the raw indeterminacies are not yet one defect type 2506
11.17.19 Threefold module composition and the concrete associator 2507
11.17.20 A canonical congruence model and the scope of the isomorphism claim 2508
11.17.21 Transport of the common ideal under exact change of context 2509
11.17.22 Falsification and refusal conditions 2510
11.17.23 Relative perturbations and the logarithmic repair 2511
11.17.24 Common ideals on a heterogeneous chain 2512
11.17.25 Why a scalar size cannot be the compositional state 2513
11.17.26 Associativity does not mean commutativity of source operations 2514
11.17.27 Binary, pathwise, and quotient meanings of "no degradation" 2515
11.17.28 A finite choice-fibre law and the missing Scholze hypothesis 2515
11.17.29 Right whiskering is not redundant 2517
11.17.30 Universal property of the decorated transfer category 2518
11.17.31 Morphisms between defect equipments 2519
11.17.32 Tightening non-minimal certificates 2521
11.17.33 Coordinator-facing C1–C6 conclusion 2522
11.17.34 Forward-reference register 2522
11.17.35 Bibliography and source-request audit 2523
11.17.36 No-circularity and dependency audit 2524
11.17.37 Rigour and structure ledger 2524
11.17.38 Final composition statement 2525

Part XII — The Grothendieck Lineage: Transferable Cores and Limits 2527

Chapter 12.1 — The generative core: the late programmes and the method behind them 2528
12.1.1 The method before the objects 2529
12.1.2 *Récoltes et Semailles*: the rising sea and the hammer 2529
12.1.3 *Esquisse d'un Programme*: finite combinatorics, reconstruction and tame topology 2532
12.1.4 *Pursuing Stacks*: homotopy types before higher machinery 2538
12.1.5 *Les Dérivateurs*: retain all diagram shapes 2540
12.1.6 The standard conjectures as a research method 2544
12.1.7 Schematic and generic points: making the general case an object 2545
12.1.8 The six verdicts together 2548
12.1.9 Boundary with the later lineage 2549
12.1.10 Downstream obligations 2550
12.1.11 Closing statement 2552

Chapter 12.2 — Weights, purity and Tannakian reconstruction 2553
12.2.1 Four source results in their own terms 2554
12.2.2 A weight filtration on context features 2558
12.2.3 Strictness as a training-time invariant 2560
12.2.4 Weight-decomposed advantage 2563
12.2.5 What can be reconstructed from a trained model 2568

12.2.6 A weight-shaped bound on holonomy . . . 2570
12.2.7 Decomposition under aggregation . . . 2571
12.2.8 Interactions, degenerations and limits . . . 2574
12.2.9 Ledger of claims . . . 2576
12.2.10 Downstream obligations . . . 2577

Chapter 12.3 — Almost mathematics, almost purity and heights 2579
12.3.1 The source mathematics . . . 2579
12.3.2 A numerical almost base . . . 2582
12.3.3 Quantisation, truncation and iterative solves . . . 2583
12.3.4 The §Q7 square and a computable gradient-bias bound . . . 2586
12.3.5 Numerical almost purity . . . 2589
12.3.6 Heights on experts, covers and claim records . . . 2591
12.3.7 Arakelov-style joint cost . . . 2596
12.3.8 Composition laws and operational consequences . . . 2597
12.3.9 Limits . . . 2599
12.3.10 Transfer ledger . . . 2600
12.3.11 Downstream obligations . . . 2600

Chapter 12.4 — $\mathbf{A}^1$-invariance, transfers, slices and univalence 2603
12.4.1 The topology Voevodsky needed . . . 2603
12.4.2 Decision residue and Nisnevich routing . . . 2605
12.4.3 $\mathbf{A}^1$-homotopy and declared training invariance . . . 2610
12.4.4 Finite correspondences and transfers . . . 2613
12.4.5 The slice filtration and a bounded curriculum . . . 2618
12.4.6 Univalent identity for typed claim records . . . 2620
12.4.7 The five transfers as one discipline . . . 2624
12.4.8 Rigour and falsification summary . . . 2625
12.4.9 Downstream obligations . . . 2626

Chapter 12.5 — Tilting, the pro-étale site, diamonds and condensed structure 2629
12.5.1 Perfectoid spaces: reversible change of characteristic . . . 2629
12.5.2 Almost purity and the pro-étale site . . . 2631
12.5.3 Condensed mathematics: the foundational repair . . . 2633
12.5.4 Solid modules and infinite context . . . 2637
12.5.5 Precision regimes and the limit of the tilting analogy . . . 2639
12.5.6 Diamonds, the v-topology and quotients for sharing . . . 2642
12.5.7 Prisms and two-variable representation . . . 2644
12.5.8 Composition of the repairs . . . 2647
12.5.9 The five source programmes under one evaporation ledger . . . 2648
12.5.10 Consequences for the existing mathematical stack . . . 2649
12.5.11 Downstream obligations . . . 2650

Chapter 12.6 — ∞-topoi, higher descent and obstruction towers 2652
12.6.1 The source mathematics: five cores and five evaporation tests . . . 2652
12.6.2 The router as a cocartesian fibration . . . 2655
12.6.3 Higher descent and the obstruction tower . . . 2659
12.6.4 Ambidexterity: when averaging and gluing coincide . . . 2662
12.6.5 Cotangent complexes for architecture change . . . 2664

12.6.6 Stable and operadic boundaries . . . 2667
12.6.7 Monadicity and the sheafification claim . . . 2667
12.6.8 Hyperdescent, spectral sequences and the coherence fringe . . . 2668
12.6.9 Beck–Chevalley coherence for routing over covers . . . 2669
12.6.10 Finite-group ambidexterity and the characteristic obstruction . . . 2670
12.6.11 Two local sheafification verdicts . . . 2671
12.6.12 Independence of obstruction stages and dimensional cost . . . 2671
12.6.13 Integrated consequences and exact boundaries . . . 2672
12.6.14 Downstream obligations . . . 2674

Chapter 12.7 — Species, mutations and multiradial containers 2676
12.7.1 Six source items and the public evaporation test . . . 2676
12.7.2 Anabelioids, reconstruction and the hypothesis that fails . . . 2678
12.7.3 Frobenioids and the complete specification package . . . 2678
12.7.4 The species lattice and univalent identity . . . 2680
12.7.5 Model theatres: complete computational contexts . . . 2683
12.7.6 Structural and scalar fibrations . . . 2685
12.7.7 Why addition and multiplication cannot both survive a non-trivial scalar mutation . . . 2687
12.7.8 Mutation checking and the finite certification boundary . . . 2688
12.7.9 Multiradial containers . . . 2689
12.7.10 Alien copies and non-circular evaluation . . . 2692
12.7.11 Exact degenerations and boundaries . . . 2695
12.7.12 Failure modes and mechanisms . . . 2696
12.7.13 Four counterfactual verdicts . . . 2696
12.7.14 Foundation for the species/mutation engine . . . 2697
12.7.15 Downstream obligations . . . 2698
12.7.16 Final rigour ledger . . . 2699

Chapter 12.8 — The seven transfers as one structure 2700
12.8.1 Public evaporation of the seven axes . . . 2700
12.8.2 The typed seven-axis object . . . 2701
12.8.3 Compatibility matrix . . . 2704
12.8.4 Filtered higher descent . . . 2706
12.8.5 Almost-strict morphisms . . . 2707
12.8.6 Condensed almost modules . . . 2707
12.8.7 Nisnevich hyperdescent . . . 2708
12.8.8 Univalent claim identity and species separation . . . 2709
12.8.9 Weight, slice and Postnikov tri-filtration . . . 2709
12.8.10 Restricted tilt and descent . . . 2710
12.8.11 Four no-go theorems . . . 2711
12.8.12 Interoperability and tri-filtration checkers . . . 2715
12.8.13 The minimal load-bearing lineage core . . . 2716
12.8.14 Dependency on Part XIII algorithms . . . 2718
12.8.15 Research agenda for the open interactions . . . 2719
12.8.16 Open cells declared under §Q3 . . . 2720
12.8.17 Counterfactual and compatibility ledger . . . 2720
12.8.18 Downstream obligations . . . 2722

Part XIII — Algorithms of the Grothendieck Lineage and the Machinery of Replacement **2725**

Chapter 13.1 — Weight-filtered representation and advantage **2727**
13.1.1 Contract surface and canonical authorities . . . 2727
13.1.2 Constructing the filtration . . . 2730
13.1.3 Strict filtered layers . . . 2732
13.1.4 Weight-filtered advantage and DGSPO . . . 2735
13.1.5 Purity gate and low-weight reward-hacking detector . . . 2738
13.1.6 Filtered MoE routing . . . 2741
13.1.7 Tannakian symmetry audit . . . 2744
13.1.8 Configuration, API and checkpoint semantics . . . 2746
13.1.9 Observability, failure handling and compliance . . . 2748
13.1.10 Conformance fixtures and backward checks . . . 2750
13.1.11 Public transplant ledger . . . 2752
13.1.12 Downstream obligations . . . 2753

Chapter 13.2 — The almost-isomorphism calculus for numerics **2755**
13.2.1 Scalar and tolerance authority . . . 2755
13.2.2 Certificate algebra . . . 2758
13.2.3 Quantisation as a typed quotient and transport . . . 2761
13.2.4 Low-rank truncation . . . 2763
13.2.5 The truncated gluing solve and its actual backward . . . 2764
13.2.6 Distributed certificate composition . . . 2768
13.2.7 Height-bounded expert and router search . . . 2769
13.2.8 Runtime, API, serialisation and audit . . . 2771
13.2.9 A worked forward/backward transaction . . . 2773
13.2.10 Transfer, grade and deletion ledger . . . 2776
13.2.11 Decision stability is a separate contract . . . 2776
13.2.12 Certificate release and conformance . . . 2778
13.2.13 Downstream obligations . . . 2780

Chapter 13.3 — Homotopy-invariant training and transfer correspondences **2782**
13.3.1 Shared result, certificate and evaporation types . . . 2782
13.3.2 Residue-preserving Nisnevich routing . . . 2783
13.3.3 Mayer–Vietoris descent engine . . . 2787
13.3.4 Declared intervals and homotopy-invariant training . . . 2790
13.3.5 Sparse finite correspondences between experts . . . 2792
13.3.6 Finite slice curricula with extension gates . . . 2794
13.3.7 Univalent checking for the typed claim registry . . . 2796
13.3.8 Integrated training and update transaction . . . 2798
13.3.9 Configuration, API and observability . . . 2800
13.3.10 Validation and public falsification . . . 2804
13.3.11 Downstream obligations . . . 2806

Chapter 13.4 — Tilting transfer, prismatic duality and condensed serving **2809**
13.4.1 Condensed state API on finite and profinite probes . . . 2810
13.4.2 Topology-aware linear algebra and exactness . . . 2814
13.4.3 Solid completion for infinite-context aggregation . . . 2815

13.4.4 Restricted tilt of precision regimes . . . 2818
13.4.5 Diamond quotient engine for typed sharing . . . 2820
13.4.6 Prismatic-style two-coordinate representation contract . . . 2822
13.4.7 Serving, serialisation and distributed failure atomicity . . . 2824
13.4.8 Worked end-to-end certified transaction . . . 2827
13.4.9 Public transfer, grade and counterfactual ledger . . . 2828
13.4.10 Cross-module certificate composition . . . 2829
13.4.11 Downstream obligations . . . 2832

Chapter 13.5 — Higher-descent gluing and the obstruction-tower solver 2834
13.5.1 Public evaporation tests and executable consequences . . . 2834
13.5.2 Cocartesian routing over context refinement . . . 2835
13.5.3 Higher-coherent descent data . . . 2838
13.5.4 The Postnikov obstruction-tower solver . . . 2842
13.5.5 Ambidexterity and the norm engine . . . 2845
13.5.6 Cotangent and deformation engine for architecture mutation . . . 2848
13.5.7 Monadicity and sheafification audit . . . 2850
13.5.8 Distributed higher descent and evidence transactions . . . 2851
13.5.9 End-to-end higher-descent transaction . . . 2854
13.5.10 Cross-module guarantees, negative verdicts and release gates . . . 2857
13.5.11 Deterministic conformance corpus . . . 2858
13.5.12 Downstream obligations . . . 2859

Chapter 13.6 — The species/mutation engine and multiradial evaluation 2863
13.6.1 Public evaporation test for the executable transplants . . . 2863
13.6.2 Normative runtime types . . . 2865
13.6.3 Species-lattice compiler and semantic equivalence . . . 2867
13.6.4 Mutation spans, composition and preparation . . . 2869
13.6.5 Structural/value link transport and the decidable checker . . . 2871
13.6.6 Multiradial container and evaluation engine . . . 2872
13.6.7 Alien-copy executor . . . 2875
13.6.8 Integration with self-evolution, claims and evaluation . . . 2877
13.6.9 Distributed orchestration, byte accounting and replay . . . 2878
13.6.10 Complete end-to-end mutation and evaluation transaction . . . 2880
13.6.11 Registry evolution and certificate invalidation . . . 2882
13.6.12 Computing and checking a non-trivial structural invariant . . . 2883
13.6.13 Aggregation policy and dependence-aware outputs . . . 2884
13.6.14 Adversarial schedules and atomic visibility . . . 2884
13.6.15 Transaction trace obligations . . . 2885
13.6.16 Module completion matrix . . . 2886
13.6.17 Counterfactual verdicts and accurate boundaries . . . 2887
13.6.18 Security, privacy and deterministic evidence custody . . . 2887
13.6.19 Downstream obligations . . . 2887
13.6.20 Final rigour and status ledger . . . 2889

Chapter 13.7 — The unified lineage stack: costs, degenerations and failure modes 2890
13.7.1 Evaporation, tiers and dependency-changing fusion . . . 2890
13.7.2 Module A: weight-filtered representation and objective . . . 2893
13.7.3 Module B: almost-isomorphism calculus as a certificate membrane . . . 2894

13.7.4 Module C: residue-preserving route, transfer and slice transaction 2895
13.7.5 Module D: condensed state, solid completion and precision transport . . 2897
13.7.6 Module E: higher-coherent descent and explicit combination 2898
13.7.7 Module F: species mutation, invariant evaluation and publication 2900
13.7.8 One forward and training programme 2901
13.7.9 Normative whole-stack cost account 2903
13.7.10 Exact local and whole-programme degenerations 2905
13.7.11 Interaction failures that do not belong to any isolated module 2907
13.7.12 Counterfactual verdicts 2909
13.7.13 Compliance, unresolved cells and release 2910
13.7.14 Downstream obligations 2914

Chapter 13.8 — Executable reference implementation of the lineage algorithms 2917
13.8.1 The evaporation test, applied to executables 2917
13.8.2 Environment, determinism and how to reproduce 2918
13.8.3 The verdict lattice 2919
13.8.4 The reference fixture 2920
13.8.5 Weight-filtered representation and advantage 2922
13.8.6 The almost-isomorphism certificate algebra 2924
13.8.7 The normative solve, its matched reverse and the surrogate budget 2928
13.8.8 Higher descent and the obstruction tower 2934
13.8.9 Routing as a certified cover 2936
13.8.10 Tilting, the precision regime and the realised-error audit 2938
13.8.11 Species, admissible mutation and the refusal of illicit identification . . . 2939
13.8.12 The end-to-end composition and the conformance harness 2942
13.8.13 Structural costs and the open-cell ledger 2945
13.8.14 Counterfactual verdicts 2948
13.8.15 Downstream obligations 2948

Chapter 13.9 — Validation instruments and falsification gates for the lineage Parts 2950
13.9.1 Evaporation, tiers and authority 2950
13.9.2 Falsification register 2952
13.9.3 Instrument definitions 2960
13.9.4 Degeneration conformance 2963
13.9.5 Matched-baseline gate 2965
13.9.6 Retirement and deletion 2966
13.9.7 Closing MG's open compliance cells 2968
13.9.8 Execution status 2970
13.9.9 Counterfactual and contrary verdicts 2972
13.9.10 Downstream obligations 2972

Chapter 13.10 — One forward map: the normative functor of points and the uniqueness gate 2975
13.10.1 What this chapter is, and the one thing it forbids 2975
13.10.2 The relative viewpoint made syntactic 2976
13.10.3 The numerical contract 2979
13.10.4 The normative forward map 2981
13.10.5 Naturality in the base: what the relative presentation buys, and what it does not 2987

13.10.6 The FORWARD-UNIQUE gate . . . 2990
13.10.7 Absorption of the eight competing definitions . . . 2995
13.10.8 Rigour ledger and undetermined items . . . 2999
13.10.9 Closing . . . 3000

Chapter 13.11 — Exact degeneration: a parameter embedding with bitwise value, gradient and trajectory equality 3002
13.11.1 What is claimed, and the one thing that must be fixed before it can be claimed . . . 3002
13.11.2 The numerical contract . . . 3003
13.11.3 Programs, slots and bitwise equality . . . 3006
13.11.4 The two parameter spaces, and the embedding . . . 3008
13.11.5 The dévissage: the filtration, and why each successive quotient is easier 3011
13.11.6 The graded pieces, and value equality . . . 3012
13.11.7 Gradient equality . . . 3016
13.11.8 One optimiser step . . . 3018
13.11.9 Trajectory equality by induction . . . 3020
13.11.10 The verdict at the deployed manifest . . . 3022
13.11.11 The ledger: carried, elided, evaded, undetermined, impossible . . . 3023
13.11.12 Rigour grades, and what would falsify the chapter . . . 3027

Chapter 13.12 — The overhead theorem: one frozen ledger, an Arakelov cost pair, and the exact degeneration budget 3029
13.12.1 Scope, verdict, and the inventory that may not move . . . 3029
13.12.2 Closing the fourteen cost items before taking a quotient . . . 3031
13.12.3 The reference Transformer parameter ledger . . . 3033
13.12.4 Forward and training arithmetic from the same record . . . 3034
13.12.5 Optimiser-state bytes and the freeze . . . 3036
13.12.6 Activation liveness without a fictitious sum . . . 3037
13.12.7 Logical collective communication . . . 3038
13.12.8 Per-token serving arithmetic . . . 3039
13.12.9 Why the carried structure does not become a second attention pass . . . 3039
13.12.10 `AmbiDefect` is not free, even when its realised value is zero . . . 3041
13.12.11 The Arakelov cost type and the exact sum rule . . . 3043
13.12.12 The coordinate ratios and the overhead theorem . . . 3045
13.12.13 Failure modes and the accurate worst case . . . 3046
13.12.14 Final ledger and what has, and has not, been proved . . . 3048
13.12.15 Sensitivity, budget slack, and adversarial substitutions . . . 3049
13.12.16 Audit closure . . . 3052

Chapter 13.13 — Monodromy separation: the exact S_5 scan, its reach, and the conditional it licenses 3054
13.13.1 What this chapter assembles, and the three parts that must never merge 3054
13.13.2 Conventions carried in from Chapter 2.14, and the two this chapter adds 3055
13.13.3 Part one: the upper bound, and the exact delimitation it applies to . . . 3056
13.13.4 Part two: the hardness of the ordered S_5 product . . . 3058
13.13.5 Part three: the construction, and what it must exhibit . . . 3060
13.13.6 Part four: the conditional conclusion, and the accuracy clause . . . 3073
13.13.7 Part five: does the construction satisfy the hypotheses it invokes? . . . 3074

13.13.8 The four-way non-collapse, and the exact sentence that may be published 3077
13.13.9 Rigour ledger . . . 3078
13.13.10 Open obligations . . . 3079
13.13.11 What the chapter establishes . . . 3079

Chapter 13.14 — A separating task family: three grammars, one hard core, and the controls that can refute them 3081
13.14.1 What this chapter owes, and the five statements it must never merge . . . 3081
13.14.2 Conventions carried in by number . . . 3082
13.14.3 What a task family is, formally . . . 3083
13.14.4 Family A: consistency and coreference chains . . . 3084
13.14.5 Family B: bracketed entity state . . . 3086
13.14.6 Family C: permutation composition over embedded templates . . . 3089
13.14.7 Reduction direction, and the uniformity audit that must accompany it . . . 3091
13.14.8 The baseline side, stated as the conditional it is . . . 3092
13.14.9 Negative control one: the solvable-group variant . . . 3092
13.14.10 Negative control two: the structure-destroying variant with identical surface statistics . . . 3095
13.14.11 The falsification protocol . . . 3097
13.14.12 Reachability: how the family meets item (iii) of the separation theorem 3098
13.14.13 Instance generation and contamination resistance . . . 3100
13.14.14 Clause: what this chapter does not prove . . . 3101
13.14.15 Rigour ledger . . . 3103
13.14.16 Source-request table . . . 3104
13.14.17 Import and deletion audit . . . 3104
13.14.18 Cut ledger . . . 3104
13.14.19 Open obligations . . . 3105
13.14.20 Deliberate demotions . . . 3105

Chapter 13.15 — A machine-checking kernel specification: statement-level Lean 4 encodings, finite fixtures, and the conjunctive release gate 3107
13.15.1 What this chapter is for, and the four things it refuses to conflate . . . 3107
13.15.2 The artefact status ladder and the licence grades S0, S1, S2 . . . 3108
13.15.3 The trusted computing base, and the tactics that silently enlarge it . . . 3109
13.15.4 Encoding the exact-degeneration theorem: what a statement-level encoding requires . . . 3111
13.15.5 The reverse program, the one-step map, and the trajectory induction . . . 3114
13.15.6 Encoding the reachability theorem of Chapter 13.13 . . . 3118
13.15.7 Finite fixture semantics: what a finite model may and may not stand for 3120
13.15.8 The kernel-checkable proof-term artefact . . . 3123
13.15.9 The kernel-check pipeline, stage by stage, with its failure modes . . . 3124
13.15.10 Failure to check is automatic demotion, and the demotion has a fixed point 3127
13.15.11 §Q12: the conjunctive release gate over the finite T-graded assertions . . . 3128
13.15.12 The three gaps a kernel cannot close . . . 3132
13.15.13 Import and deletion audit . . . 3133
13.15.14 Cut ledger, graded transfer, and outstanding sources . . . 3133
13.15.15 Deliberate demotions . . . 3136

Chapter 13.16 — The replacement ledger: what Part P proves, at which grade, under which machine check, and what would refute it **3138**
13.16.1 Purpose, and the four things this chapter refuses to do . . . 3138
13.16.2 The five columns, and the rule that assigns each . . . 3139
13.16.3 Chapter 13.10: the one forward map . . . 3143
13.16.4 Chapter 13.11: exact degeneration . . . 3147
13.16.5 Chapter 13.12: the overhead theorem . . . 3153
13.16.6 Chapter 13.13: the monodromy separation . . . 3158
13.16.7 Chapter 13.14: the separating task family . . . 3163
13.16.8 Chapter 13.15: the kernel-check specification, and what it actually awards 3166
13.16.9 The licence calculus: which units, passed, license which sentences . . . 3168
13.16.10 The unproved register: the exact list, where each is stuck, and what falls with it . . . 3174
13.16.11 The falsification protocol . . . 3176
13.16.12 Graded transfer table . . . 3182
13.16.13 Source-request table . . . 3182
13.16.14 Import and deletion audit . . . 3183
13.16.15 Retrofit register . . . 3184
13.16.16 Cut ledger . . . 3184
13.16.17 Deliberate demotions . . . 3186

Part XIV — The Replacement Case **3188**

Chapter 14.1 — The replacement assessment **3189**
14.1.1 How to read the correspondence ledger . . . 3190
14.1.2 The ledger . . . 3191
14.1.3 Reading the ledger . . . 3196
14.1.4 What the ledger does not contain . . . 3196
14.1.5 The four-tier verdict . . . 3197
14.1.6 What would falsify the whole programme . . . 3203
14.1.7 Why the mathematics is not decoration . . . 3204
14.1.8 The provenance volumes: what Parts XI, XII and XIII settle . . . 3206
14.1.9 Roadmap . . . 3209
14.1.10 Closing statement . . . 3211
14.1.11 Conditional replacement calculus . . . 3212
14.1.12 Four conditional meanings of replacement . . . 3215
14.1.13 Conditional failure and salvage . . . 3216
14.1.14 Deliberate demotions . . . 3217

Appendices 3219

Appendices 3220

Appendix A0 — Complete notation 3220

A0.1 Naming grammar 3222

A0.2 Categories, sites and topoi 3224

A0.3 Sheaves, descent and cohomology 3226

A0.4 Architecture layers and runtime state 3228

A0.5 Mixture-of-experts 3230

A0.6 Reinforcement learning 3231

A0.7 Information theory, probability and statistics 3233

A0.8 Linear algebra and numerical analysis 3234

A0.9 Logical grades and document markers 3236

A0.10 Conformance of these tables to actual usage 3237

A0.11 Deliberate demotions 3240

Appendix A1 — Proofs of the principal results 3241

A1.1 The coend quotient and attention weights — proved in Chapter 2.9 3243

A1.2 The down-set Context Site — proved in Chapter 2.1 3244

A1.3 ReLU, LayerNorm and max-pooling 3245
A1.3.1 LayerNorm 3245
A1.3.2 Max-pooling 3246

A1.4 Holonomy, global sections and the growth dichotomy — final owners Chapters 2.11 and 3.6 3247

A1.5 The restriction-map collapse theorem — proved in Chapter 5.1 3248

A1.6 Heterogeneous stalks exclude the identity sheaf 3249

A1.7 The matched-budget top-k redundancy gap 3250

A1.8 The Čech–Hodge identity for grouped advantages 3251
A1.8.1 General comparison graph and leave-one-out baseline 3252
A1.8.2 Variance-optimal weights 3252

A1.9 Group centring, leave-one-out and random group size 3254

A1.10 Exact likelihood of a simultaneous revision 3256

A1.11 Nuclei and Lawvere–Tierney topologies on a localic topos 3258

A1.12 Explicit activation counterexamples 3260

A1.13 Full affine LayerNorm characterisation 3261

A1.14 Haar projection onto holonomy invariants 3262

A1.15 Euler characteristic and non-vacuous obstruction space 3263

A1.16 Gerbe-diagnostic vacuity on low-dimensional nerves 3264

A1.17 Pooled-query non-separability 3265

A1.18 Hodge decomposition and the obstruction certificate 3266

A1.19 Stability of the descent resolvent 3268

A1.20 Submodularity of coverage and the greedy guarantee 3269

A1.21 Geometric morphisms and the preserved logical fragment 3270

A1.22 Parallel-action policy gradient and abstention — derived in A7 3271

A1.23 Bias introduced by truncated importance sampling 3272

A1.24 Attention compositionality and projection structure 3273

A1.25 The read-out-only saddle at zero restrictions 3274

A1.26 Early results of Parts I and II 3275

A1.27 What remains a proof sketch 3278

A1.28 Division of labour with the expanded proof appendix 3279

A1.29 Deliberate demotions 3280

Appendix A2 — Glossary 3281

A2.1 A 3283

A2.2 B 3284

A2.3 C 3285

A2.4 D 3287

A2.5 E 3288

A2.6 F 3289

A2.7 G 3290

A2.8 H 3291

A2.9 I 3292

A2.10 J 3293

A2.11 K 3294

A2.12 L 3295

A2.13 M 3296

A2.14 N 3297

A2.15 O 3298

A2.16 P 3299

A2.17 Q 3300

A2.18 R 3301

A2.19 S 3302

A2.20 T 3304

A2.21 U 3305

A2.22 V 3306

A2.23 W 3307

A2.24 Z 3308

A2.25 Divergence register: where this book's term is not the source's term 3309

A2.26 Deliberate demotions 3313

Appendix A3 — A fully worked end-to-end example 3314

A3.1 The ground rules of this appendix 3316

A3.2 The site 3317

A3.2.1 Evidence atoms and the order between them 3317

A3.2.2 The lattice of contexts 3317

A3.2.3 The three coverage axioms, verified on this instance 3318

A3.2.4 Why the subspace lattice is not an alternative 3319

A3.3 The truth object of this site **3321**
A3.3.1 Double negation fails, and it fails on the summaries . . . 3321
A3.3.2 The double-negation sublocale forgets exactly the summaries . . . 3321
A3.3.3 The support up-set as a finite provenance index . . . 3322
A3.3.4 Forcing, and the retrieval mandate . . . 3322
A3.3.5 Withdrawn sheafification fixture . . . 3322

A3.4 The covering family and its nerve **3323**
A3.4.1 Four experts, arranged as the boundary of a tetrahedron . . . 3323

A3.5 The sheaf **3325**
A3.5.1 Stalks: heterogeneous by construction . . . 3325
A3.5.2 Interface bases from Householder reflectors . . . 3325
A3.5.3 Restrictions . . . 3325
A3.5.4 The Identity Sheaf Network is inexpressible on this instance . . . 3326

A3.6 Holonomy **3327**
A3.6.1 Transport and the cycle basis . . . 3327
A3.6.2 The commutator: this group is non-abelian . . . 3327
A3.6.3 Gauge and basepoint invariance, verified . . . 3328
A3.6.4 The fixed subspace, and H^0 . . . 3328
A3.6.5 A heterogeneous-stalk refinement of $H^0 \cong (\mathbb{R}^d)^{\mathrm{Hol}}$ . . . 3328
A3.6.6 Freeness, word growth, and the ε-counting trap . . . 3329

A3.7 The coboundary and the sheaf Laplacian **3331**
A3.7.1 δ^0 explicitly . . . 3331
A3.7.2 The Laplacian and its block structure . . . 3331
A3.7.3 The spectrum . . . 3331
A3.7.4 The kernel is the private subspace . . . 3332
A3.7.5 Two controls: the abelian variant and the constant sheaf . . . 3332

A3.8 The descent solve **3334**
A3.8.1 The observation cochain . . . 3334
A3.8.2 The coboundary trap, and the harmonic residual . . . 3334
A3.8.3 Conjugate gradients . . . 3335
A3.8.4 Cost accounting under a declared convention . . . 3335

A3.9 Cohomology in degree one: the narrow result **3337**

A3.10 Degree-two capacity, not a gerbe computation **3338**

A3.11 Attention on the instance, and the compositionality defect **3339**
A3.11.1 The attention matrix . . . 3339
A3.11.2 The defect . . . 3339

A3.12 Routing: relative max coverage **3341**
A3.12.1 The competence matrix . . . 3341
A3.12.2 Top-k versus greedy . . . 3341

A3.13 The naturality obstruction, measured **3343**

A3.14 An off-manifold rank-completion sweep 3344

A3.15 One DGSPO update, in full 3345
A3.15.1 The rollouts and the Ω-valued reward . . . 3345
A3.15.2 Gödel arithmetic on the instance . . . 3345
A3.15.3 Disintegration . . . 3346
A3.15.4 The group nerve and the coboundary of the reward . . . 3347
A3.15.5 Theorem 6.2.4, verified numerically . . . 3347
A3.15.6 The estimator family: one vector, three divisors . . . 3347
A3.15.7 The degree-weighted generalisation . . . 3349
A3.15.8 The gate and the importance correction . . . 3350

A3.16 Component-level floor checks, not an integrated degeneration 3352

A3.17 Five ways this example could have been rigged 3353

A3.18 What this instance establishes, and what it does not 3355

A3.19 Precision, reproducibility and the source 3357

A3.20 Numerical reconciliation with Appendix A4 3358

A3.21 Adversarial closure controls 3359

A3.22 The ledger of computed quantities 3360

A3.23 Closed forms behind the ledger, and an internal consistency audit 3362

A3.24 Deliberate demotions 3365

Appendix A4 — Reproducing the worked example 3366

A4.1 The environment 3368

A4.2 Listing 1 — the site, its lattice, and the three coverage axioms 3369

A4.3 Listing 2 — the Heyting algebra of the site 3371

A4.4 Listing 3 — the cellular sheaf 3373

A4.5 Listing 4 — the subspace alternative fails distributivity 3375

A4.6 Listing 5 — holonomy, the fixed subspace, and word growth 3376

A4.7 Listing 6 — the coboundary, the sheaf Laplacian, and the spectrum 3378

A4.8 Listing 7 — the abelian variant and the constant sheaf 3380

A4.9 Listing 8 — the descent solve by guarded conjugate gradients 3381

A4.10 Listing 9 — cohomology dimensions and the scalar nerve complex 3384

A4.11 Listing 10 — degree-two capacity, without a planted gerbe 3386

A4.12 Listing 11 — the compositionality defect 3387

A4.13 Listing 12 — relative max coverage against top-k 3389

A4.14 Listing 13 — the naturality residual with a registered invariance class 3391

A4.15 Listing 14 — one DGSPO update, part one: the guarded advantage identity 3393

A4.16 Listing 15 — one DGSPO update, part two: the degree-weighted generalisation 3395

A4.17 Listing 16 — the degeneration check 3397

A4.18 Listing 17 — invariance, inexpressibility, and off-manifold rank completion 3398

A4.19 Listing 18 — calibration: Haar NAb, packing numbers, Gödel arithmetic 3401

A4.20 Listing 19 — adversarial core closure tests 3403

A4.21 The assertion suite 3406

A4.22 Diffing against a fresh run 3408

A4.23 Stage-4 supplement — worked finite transactions for the five engines 3409
A4.23.1 Listing 20 — weight-graded compilation 3409
A4.23.2 Listing 21 — almost-descent with separate ledgers 3411
A4.23.3 Listing 22 — solid successor, tail and named untilt 3413
A4.23.4 Listing 23 — canonical claim, proof term, transport, invariance and cd audit 3414
A4.23.5 Listing 24 — coherent route, norm defect, interaction tower and BC square 3416

A4.24 Supplementary assertion suite 3419
A4.24.1 Listing 25 — atomic cross-engine receipt and single-fault meets 3420

A4.25 Deliberate demotions 3424

Appendix A5 — Complete algorithm compendium 3425

A5.1 Uniform contract and type vocabulary 3427

A5.2 Backward and trainability contract 3428

A5.3 Shared conformance contract 3430

A5.4 Canonical context-site construction 3432

A5.5 Greedy minimal-sufficient-cover generation 3434

A5.6 Differentiable probabilistic cover 3436

A5.7 Restriction-map application 3438

A5.8 Stiefel retraction and integrity check 3440

A5.9 Naturality-residual measurement 3442

A5.10 Regularised gluing by preconditioned conjugate gradients 3444

A5.11 Consistency and transmission diagnostic 3447

A5.12 Attention idempotence diagnostic 3449

A5.13 Algebraic holonomy certification 3450

A5.14 Beck–Chevalley deviation probe 3452

A5.15 Obstruction interface — under redesign 3454

A5.16 Link-explicit expert dispatch 3455

A5.17 Link-aware expert combine 3457

A5.18 Gerbe interface — under redesign 3459

A5.19 Exact simultaneous revision-round likelihood 3460

A5.20 Degree-weighted cover advantage 3462

A5.21 Exact sequence-ratio primitive 3464

A5.22 Sequence-level truncated importance sampling 3467

A5.23 Dynamic group admission 3469

A5.24 Frozen-route replay 3470

A5.25 Externally committed site protocol 3472

A5.26 Abstention calibration and decision 3474

A5.27 Self-edit certification gate 3476

A5.28 Reference-policy reset transaction 3479

A5.29 Revisable decoder state cache 3481

A5.30 Cross-algorithm composition sketch 3483

A5.31 Stage-4 compendium supplement — the five engine transactions 3485
A5.31.1 Deterministic parallel scheduling, cost charging and restart 3494

A5.32 Deliberate demotions 3497

Appendix A6 — Experimental protocol and conformance harness 3498

A6.1 Metric invariance registry 3500

A6.2 Preregistration and data custody 3503
A6.2.1 The signed registers and the harness invocation . . . 3503

A6.3 Conventions for every confirmatory experiment 3505

A6.4 Claims and killing ablations 3506
A6.4.1 C1 — Context-order utility . . . 3506
A6.4.2 C2 — Coverage routing . . . 3506
A6.4.3 C3 — Learned restrictions . . . 3506
A6.4.4 C4 — Separate effects of F1 and F2 . . . 3506
A6.4.5 C5 — Forward gluing . . . 3506
A6.4.6 C6 — Naturality residual . . . 3507
A6.4.7 C7 — Obstruction head . . . 3507
A6.4.8 C8 — Gerbe diagnostic . . . 3507
A6.4.9 C9 — Holonomy capability wager . . . 3507
A6.4.10 C10 — Exact round likelihood . . . 3507
A6.4.11 C11 — Cover baseline . . . 3508
A6.4.12 C12 — Frozen route replay . . . 3508
A6.4.13 C13 — Dynamic sampling . . . 3508
A6.4.14 C14 — Calibrated abstention . . . 3508
A6.4.15 C15 — Beck–Chevalley probe . . . 3508
A6.4.16 C16 — Self-edit gate . . . 3509

A6.5 Baseline suite and fairness 3510

A6.6 Datasets and task selection 3513
A6.6.1 Formal mathematics protocol . . . 3513
A6.6.2 Code protocol . . . 3514
A6.6.3 Retrieval-grounded question answering protocol . . . 3514
A6.6.4 Route-dependent and multi-context protocol . . . 3514
A6.6.5 Safety, shift and abstention protocol . . . 3515

A6.7 Instrument calibration and positive controls 3516

A6.8 Seeds, variance and statistical models 3517
A6.8.1 Power and stopping calculation . . . 3517

A6.9 Falsification ladder 3519

A6.10 Compute and cost accounting 3520

A6.11 Independent conformance harness 3521

A6.12 Distributed execution protocol 3522

A6.13 Incident classification and stop rules 3523

A6.14 Reporting templates **3524**

A6.15 Adversarial protocol audit **3525**

A6.16 Decision rules and publication **3526**

A6.17 Subordination to the signed statistical plan **3527**

A6.18 Deliberate demotions **3530**

Appendix A7 — Long derivations **3531**

A7.1 Variation of the gluing functional **3533**
A7.1.1 Differentiating through the minimiser 3534
A7.1.2 Why F1 and F2 are jointly required 3534
A7.1.3 Stiefel-constrained criticality 3535

A7.2 Čech–Hodge computation of grouped advantage **3536**
A7.2.1 Complete graph 3536
A7.2.2 Graph baseline construction 3537
A7.2.3 Variance-optimal cover weights 3538

A7.3 Spectrum of the group-nerve Laplacian **3539**

A7.4 Exact mutual information and scale-sensitive InfoNCE **3541**

A7.5 Naturality obstruction for pointwise activation **3543**
A7.5.1 Gauge dependence 3544

A7.6 Top-k redundancy gap **3545**
A7.6.1 Noisy duplicates 3545
A7.6.2 What the theorem does not cover 3545

A7.7 Tetrahedral-boundary topology and the withdrawn gerbe datum **3547**

A7.8 Parallel revision policy gradient **3548**
A7.8.1 Round subsampling 3549
A7.8.2 Schedule-conditioned versus marginal likelihood 3549

A7.9 Bias from truncated importance sampling **3551**

A7.10 Incentive compatibility of abstention **3553**

A7.11 Harmonic residual, projections and numerical error **3554**
A7.11.1 Gauge invariance 3555
A7.11.2 Refinement 3555

A7.12 Basepoint, gauge and holonomy presentation **3557**

A7.13 Beck–Chevalley deviation and link dependence **3559**

A7.14 Additional analysis of the F1/F2 annealing conflict **3560**

A7.15 Submodular cover and the routing relaxation **3561**
A7.15.1 Probabilistic coverage 3562

A7.16 Joint critical points of sections and restrictions **3563**

A7.17 Non-scalar reward and scalar policy gradients **3565**
A7.17.1 Vector baselines 3566

A7.18 Site commitment and the trivial-site game **3567**

A7.19 Naturality beyond ReLU **3569**

A7.20 One-way conformance of local specialisations **3571**

A7.21 Clipping geometry in DGSPO **3572**

A7.22 Pooled-query non-separability **3574**

A7.23 Aggregation idempotence calculation **3575**

A7.24 Structural self-edit guarantees and counterexamples **3577**

A7.25 Retraction geometry and optimiser state **3579**

A7.26 Detailed cohomology of $\partial\Delta^3$ **3581**

A7.27 Spectral perturbation of cover advantages **3583**

A7.28 Batch scaling versus group standardisation **3585**

A7.29 Curvature of the InfoNCE scale objective **3587**

A7.30 Remaining gaps and boundaries of the derivations **3588**

A7.31 Three quantitative backward-error inequalities **3589**
A7.31.1 Inequality one: residual to state 3590
A7.31.2 Inequality two: implicit derivative to oracle 3591
A7.31.3 Inequality three: exact programme reverse to implicit surrogate 3592
A7.31.4 Executable certificate construction 3594
A7.31.5 Certifying constants and turning the inequalities into stop rules 3595

A7.32 Deliberate demotions **3600**

Appendix A8 — Proofs for the lineage Parts **3601**

A8.1 What this appendix deliberately does not repeat **3603**

A8.2 Standing hypotheses and citation discipline **3604**

A8.3 Tame stratification of routers **3605**

A8.4 Filtered descent: strictness is the load-bearing hypothesis 3608

A8.5 Almost-strictness and graded exactness 3610

A8.6 Convergence of the obstruction tower 3613

A8.7 Cohomological dimension of the finite decision site 3615

A8.8 Order independence of iterated filtrations 3617

A8.9 The normative gluing operator and its backward map 3618

A8.10 Cross-theatre transport and multiradial comparison 3621

A8.11 Proof-to-main-text traceability 3623

A8.12 What remains open after this appendix 3625

A8.13 Downstream obligations 3626

Appendix A9 — Attribution register: imported results, source identity, and release caps 3628

A9.1 Purpose and governing policy 3629

A9.2 The complementary boundary: unregistered material and its author 3631

A9.3 Part II: sites, topoi, higher categories, and condensed coefficients 3632

A9.4 Part II: cohomology, Galois theory, motives, and anabelian reconstruction 3636

A9.5 Part III: separation complexity, approximation, and architecture theorems 3645

A9.6 Part IV: reinforcement learning, calibration, and sequential statistics 3649

A9.7 Lineage: Deligne, Voevodsky, Scholze, Lurie, and Mochizuki 3652

A9.8 Bibliography work orders and release-facing residual risks 3658

A9.9 Deliberate demotions 3661

Appendix A10 — Expanded proofs: complete arguments for exact degeneration, monodromy reachability, and bounded overhead 3662

A10.1 Scope, and what an expanded proof is required to contain 3663

A10.2 Part A — the exact degeneration theorem, proved in full 3665
A10.2.1 The hypothesis register 3665
A10.2.2 The arithmetic identities the dévissage consumes 3666
A10.2.3 Three corrections to the mainline illustrations 3668

A10.2.4 The reachable set, and the dévissage criterion with its induction written out . . . 3670
A10.2.5 The eight graded pieces . . . 3671
A10.2.6 Value equality, and then gradient equality . . . 3675
A10.2.7 One optimiser step, and then the trajectory . . . 3677

A10.3 Part B — the reachability half of the monodromy separation theorem 3679
A10.3.1 The hypothesis register for Part B . . . 3679
A10.3.2 The scan, its invariants, and its correctness . . . 3680
A10.3.3 Boolean depth, uniformity, and the fixed-depth claim . . . 3681

A10.4 Part C — the overhead theorem, derived from the frozen inventory 3684
A10.4.1 The hypothesis register for Part C . . . 3684
A10.4.2 The finite coordinates . . . 3685
A10.4.3 The archimedean coordinates . . . 3685

A10.5 Part D — the item named NB: search, verdict, and what is delivered instead 3688
A10.5.1 The search . . . 3688
A10.5.2 The verdict . . . 3688
A10.5.3 What is delivered instead: the one NB-labelled assertion in the book, proved . . . 3689
A10.5.4 Budget reallocation, recorded . . . 3690

A10.6 Cross-cutting registers 3691
A10.6.1 Cut ledger . . . 3691
A10.6.2 Remediation register . . . 3691
A10.6.3 Graded transfer table . . . 3691
A10.6.4 Source-request table . . . 3692
A10.6.5 Import and deletion audit . . . 3692
A10.6.6 The complete GAP register . . . 3692

A10.7 Deliberate demotions 3694

Appendix A11 — Normative pseudocode: typed operators, zero cases, and executable vectors 3696

A11.1 Authority, scope, and the prohibition on a second forward map 3697

A11.2 Detached coverage audit and the only executable fallback 3700

A11.3 L0 context-site construction from admitted atoms 3703

A11.4 L1 sheaf construction on the frozen L0 base 3706

A11.5 Frozen-base L2 descent attention followed by L3 nucleus closure 3709

A11.6 L4 sequence-level reconcile with a fixed realised reverse 3713

A11.7 L5 obstruction stage: provenance first, abstention first 3717

A11.8 L6 geometric forcing and principal support sieves 3720

A11.9 L7 holonomy read-out with exact witness discipline 3723

A11.10 L8 base change between query transactions 3726

A11.11 The L5/L6/L7 antichain, append-only bus, and output gate 3729

A11.12 Cross-operator conformance ledger 3732

A11.13 Formal algorithm obligations 3733

A11.14 Deliberate demotions 3737

Appendix A12 — Pre-registered statistical analysis plan 3739

A12.1 Scope, authority, and the closed claim universe 3740

A12.2 Canonical records and reduction from raw logs 3742

A12.3 Common estimators, uncertainty, and computational determinism 3744

A12.4 Registered endpoint catalogue 3746

A12.5 Exact endpoint computations 3749

A12.6 Multiplicity and the twenty confirmatory hypotheses 3753

A12.7 Training, evaluation, stopping, ties, crashes, and outliers 3756

A12.8 Falsification predicates and decision states 3758

A12.9 Reporting contract and negative-results obligation 3760

A12.10 Claims that are not empirical until relabelled or redesigned 3763

Appendix A13 — Formalisation source appendix: Lean 4 modules, dependency closure, and the complete trust ledger 3764

A13.1 Scope, ownership, and the status sentence that governs every listing 3765

A13.2 Exact version lock and source tree 3767

A13.3 Common prelude: status tags and an injective map 3769

A13.4 PB source module one: homogeneous finite programs and dévissage 3770

A13.5 PB source module two: forward, reverse, one-step, and trajectory squares 3772

A13.6 PB source module three: a finite smoke fixture and nothing more 3775

A13.7 PD source module one: the exact five-point permutation carrier 3777

A13.8 PD source module two: sequential reference prefixes and exact acceptance 3779

A13.9 PD source module three: semantic reachability and abstract schedule budget 3781

A13.10 Audit driver: the declarations whose axiom closure must be printed 3783

A13.11 Module dependency graph and the exact Mathlib surface 3784

A13.12 Complete axiom, `sorry`, and decision-procedure ledger 3786

A13.13 Declaration-by-declaration verification table 3789

A13.14 Build and reproduction sequence 3792
A13.14.1 Common failure modes and their exact meanings 3794

A13.15 Formalisation gap: Chapter 13.11 versus the listed theorem types 3796
A13.15.1 PB gaps, item by item 3796

A13.16 Formalisation gap: Chapter 13.13 versus the listed theorem types 3798
A13.16.1 PD gaps, item by item 3798

A13.17 Review protocol, falsification points, and promotion discipline 3800

A13.18 Deliberate demotions 3801

Appendix A14 — Deterministic task generators, negative controls, and contamination-resistant evaluation 3803

A14.1 Scope and the operational boundary 3804

A14.2 Canonical octets and dataset framing 3806

A14.3 Counter-mode pseudorandomness and stream splitting 3808

A14.4 Balanced labels and the exact label-null 3810

A14.5 Exact finite-group substrate 3812

A14.6 Difficulty as a located, monotone resource envelope 3813

A14.7 Family one: exact fixed-group identity words 3815
A14.7.1 Word-family difficulty and admissible scoring 3816

A14.8 Family two: deterministic Barrington compilation 3817

A14.9 Family three: exact prefix-scan conformance 3820

A14.10 Family four: degree-one collisions and class-two order 3822

A14.11 Family five: exact relation lattices and torsion 3825

A14.12 Family six: monodromy-rigidity records 3827

A14.13 Cross-family control matrix 3829

A14.14 Pre-registered sample sizes, accuracy, and gap rules 3831

A14.15 Anti-contamination protocol 3833
A14.15.1 Generation-time-only handling . . . 3834
A14.15.2 Canary strings . . . 3834

A14.16 Contamination predicate and post-suspicion procedure 3835
A14.16.1 Detection workflow after a suspected leak . . . 3836

A14.17 Seed publication and benchmark retirement 3837

A14.18 Conformance, independent implementation, and release checklist 3838

A14.19 Operational conclusion 3840

Appendix A15 — Negative results: every cut, every demotion, and the hypothesis that failed in each case 3841

A15.1 What this chapter is, and the two ways it could be misleading 3842

A15.2 The evidence rule and the citation convention 3843

A15.3 The evaporation test: exact statement, and what it cannot decide 3844

A15.4 The book-wide restatement census 3846

A15.5 The six evaporated postulates 3847
A15.5.1 E1 — The universal central-extension postulate . . . 3847
A15.5.2 E2 — Tannakian or anabelian reconstruction of a trained model . . . 3848
A15.5.3 E3 — A global perfectoid tilt . . . 3848
A15.5.4 E4 — A motive of a neural model . . . 3849
A15.5.5 E5 — Unrestricted theatre identification . . . 3849
A15.5.6 E6 — Prismatic comparison . . . 3850

A15.6 The decoration rate 3851
A15.6.1 A second caliber: the audited lineage subset . . . 3853
A15.6.2 A third caliber, and the harshest reading . . . 3853
A15.6.3 The consumer ratchet, the gate that was failing, and what happened to it 3854

A15.7 §F4 discharged in one place 3855
A15.7.1 The obligation, and where it is stated . . . 3855
A15.7.2 Two defects found while discharging . . . 3855
A15.7.3 The discharge table . . . 3856

A15.8 Cut ledger: the one hundred cuts of the second restatement audit 3861
A15.8.1 Chapter 11.12 — ten cuts: the identification failures . . . 3861
A15.8.2 Chapter 11.13 — eleven cuts: almost mathematics and tolerance . . . 3862
A15.8.3 Chapter 11.14 — eight cuts: perfectoid and pro-étale . . . 3862

A15.8.4 Chapter 11.15 — nine cuts: motivic vocabulary 3863
A15.8.5 Chapter 11.16 — nine cuts: higher category vocabulary 3863
A15.8.6 Chapter 11.17 — twenty-two cuts and three demotions: the defect calculus 3863
A15.8.7 The three demotions 3864

A15.9 The twenty-two cuts and the forty grade findings of the first restatement audit 3866

A15.10 Two pairs are excluded, not reconciled 3868
A15.10.1 G against C: finite tame compilation against unrestricted infinite-context topology 3868
A15.10.2 V against S: global record identification against nominal species separation 3869

A15.11 Fourteen of the twenty-one pairs hold only under a named finite witness 3871

A15.12 Four no-go theorems 3873

A15.13 Import and deletion audit 3875

A15.14 The undetermined register and its source requests 3877

A15.15 The eight unapplied forward-map demotions 3881

A15.16 What this chapter does not claim 3882

A15.17 The tally 3883

A15.18 Deliberate demotions 3884

Appendix A16 — Master bibliography 3886

A16.1 Topos theory and category theory 3888

A16.2 Sheaves, cohomology and cellular sheaf theory 3890

A16.3 Geometric group theory 3892

A16.4 Transformers, attention and diffusion 3893

A16.5 Mixture of experts 3894

A16.6 Reinforcement learning and verifiable reasoning 3895

A16.7 Self-improvement and evolution 3897

A16.8 Categorical and geometric deep learning 3898

A16.9 Safety and alignment under fine-tuning 3899

A16.10 Verification repair sources 3900

A16.11 Stage Four attribution-register reconciliation 3904

A16.12 Citation and verification policy **3908**

A16.13 Verification note **3909**

A16.14 Deliberate demotions **3911**

List of Figures

1 **FM.1** Master map of the mathematical programme. . . . civ

2 **1.1.1** Five structural design hypotheses. . . . 4

3 **1.3.1** Three methodological transfers. . . . 31

4 **2.1.1** Context sites and flat token indexing. . . . 50
5 **2.1.2** Boolean and Heyting truth values. . . . 60
6 **2.1.3** Nucleus tests for common nonlinearities. . . . 66
7 **2.1.4** Cited M transport of the geometric fragment. . . . 73

8 **2.5.1** Čech complex of a cover. . . . 178
9 **2.5.2** H^1 obstruction from measured edge data. . . . 183
10 **2.5.3** Two-dimensional nerves for H^2. . . . 188

11 **2.9.1** Coend as a coequaliser. . . . 299
12 **2.9.2** Proposed T-Block transport module. . . . 305

13 **2.11.1** Schematic holonomy capacity trade-off. . . . 349
14 **2.11.2** Holonomy growth gap. . . . 352
15 **2.11.3** Holonomy diagnostics and ternary growth verdict. . . . 357

16 **2.15.1** The six-functor adjunctions. . . . 489
17 **2.15.2** Beck–Chevalley comparison square. . . . 499
18 **2.15.3** Finite torsor obstruction. . . . 509

19 **3.1.1** The whole GM-$\Omega\infty$ model in one figure. . . . 602
20 **3.1.2** Nine-layer architecture stack. . . . 606
21 **3.1.3** Single-query architecture dataflow. . . . 617
22 **3.1.4** The end-to-end algorithmic pipeline. . . . 628

23 **3.2.1** Evidence-atom context site. . . . 634
24 **3.2.2** Join-coverage axioms and counterexample. . . . 641

25 **3.3.1** Heterogeneous stalks and restrictions. . . . 654
26 **3.3.2** Discrete and indiscrete coends. . . . 663
27 **3.3.3** Transported barycentric attention. . . . 665

28 **3.4.1** Nucleus lattice and naturality defect. . . . 690
29 **3.4.2** Spectral descent solver. . . . 721

30 **3.5.1** Orthogonal obstruction-head decomposition. . . . 735
31 **3.5.2** Kripke–Joyal support sieves. . . . 751

32 **3.6.1** Holonomy gap and counting artefact. . . . 775
33 **3.6.2** Cited M geometric and non-geometric site transport. . . . 782

34 **4.1.1** Experts as open subtoposes. . . . 937
35 **4.1.2** Three expert formalisations. . . . 938

36 **4.2.1** Coverage and top-k routing. . . . 968
37 **4.2.2** Schematic detached RMC audit. . . . 974

38 **4.3.1** Historical link-coherence proposal. . . . 997
39 **4.3.2** Structural diagnostic prerequisites. . . . 1005

40 **4.4.1** Graded architecture specialisations. . . . 1015
41 **4.4.2** MoE techniques in topos notation. . . . 1023

42 **4.5.1** Schematic overlap-cost accounting. . . . 1036
43 **4.5.2** Restriction-map collapse ray. . . . 1044

44 **5.1.1** Schematic restriction-map degeneracies. . . . 1206
45 **5.1.2** Four partial anti-collapse mechanisms. . . . 1216

46 **5.2.1** Pretraining objectives on two structural axes. . . . 1252
47 **5.2.2** Schematic four-phase curriculum. . . . 1259

48 **5.3.1** Cited M geometric-fragment transfer conditions. . . . 1281
49 **5.3.2** Schematic forgetting asymmetry. . . . 1285

50 **5.4.1** Compiled site-construction pipeline. . . . 1306
51 **5.4.2** Site-trivialisation attack routes. . . . 1315

52 **6.1.1** Policy-optimisation ratio hierarchy. . . . 1333
53 **6.1.2** Importance-ratio levels for revisable decoding. . . . 1348

54 **6.2.1** Čech–Hodge advantage normalisations. . . . 1359
55 **6.2.2** Sampling groups and cover-weighted baselines. . . . 1363

56 **6.3.1** DGSPO update pipeline. . . . 1403
57 **6.3.2** Descent gate for rollout groups. . . . 1409

58 **6.4.1** Self-evolution control loop. . . . 1433
59 **6.4.2** Stratified self-edit language. . . . 1437

60 **6.5.1** Reinforcement-learning attack taxonomy. . . . 1461
61 **6.5.2** Local and global verifier failure. . . . 1465

62 **6.7.1** Rollout–learner trust boundary. . . . 1513
63 **6.7.2** Sources of likelihood mismatch. . . . 1519

64 **7.1.1** Versioned descent transaction. . . . 1541
65 **7.1.2** Heterogeneous-stalk memory layout. . . . 1546

66 **7.2.1** Fused restriction kernel. . . . 1568
67 **7.2.2** Symbolic structural-kernel roofline. . . . 1571

68 **7.3.1** Parallelism axes and ownership. . . . 1587
69 **7.3.2** Distributed communication schedules. . . . 1591

70 **7.4.1** Serving-state and cache architecture. . . . 1607
71 **7.4.2** Schematic coherence-tier work. . . . 1616

72 **8.1.1** API type lattice. . . . 1633

73 **8.2.1** Reference implementation flow. . . . 1670
74 **8.2.2** Structural-path conformance harness. . . . 1681

75 **9.1.1** Validation instrument matrix. . . . 1761
76 **9.1.2** Falsification-gate dependency graph. . . . 1776

77 **9.2.1** Outcome families and structural telemetry. . . . 1791
78 **9.2.2** Metric-gaming routes and validity gates. . . . 1809

79 **10.1.1** Rigour and dependency ledger. . . . 1825
80 **10.1.2** Four claim-grade lanes. . . . 1827

81 **10.2.1** Comparison-baseline positioning matrix. . . . 1862
82 **10.2.2** Priority risk register. . . . 1869
83 **10.2.3** Gated implementation roadmap. . . . 1874

84 **11.1.1** Grothendieck-programme transfer matrix. . . . 1889

85 **11.2.1** Probe copresheaves and representability. . . . 1925

86 **11.3.1** Motivic requirements and current limits. . . . 1958

87 **11.5.1** IUT dependency schematic. . . . 2024

88 **11.7.1** Frobenius-like and étale-like data. . . . 2098
89 **11.7.2** Transfer tiers, claim grades and the mandates each transfer binds. 2118

90 **11.9.1** Three typed ambiguity structures. . . . 2172
91 **11.9.2** Layerwise typed-defect propagation. . . . 2177
92 **11.9.3** Instrument invariance audit for presentation and refinement transformations. . . . 2181

93 **12.1.1** Two ambient enlargements and their different verdicts. . . . 2531
94 **12.1.2** Tame and non-tame router boundaries. . . . 2537
95 **12.1.3** Coherence of diagram extension. . . . 2543

96 **12.2.1** Weight-filtered training and diagnosis. . . . 2554
97 **12.2.2** Strictness and hidden weight descent. . . . 2563
98 **12.2.3** Conditional decomposition of expert aggregation. . . . 2574

99 **12.3.1** Almost envelope for numerical defects. . . . 2586
100 **12.3.2** Refinement and essential conditioning. . . . 2592

101 **12.4.1** Nisnevich routing square. . . . 2607
102 **12.4.2** Univalent claim transport. . . . 2624

103 **12.5.1** Tilting with a genuine return path. 2631

104 **12.6.1** Cocartesian routing over context refinement. 2657
105 **12.6.2** Postnikov obstruction tower for expert gluing. 2661

106 **12.7.1** Species information lattice. 2683
107 **12.7.2** Typed mutation span. 2685
108 **12.7.3** Universal multiradial comparison. 2692

109 **12.8.1** Seven-axis lineage object. 2704
110 **12.8.2** Restricted tilt and descent. 2711

111 **13.1.1** Weight-filtered learning stack. 2735
112 **13.1.2** Three-valued purity decision. 2741
113 **13.1.3** Weight-constrained coverage routing. 2744

114 **13.5.1** Cocartesian routing over refinements. 2839
115 **13.5.2** Finite higher-descent storage. 2842
116 **13.5.3** Distributed higher-descent transaction. 2854

117 **13.6.1** Multiradial runtime. 2875
118 **13.6.2** Alien evaluation boundary. 2877
119 **13.6.3** Failure-atomic mutation transaction. 2880

120 **14.1.1** Component correspondence ledger. 3191
121 **14.1.2** Four-level evidence verdict. 3198

122 **A3.1** Worked context site. 3320

123 **A3.2** Worked sheaf spectrum and solver trace. 3333

124 **A3.3** Worked Čech–Hodge policy update. 3351

List of Tables

FM.1 Claim grades and what each may be used for ci
FM.2 Status tokens for obligations that are not discharged ci
FM.3 The six fusion principles and their dependents ciii
FM.4 Part synopsis, volumes and dependencies cvi

1.1.1 The five deficits, surviving claims and grades 3
1.1.2 Observables licensed by each deficit 11
1.1.3 Repair map 12

1.2.1 The principal claims and their grades 19

1.3.1 The status vocabulary and what each word licenses 29
1.3.2 Rendering contract 33
1.3.3 Module authority index 33
1.3.4 Compliance matrix 34
1.3.5 Backward authority and blocking items 35
1.3.6 Fusion principles and the module obligations they change 37
1.3.8 Evaporation verdicts on the imports of this document 42
1.3.9 Where the transfer test is stated 45

2.1.1 Nucleus conditions for common activations on $\overline{\mathbb{R}}$ 64
2.1.2 Kripke–Joyal forcing clauses in $\mathrm{Sh}(C, J)$ 67
2.1.3 The three architectural site morphisms 72
2.1.4 The topology ladder and its architectural rungs 78
2.1.5 Rigour grades, Chapter 2.1 80

2.2.1 Topology selection by required descent 87
2.2.2 Incremental costs of the changed GM-$\Omega\infty$ components 106

2.3.1 Data required to distinguish local existence from gluing 115
2.3.2 Formalization status by mechanism 133

2.4.1 One-categorical statements, infinity-categorical scope 141
2.4.2 From one-topos to infinity-topos: cost and gain 155
2.4.3 Functor and bundle pictures for GM-Ω^{∞} 156
2.4.4 Named GM-Ω^{∞} component changes 171
2.4.5 Dominant costs and cache validity 172
2.4.6 Six-part contracts for the new modules 174
2.4.7 Decisions forced by the infinity-topos audit 175
2.4.8 Grades and boundaries 176

2.5.1 The four-cycle example in its two regimes 196
2.5.2 Negative and conditional verdicts 202

2.5.3 Rigour grades, Chapter 2.5 204

2.6.1 Result-by-result grades 234

2.7.1 Almost-purity hypotheses of Faltings and Scholze 249
2.7.2 Scalar regimes and conclusions they license 264
2.7.3 Required mathematical and architectural obligations 265
2.7.4 Negative, conditional and open conclusions 266

2.9.1 Compositionality defect of attention patterns 303
2.9.2 Rigour grades, Chapter 2.9 315

2.10.1 Rigour grades, Chapter 2.10 335

2.11.1 Verification of Theorem 2.11.12 on the five-cycle 344
2.11.2 Classification by the fixed representation 344
2.11.3 Word-ball cardinalities in $SO(3)$ 356
2.11.4 Rigour grades, Chapter 2.11 367

2.12.1 The SGA 1 axioms and their consumers 381
2.12.2 Invariance and budget classes for L7-GAL outputs 399
2.12.3 Results imported, conditional, or modelled 406

2.13.1 Result-by-result grades 448

2.15.1 Tannakian hypotheses against neural representation spaces 506
2.15.2 The anabelian dictionary, literal and surrogate 511
2.15.3 Rigour grades, Chapter 2.15 516

2.17.1 Scope and policy record for retained names 589
2.17.2 Cut table for rejected transfers and overclaims 591
2.17.3 Load-bearing import and exact later consumer 592
2.17.4 Every M grade and its required A16 repair 593
2.17.5 Consumer of every Chapter 2.17 declaration 594

3.1.1 The tower, layer by layer 605
3.1.2 The nine layers of GM-$\Omega\infty$ 607
3.1.3 Translation between standard vocabulary and GM-$\Omega\infty$ objects 608
3.1.4 The three disciplines across the layer tower 613
3.1.5 Type signatures of the nine layers 615
3.1.6 Layer invariants and their enforcement mechanism 618
3.1.7 One-way specialisation of each layer 620
3.1.8 Invariance class of every Part III instrument 623
3.1.9 Invariance class of the added instruments 625
3.1.10 Rigour grades, Chapter 3.1 629

3.2.1 Rigour grades, Chapter 3.2 650

3.3.1 Rigour grades, Chapter 3.3 676

3.4.1 Rigour grades, Chapter 3.4 722

3.5.1 Kripke–Joyal forcing clauses . . . 745
3.5.2 Rigour grades, Chapter 3.5 . . . 754

3.6.1 L7 reportable cases . . . 778
3.6.2 The predicted transfer asymmetry and how to measure it . . . 781
3.6.3 The collected falsification gates of Part III . . . 787
3.6.4 Invariance ledger for the L7 and L8 instruments . . . 787
3.6.5 Rigour grades, Chapter 3.6 . . . 788

3.7.1 Placement and ownership of the two organs . . . 793
3.7.2 Normative tensor contract of the T-Unit . . . 795
3.7.3 Backward ownership and trainability of both organs . . . 800
3.7.4 Cost rows of the T-Unit path . . . 802
3.7.5 T-Unit failures, detections and mandatory responses . . . 802
3.7.6 Alien Engine failure analysis . . . 809
3.7.7 Invariance and budget of exported quantities . . . 811
3.7.8 Pre-registered falsification gates for the two organs . . . 813
3.7.9 Module compliance declaration for the two organs . . . 814
3.7.10 Dévissage ledger, Chapter 3.7 . . . 816
3.7.11 Required link kinds and their principal budgets . . . 819
3.7.12 Minimum ablation matrix for the two organs . . . 822

3.8.1 Configuration invariance classes . . . 826
3.8.2 Invariance classes for the new configuration fields . . . 828
3.8.3 Global dimensions . . . 829
3.8.4 Parameter categories and checkpoint policy . . . 832
3.8.5 Rank phase gates . . . 835
3.8.6 Normative research schedule . . . 835
3.8.7 Local-specialisation compliance . . . 839
3.8.8 Normative cost-comparison key . . . 840
3.8.9 Checked arithmetic of the reference configuration . . . 840
3.8.10 Mandatory cost record . . . 840
3.8.11 Configuration and schedule failures . . . 841
3.8.12 Block-level bill of materials . . . 843
3.8.13 Site curriculum . . . 845
3.8.14 What each field of the report record decides . . . 847
3.8.15 Initial data mixture . . . 849
3.8.16 Objective activation . . . 850

3.10.1 Placement and ownership of the almost-descent operations . . . 894
3.10.2 Exact declaration trace for the six algorithms . . . 895
3.10.3 Ideal-audit outcomes and licensed operations . . . 898
3.10.4 `PurityRefine` failure taxonomy . . . 902
3.10.5 Decision cases and required ambiguity factors . . . 904
3.10.6 Invariance or budget for every exported quantity . . . 908
3.10.7 Pre-registered falsification tests . . . 911
3.10.8 Endpoint contract . . . 912
3.10.9 Dévissage ledger, Chapter 3.10 . . . 913

4.1.1 Three candidate formalisations of the expert . . . 946
4.1.2 Cost of the coverage instrument and defect program . . . 962

4.2.1 Invariance class of each routed quantity . . . 981
4.2.2 Deployed task routing and detached sequence audit . . . 986

4.3.1 Asymptotic descent operations . . . 1012

4.4.1 Gate conventions as separate degeneration targets . . . 1016
4.4.2 The degeneration ladder . . . 1017
4.4.3 Summary of the practice mapping . . . 1028
4.4.4 Contemporary sparse models placed on the ladder . . . 1029

4.5.1 Disposition of the composite objective . . . 1046
4.5.2 The monitoring dashboard . . . 1048
4.5.3 Imported ideas, restated without their source vocabulary . . . 1051

4.6.1 Ownership across L0 and L8 . . . 1056
4.6.2 Invariance and budget register . . . 1070
4.6.3 Pre-registered falsification cases . . . 1075
4.6.4 Dévissage ledger, Chapter 4.6 . . . 1087
4.6.5 API outcomes . . . 1089

4.7.1 L0–L8 placement and ownership . . . 1094
4.7.2 Exact mathematical trace of the five algorithms . . . 1096
4.7.3 MJ invariance and budget register . . . 1113
4.7.4 Complexity and communication bounds . . . 1114
4.7.5 Pre-registered falsification tests . . . 1118
4.7.6 Dévissage of the Motivic Justification unit . . . 1133
4.7.7 Nine-obligation compliance record . . . 1135
4.7.8 Minimum conformance corpus . . . 1139
4.7.9 Promotion evidence and invalid substitutes . . . 1152

4.8.2 Pre-registered falsification matrix . . . 1172
4.8.3 Dévissage ledger, Chapter 4.8 . . . 1176

5.1.3 Optimiser operations by parameter type . . . 1232
5.1.4 Per-step cost of the admissible objective . . . 1239
5.1.5 Collapse telemetry, with the failure each detects . . . 1244

5.2.1 The three paradigms, corrected . . . 1258

5.3.1 Per-episode cost of openness-controlled adaptation . . . 1292

5.4.1 Analytic cost model for site construction . . . 1312

6.1.1 The five design slots of policy optimisation . . . 1334
6.1.2 Policy-optimisation algorithms and their slots . . . 1342
6.1.3 Evidence required for common landscape statements (D) . . . 1356

6.2.2 Claims licensed by each filtered-Hodge status (D) . . . 1392

6.3.1 Practical sources of the local reward r_U . . . 1404
6.3.2 Components of the DGSPO objective . . . 1413
6.3.3 DGSPO hyper-parameters with justification . . . 1418

6.4.2 Reward sources by stage . . . 1449

6.5.1 Secondary failure modes . . . 1468
6.5.2 Attacks mapped to Goodhart variants . . . 1470

6.6.1 Calibration failures and responses . . . 1494

6.7.1 Trust boundaries in the RL system . . . 1514
6.7.2 Falsification ladder and exit criteria . . . 1525
6.7.3 Costs that must appear in every DGSPO report . . . 1526
6.7.4 Minimum release acceptance matrix . . . 1531

7.1.2 Backward authority for each execution mode . . . 1550
7.1.6 Typed trace and implementation-species mutation record . . . 1555
7.1.7 `RouteRecord-v2` packed little-endian schema . . . 1556
7.1.8 Two-component failures and mandatory responses . . . 1562
7.1.9 Adversarial fixtures for two-component accounting . . . 1564

7.2.2 Traffic of one Laplacian application . . . 1569
7.2.3 The bucket trade-off over the active range . . . 1570
7.2.4 Roofline compliance record . . . 1572
7.2.5 Precision regime by stage . . . 1572
7.2.6 Restriction storage at the normative configuration . . . 1573
7.2.7 Known liveness phases, not a peak bound . . . 1573
7.2.8 Exact limits of two-witness truncation evidence . . . 1576
7.2.9 Minimum adversarial fixtures for the three-witness gate . . . 1581
7.2.10 Update-to-witness invalidation matrix . . . 1582
7.2.11 Measured-versus-theoretical fields . . . 1585

7.3.1 Communication of the reference configuration . . . 1589
7.3.2 Buffers per sequence and per MoE block . . . 1590
7.3.3 Distributed backward contract for L4 . . . 1592
7.3.4 Checkpoint manifest fields, all mandatory . . . 1592
7.3.5 Distributed evidence failures and non-negotiable outcomes . . . 1600
7.3.6 Microbatch and straggler conformance cases . . . 1602

7.4.1 Serving determinism classes . . . 1610
7.4.2 Solve outcomes . . . 1611
7.4.3 Cache classes . . . 1612
7.4.4 Serving/cache lower-bound inventory . . . 1614
7.4.5 Coherence tiers without fabricated multipliers . . . 1615
7.4.6 Envelope failures and contract-preserving outcomes . . . 1622
7.4.7 Deadline and projection outcomes . . . 1624
7.4.8 Adverse serving observations and controlling evidence . . . 1626

8.1.1 Conformance levels and their obligations . . . 1631
8.1.2 The normative and binding split . . . 1632

8.1.3 Object kinds of the API and their comparisons 1635
8.1.4 Three former operations, one interface 1640
8.1.5 The six Section invariants 1643
8.1.6 The three solve modes 1647
8.1.7 Admissible certificate solvers 1648
8.1.8 Reported quantities and their invariance 1653
8.1.9 Error taxonomy 1656
8.1.10 API calls recovering a top-k MoE Transformer 1658

8.2.1 The eight executable obligations 1669
8.2.2 The three solve conformance traces the reference emits 1675
8.2.3 Conformance traces on the twelve-token fixture 1676
8.2.4 The obstruction on $\partial\Delta^3$ 1679
8.2.5 Deviation from the reference block 1682
8.2.6 Suite, assertion, and the fixture on which it is evaluated 1683
8.2.7 Parameter groups 1685
8.2.8 Shapes produced by Algorithm 8.2.14 1687
8.2.9 Configuration of the miniature reference 1687
8.2.10 Scope of the miniature reference 1689

8.3.1 Normative MoE shapes 1717
8.3.2 Contract violations 1720
8.3.3 Contract falsification programme 1722
8.3.4 Required fields by claim kind 1724
8.3.5 The three generated views and their acceptance rule 1726
8.3.7 Active executable authorities 1730
8.3.8 Corpus locations bearing on Record 8.3.17 1732
8.3.9 Correction register entry for Record 8.3.17 (reason 1733
8.3.10 Worked record fields 1735
8.3.11 Typed contract errors 1739

9.1.1 Shared decision regions and their typed outputs 1760
9.1.2 Collapse modes 1774
9.1.3 Hard gates, emitted failure outputs and what is abandoned 1777
9.1.4 Minimum decisive pilots 1780
9.1.5 Binding of instruments to named mathematical obligations 1782
9.1.6 Minimum PC evidence packet 1786
9.1.7 Contradictions detected before claim emission 1787

9.2.1 Metric families and evidential status 1792
9.2.2 Typed metric registry 1792
9.2.3 Transformation contract of reported families 1795
9.2.4 Telemetry cost 1808
9.2.5 Structural alerts 1814
9.2.6 Binding cards for the closed endpoint register 1818
9.2.7 Required decomposition in a theorem-facing decision record 1822

10.1.1 Invariance companion ledger for the principal claims 1828
10.1.2 Adjudicated failures and unresolved proof obligations 1830

10.1.3 Foundations, logic and learning objectives . . . 1831
10.1.4 Sheaves, cohomology and obstruction . . . 1833
10.1.5 Holonomy, position structure and capability . . . 1834
10.1.6 Topos-MoE . . . 1835
10.1.7 Reinforcement learning and self-evolution . . . 1837
10.1.8 Adaptation, stacks, motives and systems . . . 1841
10.1.9 Backward-contract and cost status by component . . . 1842
10.1.10 Open release obligations . . . 1843
10.1.11 §M principles with mandatory downstream consequences . . . 1845
10.1.12 Evidence required for ledger transitions . . . 1846
10.1.13 Corrections to earlier formulations . . . 1849
10.1.14 Obligation closure criteria . . . 1854
10.1.15 Meaning and admission rule of the five release labels . . . 1856
10.1.16 Actual S-tier distribution at the recorded snapshot . . . 1857
10.1.17 Distribution of principal claims . . . 1858
10.1.18 Release-critical claim families and their present basis . . . 1858

10.2.1 Positioning summary . . . 1866
10.2.2 Ranked risk register . . . 1866
10.2.3 Escalation policy . . . 1872
10.2.4 Final disposition criteria . . . 1877
10.2.5 Prior work, restatement and new work . . . 1879
10.2.6 Release-critical risks with exact triggers . . . 1882

11.1.2 The same retrofit authorisation instantiated seven times . . . 1911
11.1.3 Transfer verdicts on Grothendieck's programme . . . 1912
11.1.4 External consumers of every numbered item of Chapter 11.1 . . . 1914

11.2.1 What changes from categories to infinity-categories . . . 1923
11.2.2 Étale and pro-étale probes . . . 1931
11.2.3 Categorical isomorphisms and analogies . . . 1937
11.2.4 Context sheaf and probe functor compared . . . 1943
11.2.5 Complete record templates for Chapter-99 quantities . . . 1946
11.2.6 Grades for the results of this chapter . . . 1949

11.3.1 Falsifiable reconciliation with Chapter 2.16 . . . 1966
11.3.2 Tier and grade for every substantive claim of this chapter . . . 1979

11.4.1 Grade, source gap, and named consumer . . . 2010
11.4.2 Required bibliographic sources, Chapters 11.3–11.4 . . . 2011

11.5.1 Primary sources for Part XI . . . 2017
11.5.2 The two kinds of data, side by side . . . 2028
11.5.3 From exposition to programme . . . 2044
11.5.4 Items of Chapter 11.5 with tier and grade . . . 2044

11.7.1 Enforceable interface summary . . . 2085
11.7.2 What each operation does to each level . . . 2093
11.7.3 A structural correspondence, and nothing more . . . 2104
11.7.4 Principle, affected chapter, and the change required . . . 2109

11.7.5 Seven-lineage retrofit coverage . . . 2113
11.7.6 Summary . . . 2118
11.7.7 Every item of this Part . . . 2120

11.8.1 Occurrences in the audited file set . . . 2138
11.8.2 Deletion test, item by item . . . 2139
11.8.3 Open obligations of Part XI . . . 2144

11.9.3 Declared perturbations . . . 2180

11.11.1 Replication spread of the perturbation witness . . . 2237
11.11.2 Witness and certificate as dimension grows . . . 2238
11.11.3 Adversarial search against uniform sampling . . . 2240
11.11.4 Basepoint budget of NAb_n on the worked instance . . . 2252
11.11.5 The assembled admissible report for the worked instance . . . 2253
11.11.6 Cycle-reporting burden under the full link-frame quotient . . . 2256
11.11.7 The consumption schedule . . . 2257
11.11.8 Accounting cells and forbidden conversions . . . 2262
11.11.9 Operation-by-operation charge estimators . . . 2264
11.11.10 Certificate fields read, written and returned . . . 2272
11.11.11 Worked allocation ledger . . . 2274

11.12.1 Direction-of-dependence declaration . . . 2281
11.12.2 Cut ledger, Chapter 11.12 . . . 2285
11.12.3 Non-transfer and aspiration ledger . . . 2305
11.12.4 Imported-result deletion test . . . 2306
11.12.5 Deligne-kernel retrofit register . . . 2306
11.12.6 Candidate-by-candidate graded transfer table . . . 2311
11.12.7 T-graded proof audit . . . 2314
11.12.8 Declaration-family inventory . . . 2315
11.12.9 Source requests . . . 2315
11.12.10 Expected conformance statuses for Example 11.12.7 . . . 2317
11.12.11 Mandatory finite conformance suite . . . 2318
11.12.12 Fail-closed status semantics . . . 2319

11.13.1 Direction-of-dependence ledger . . . 2324
11.13.2 Deletion test of imported authority . . . 2352
11.13.3 Faltings-kernel retrofit register . . . 2353
11.13.4 Graded transfer table, Chapter 11.13 . . . 2356
11.13.5 T-graded theorem audit . . . 2363
11.13.6 Source requests . . . 2363
11.13.7 Explicit cuts and the restatements that killed them . . . 2364

11.14.1 (cut register) . . . 2371
11.14.2 (operation-type discriminator) . . . 2386
11.14.3 (certificate strength and exact conclusion) . . . 2389
11.14.4 (retrofit register) . . . 2394
11.14.5 (graded transfer table) . . . 2395
11.14.6 (item-level grade and deletion audit) . . . 2397
11.14.7 (required Appendix A16 source requests) . . . 2402

11.15.1 (cut register) . . . 2407
11.15.2 (four independent claim coordinates) . . . 2431
11.15.3 (Chapter 2.17 withdrawal stress test) . . . 2434
11.15.4 (retrofit register) . . . 2437
11.15.5 (graded transfer table) . . . 2439
11.15.6 (item-level grade audit) . . . 2440
11.15.7 (declaration-to-consumer graph) . . . 2441
11.15.8 (required Appendix A16 source requests) . . . 2442

11.16.1 Direction-of-dependence ledger . . . 2445
11.16.2 Cut ledger, Chapter 11.16 . . . 2448
11.16.3 Coherence audit statuses . . . 2453
11.16.4 Independence witnesses . . . 2464
11.16.5 Minimal conformance corpus . . . 2466
11.16.6 Imported-result deletion test . . . 2468
11.16.7 Lurie-transfer retrofit register . . . 2468
11.16.8 Graded transfer table, Chapter 11.16 . . . 2475
11.16.9 T-graded theorem audit . . . 2477
11.16.10 Declaration-family grade inventory . . . 2477
11.16.11 Source-request table . . . 2479

11.17.1 Clause-by-clause discharge of the current C1–C6 obligation . . . 2481
11.17.2 Seven-instance hypothesis matrix . . . 2498
11.17.3 Marked forward references . . . 2523
11.17.4 Appendix A16 source requests . . . 2523
11.17.5 Declaration, grade, and first consumer ledger . . . 2524

12.1.1 Evaporation ledger for the six late-programme transfers . . . 2549
12.1.E Downstream obligations, Chapter 12.1 . . . 2550

12.2.A Source results and retained consequences . . . 2557
12.2.B Tannakian hypotheses for trained feature objects . . . 2569
12.2.C Degenerations and exact floors . . . 2575
12.2.D Failure mechanisms and mandated responses . . . 2576
12.2.E Required exports to existing host chapters . . . 2577

12.3.1 Heights and finiteness hypotheses by object type . . . 2596
12.3.2 Admission tests for almost numerical claims . . . 2599
12.3.3 Source items, evaporation outcomes and retained grades . . . 2600
12.3.E Downstream obligations, Chapter 12.3 . . . 2601

12.4.E Downstream obligations, Chapter 12.4 . . . 2626

12.5.D Existing constructions repaired by condensed coefficients . . . 2636
12.5.A Source specialisations and hypotheses . . . 2645
12.5.B Representation-prism faces, all Tier S and grade D . . . 2646
12.5.C Evaporation and counterfactual ledger, Chapter 12.5 . . . 2649
12.5.E Downstream obligations, Chapter 12.5 . . . 2650

12.6.1 Evaporation verdicts, Chapter 12.6 . . . 2673

12.6.2 Counterfactual verdicts, Chapter 12.6 . . . 2673
12.6.E Downstream obligations, Chapter 12.6 . . . 2674

12.7.A Evaporation verdicts, Chapter 12.7 . . . 2677
12.7.B Degenerations of the species apparatus . . . 2695
12.7.C Failure modes and enforcing mechanisms . . . 2696
12.7.D Counterfactual verdicts, Chapter 12.7 . . . 2697
12.7.E Downstream obligations, Chapter 12.7 . . . 2698

12.8.1 Evaporation verdicts for the seven lineage axes . . . 2700
12.8.2 Compatibility of the lineage axes . . . 2704
12.8.3 Additional witnesses for conditionally compatible pairs . . . 2705
12.8.4 Deletion witnesses for the minimal lineage core . . . 2717
12.8.5 Part XII to Part XIII dependency boundary . . . 2718
12.8.6 Open interactions and decisive falsifiers . . . 2719
12.8.7 Undischarged obligation cells of the seven-axis synthesis . . . 2720
12.8.8 Counterfactual verdicts, Chapter 12.8 . . . 2721
12.8.E Downstream obligations, Chapter 12.8 . . . 2722

13.1.A Tensor, ownership and update contract . . . 2729
13.1.B Normative configuration cells and present status . . . 2747
13.1.C Required observables and operational actions . . . 2749
13.1.D Module-completeness matrix . . . 2750
13.1.E Downstream obligations, Chapter 13.1 . . . 2753

13.2.1 Scalar-regime statuses and licensed conclusions . . . 2758
13.2.2 Certificate operations and exact finite bounds . . . 2761
13.2.3 Quantisation and truncation admission hypotheses . . . 2764
13.2.4 Gluing forward/backward modes . . . 2768
13.2.5 Endpoint inputs, outputs and refusal states . . . 2772
13.2.6 Worked transaction budget ledger . . . 2775
13.2.7 Algorithmic transplants, evaporation outcomes and grades . . . 2776
13.2.8 Minimum conformance corpus for this calculus . . . 2779
13.2.E Downstream obligations, Chapter 13.2 . . . 2780

13.3.1 Evaporation tests and retained grades, Chapter 13.3 . . . 2783
13.3.2 Nisnevich router cost, communication and failure semantics . . . 2787
13.3.3 Descent engine: complexity and controls . . . 2789
13.3.4 Interval module contract . . . 2791
13.3.5 Correspondence costs, communication and failure modes . . . 2794
13.3.6 Slice accounting and OPEN convergence cells . . . 2796
13.3.7 Claim-equivalence module contract . . . 2798
13.3.8 Normative configuration fields and forbidden defaults . . . 2802
13.3.9 Integrated failure atomicity and public verdicts . . . 2804
13.3.E Downstream obligations, Chapter 13.3 . . . 2806

13.4.1 Runtime statuses used by every module . . . 2810
13.4.2 Precision-transfer branch obligations . . . 2820
13.4.3 Module cost, communication and conservative degeneration . . . 2826
13.4.4 Evaporation outcomes and retained grades . . . 2829

13.4.5 Cross-module composition gates . . . 2832
13.4.E Downstream obligations, Chapter 13.4 . . . 2832

13.5.1 Evaporation verdicts and the contracts they change . . . 2835
13.5.2 Cocartesian-router states and permitted conclusions . . . 2838
13.5.3 Obstruction-tower stopping semantics . . . 2845
13.5.4 Counterfactual verdicts, Chapter 13.5 . . . 2857
13.5.5 Module contracts, exact degenerations and primary failures . . . 2857
13.5.6 Exact conformance fixtures for the higher-descent runtime . . . 2858
13.5.E Downstream obligations, Chapter 13.5 . . . 2860

13.6.A Evaporation verdicts, Chapter 13.6 . . . 2864
13.6.B Normative runtime schemas and shape obligations . . . 2866
13.6.C Coverage of the species engine . . . 2886
13.6.D Counterfactual verdicts, Chapter 13.6 . . . 2887
13.6.E Downstream obligations, Chapter 13.6 . . . 2888

13.7.1 Ownership map of the unified stack . . . 2892
13.7.2 Unified reference cost, memory and communication ledger . . . 2905
13.7.3 Per-layer and whole-programme degeneration hypotheses . . . 2906
13.7.4 Composite interaction failures and preventing mechanisms . . . 2908
13.7.5 Contrary verdicts returned by the unified stack . . . 2909
13.7.6 Compliance matrix for the unified lineage stack . . . 2911
13.7.7 Explicit OPEN obligation ledger . . . 2913
13.7.E Downstream obligations, Chapter 13.7 . . . 2914

13.8.1 Evaporation verdicts, Chapter 13.8 . . . 2918
13.8.2 Solve conformance trace status on the reference fixture . . . 2934
13.8.3 Conformance checks, execution status and observed result . . . 2944
13.8.4 Open codes addressed by Chapter 13.8 . . . 2947
13.8.5 Counterfactual verdicts, Chapter 13.8 . . . 2948
13.8.E Downstream obligations, Chapter 13.8 . . . 2948

13.9.1 Evaporation test for the validation apparatus . . . 2951
13.9.2 Grade and status discipline . . . 2952
13.9.3 Falsification register for load-bearing Part XII claims . . . 2953
13.9.4 Falsification register for load-bearing Part XIII claims . . . 2957
13.9.5 Normative lineage instrument catalogue . . . 2961
13.9.6 Specialised degeneration predicates for MA–MG . . . 2965
13.9.7 Pre-registered retirement rules for every Part XIII M-unit . . . 2967
13.9.8 Instruments and settling observations for MG O1–O14 . . . 2968
13.9.9 Gates specified here but not executed . . . 2970
13.9.10 Pre-committed adverse verdicts . . . 2972
13.9.E Downstream obligations, Chapter 13.9 . . . 2973

13.10.1 The frozen shape schema . . . 2979
13.10.2 Base-change status of every operator of the normative map . . . 2987
13.10.4 Rigour grades, Chapter 13.10 . . . 2999
13.10.5 UNDETERMINED register for this chapter . . . 3000

13.11.1 The slot ledger . . . 3024
13.11.2 The complete UNDETERMINED list, with obstructions . . . 3026

13.12.1 Disposition of D1–D14 before the overhead calculation . . . 3032
13.12.2 Coordinatewise overhead of the degeneration . . . 3045
13.12.3 Frozen inventory of the compiled embedding . . . 3048

13.13.1 Leaf letters of the worked instance . . . 3065
13.13.2 Upsweep aggregates and prefixes . . . 3066
13.13.3 Hypotheses of the reference side . . . 3075
13.13.4 Hypotheses of the hardness side . . . 3075
13.13.5 Hypotheses of the architectural construction . . . 3075

13.14.1 Graded transfer table, Chapter 13.14 . . . 3082
13.14.2 Control cells and predictions . . . 3096
13.14.3 Level-d dimensions and monotone measures . . . 3101
13.14.4 Alternative explanations and their controls . . . 3102
13.14.5 Grade of every numbered statement in this chapter . . . 3103
13.14.6 Bibliography records required for promotion . . . 3104
13.14.7 Claims withdrawn and their replacements, Chapter 13.14 . . . 3105

13.15.2 Proof-term inventory for the PD encoding . . . 3120
13.15.3 The fixture register . . . 3122
13.15.4 Failure modes of the checking kernel . . . 3126
13.15.6 The Q12−LEAN conjunct inventory . . . 3129
13.15.7 Cut ledger, Chapter 13.15 . . . 3133
13.15.8 Graded transfer table, Chapter 13.15 . . . 3134
13.15.10 Remediation register . . . 3135

13.16.1 Dispatch vocabulary and registered grades . . . 3139
13.16.2 Grades and machine checks, Chapter 13.10 . . . 3143
13.16.3 Grades, Chapter 13.11 . . . 3147
13.16.4 Grades and machine checks, Chapter 13.12 . . . 3153
13.16.5 Grades and machine checks, Chapter 13.13 . . . 3158
13.16.6 Grades and machine checks, Chapter 13.14 . . . 3164
13.16.7 Grades and artefact status, Chapter 13.15 . . . 3167
13.16.8 Machine-check census . . . 3167
13.16.9 Verification units and verdicts . . . 3168
13.16.13 The unproved register for Part P . . . 3174
13.16.14 Graded transfer table, Chapter 13.16 . . . 3182
13.16.17 Claims withdrawn and their replacements, Chapter 13.16 . . . 3185

14.1.1 The correspondence ledger, part one . . . 3192
14.1.2 The correspondence ledger, part two . . . 3194
14.1.3 Gates on the replacement research target . . . 3210
14.1.4 Conditions of the replacement conjunction . . . 3213
14.1.5 Conditional conclusions and their distinct antecedents . . . 3215
14.1.6 Present conditional verdicts . . . 3217

A0.1 Naming conventions by mathematical shape . . . 3222

A0.2 Category-theoretic and topos-theoretic notation . . . 3224
A0.3 Sheaf and cohomological notation . . . 3226
A0.4 Layer symbols L0–L8 . . . 3228
A0.5 Runtime tensor and record notation . . . 3228
A0.6 MoE notation . . . 3230
A0.7 RL and DGSPO notation . . . 3231
A0.8 Information-theoretic and statistical notation . . . 3233
A0.9 Linear-algebraic and numerical notation . . . 3234
A0.10 Specification-level notation . . . 3236
A3.1 The context lattice of the worked instance . . . 3318
A3.2 The nerve of the covering family . . . 3323
A3.3 Spectral summary of $L_{\mathcal{F}}$ . . . 3331
A3.4 Conjugate-gradient trace, $\tau = 1$ . . . 3335
A3.5 Operation counts for the guarded solve . . . 3336
A3.6 Compositionality defect against temperature . . . 3339
A3.7 Greedy trace for relative max coverage . . . 3342
A3.8 Naturality residual . . . 3343
A3.9 Off-manifold rank completion . . . 3344
A3.10 Gödel arithmetic of the worked example . . . 3346
A3.11 The estimator family on the worked instance . . . 3348
A3.12 Exact controls added after independent review . . . 3359
A3.13 Master ledger . . . 3360
A3.14 Redundancy audit . . . 3363
A6.1 Invariance and transformation budgets . . . 3500
A6.2 The signed preregistration objects . . . 3503
A6.5 Required modern baseline arms . . . 3511
A6.6 Staged pass and stop conditions . . . 3519
A6.7 Ownership map . . . 3528
A8.1 Proof-to-main-text traceability map . . . 3623
A8.2 Downstream obligations, Appendix A8 . . . 3626

A11.1 Global shapes and meanings 3698

A11.2 Executed grades 3732

A12.1 Endpoints, estimators and refuting observations 3746

A12.2 The single confirmatory family 3753
A12.3 Exact Holm critical values 3754

A12.4 Required primary-results table shape 3761

A13.1 Direct `sorry` register, complete for the displayed source 3786
A13.2 Explicit decision and generated-decision dependencies 3787

A13.3 All explicit local declarations 3789

A15.1 The six primary registers consolidated by this chapter 3843

A15.2 Restatement-test census 3846

A15.3 The six evaporated postulates 3850

A15.5 Graded transfer table of the seven lineages 3853
A15.6 Consumer-ratchet state 3854

A15.8 The eight identification cuts of Chapter 11.12 3861
A15.9 The three demotion verdicts 3864

A15.10 The fourteen conditionally compatible pairs 3871

A15.11 Remediation register 3875

A15.12 Undetermined register and source-request table 3877

A15.13 Total negative results recorded by this chapter, by kind 3883

The claim vocabulary

Two vocabularies run through the book and they are not interchangeable. The first grades a claim that has been made. The second marks an obligation that has not been discharged. A reader who tracks both can determine the standing of any statement without trusting the surrounding prose.

Table FM.1 — Claim grades and what each may be used for.

Grade	Meaning	May it discharge a proof obligation?
T	Theorem — proved here, or a correctly cited standard result with its hypotheses stated and checked	Yes
M	Measurable — a well-posed quantity with a definition and an algorithm, whose *interpretation* may still be conjectural	Only about itself
D	Design criterion — a motivated construction guided by an analogy, not a transplanted theorem	No
X	Metaphor — illustrative only	Never

The distinction between **T** and **D** is the one that matters, and it is the distinction most often blurred. A theorem about arithmetic geometry does not become a theorem about neural networks by being restated with neural vocabulary. Where a deep result supplied inspiration — the section conjecture, Tannakian reconstruction, inter-universal Teichmüller theory, motivic cohomology — we say so and grade the consequence **D** or **X**. A **D** or **X** step may never appear as a premise in a **T** argument; enforcing that rule mechanically catches errors that prose does not.

Table FM.2 — Status tokens for obligations that are not discharged.

Token	Meaning	Consequence for the reader
OPEN	A required cell of a module's contract is empty: no forward type, backward map, cost, invariance declaration, conformance test or gate	The module is not specified; nothing downstream may assume it
UNTRAINABLE	A forward pass is defined but no usable gradient contract exists at the stated cost	The component cannot be trained as written; any claim resting on learning it is void
UNDETERMINED	A well-posed question whose answer the specified procedure cannot return	The instrument abstains; abstention is a legitimate output and is never silently replaced by zero
WITHDRAWN	A claim once made and now retracted, with the refuting argument recorded	The claim may not be cited, plotted, gated on, or used as a premise

The single claim authority is the typed claim registry of Chapter 8.3: where any overview, appendix, figure or API entry appears to grade a proposition differently, the registry governs and the discrepancy is a defect. The single executable authority is the named normative

definition of each algorithm and type; every other occurrence cites it and may not restate a stronger version.

Where the generative principles are used

Part XI is not an appendix to the architecture. It supplies six principles that Parts II through X are written to embody, and the test of that claim is that deleting Part XI would break definitions, theorems and algorithms elsewhere; it would not simply shorten the book. The table below is the reader's dependency map: it names the principle, the chapter that supplies it, and the downstream object that changes when it changes. It is given here, before Part II, because a reader who meets these principles only at the end will read a substantial intervening body of material without knowing what generated them.

Table FM.3 — The six generative principles, their supplying chapters, and what depends on them.

Principle	Supplied by	Named downstream dependency
M1 Relative point of view: every construction is stated over a base, base change is a first-class operation	Chapter 11.1	Layer signatures and their cartesian/non-cartesian declaration in Chapter 3.1; terminal-site specialisation as a fibre in Chapter 3.2; the relocation definition that unifies fine-tuning, transfer and versioning in Chapter 5.3
M2 Frobenius-like versus étale-like: value-carrying and structure-carrying data are never interchangeable	Chapter 11.3	The typing of every stalk-valued and site-valued object in Parts II–III; reward versus gate typing in Chapter 6.3; the section-object field annotations in Chapter 7.1 and the API kinds in Chapter 8.1
M3 Inter-universality: experts share no ambient universe, so every comparison names its link	Chapter 11.3	The mandatory link input to `combine` in Chapter 4.1; the minimal-cover formulation in Chapter 4.2; the gerbe obstruction restated as failure of links to compose in Chapter 4.3
M4 Indeterminacy calculus: no quantity is reported without proved invariance or a stated budget	Chapters 11.2, 11.9, 11.11	The observable ledger of Chapter 3.1; the transformation registry of Chapter 9.2; the admissible-report definition of Chapter 9.1; the open-budget rows of Chapter 10.1
M5 Species and mutations: the specification is invariant under a declared class of implementation change	Chapter 11.2	The permitted-mutation class in Chapter 7.1; bit-comparability restated as invariance in Chapter 8.1; the conformance relation of Chapter 8.3
M6 Dévissage: structure is stripped one layer at a time and each degeneration names what was stripped	Chapter 11.1	The ordered stripping ledger of Chapter 3.1 and the final accounting in Part XIV

Two disciplines govern that material and are stated once. Every Grothendieck or IUT item is placed in exactly one tier: **Tier G**, transferred mathematics whose hypotheses are verified here and which is used as a theorem; **Tier S**, a pattern of reasoning imported as a design principle with no claim that any theorem transfers; **Tier A**, analogy, which may motivate and may never justify. And every such construction must survive restatement in ordinary mathematical language with the imported vocabulary removed. If it evaporates it is cut rather than softened.

FRONT MATTER · MASTER MAP OF THE MATHEMATICAL PROGRAMME

Seven lineages, seven typed repairs: what is transferred, and where the composition law stops

Each row transfers a *solution technique*, never a source theorem; the last column gives that technique's status in the composition law, evaluated only in the subcategory named there.

a Source programme → construction transferred into this book's objects → the host contract that fails without it

lineage · axis	what the source solved	construction transferred here	deleting it makes this ill-typed	composition-law scope
G **Grothendieck** Ch. 98, LA	choose the category until the theorem is trivial	versioned finite site with a tame-cell certificate	Definition 53.5 site build; Definition 70.1 compiled site	**VERIFIED** bounded probe links
W **Deligne** Ch. LB, MA	weights bound cohomology; strictness needs proof	a finite filtration on each coefficient object, with a strictness verdict	Chapter 61 advantage objects; Chapter 35 holonomy report	**VERIFIED** filtration loss only
A **Faltings** Ch. LC, MB	objects isomorphic up to an ideal of scalars	one idempotent ideal, a typed defect mask, a discrete height	Chapter 71 precision certificate; Chapter 41 bounded search	**VERIFIED** every check discharged
V **Voevodsky** Ch. LD, MC	invert a deformation; identity becomes equivalence	residue-preserving coverage, an interval, finite transfers, slices	Chapter 31 coverage; Definition 82.1 claim record	**VERIFIED** constant-degree transfers
C **Scholze** Ch. LE, MD	move to an easy setting and carry the answer back	condensed and solid coefficients, a pro-context tower	Chapter 20 state; Chapter 70 packed sections; Chapter 73 serving	**OPEN** no inverse-choice calculus
H **Lurie** Ch. LF, ME	gluing data that agree only up to homotopy	cocartesian routing fibration, gluing object, obstruction cap	Chapter 34 obstruction record; Chapter 41 refinement routing	**OPEN** mate pasting is not equipment
S **Mochizuki** Ch. 9C, LG	compare two copies that share no ring of scalars	complete species, typed mutation spans, declared links	Definition 63.21 self-evolution; Chapter 91 metric comparison	**OPEN** indeterminacies not yet one type

b How the seven depend on one another

- **G** → the finite versioned base under every other axis
- **C** → A: an ideal of the *same* condensed ring
- **A + C** → W: condensed almost modules are abelian, so a filtration can live there
- **V + H** → descent: hyperdescent for truncated decision sheaves
- **W · V · H** → independent gradings, but only given commutation witnesses
- **S** → every cross-execution comparison: no declared link, no comparison

c What the evaporation test cut, and the two excluded pairs

Removed because nothing well-posed survived restatement: the universal central extension postulate; Tannakian or anabelian reconstruction of a trained model; a global perfectoid tilt; a motive of a neural model; unrestricted theatre identification; prismatic comparison.

Two pairs are excluded, not reconciled. Finite tame compilation against unrestricted infinite-context topology; global record identification against nominal species separation. Fourteen of the twenty-one pairs are conditional on a named finite witness.

Source mathematics enters as T; the decision to adopt it here is D.

transferred construction, host object · scope verified in the named subcategory · open: technique adopted, calculus not yet constructed · exclusion guard: the pair may not be active together

Figure FM.1. Master map of the mathematical programme.

Inter-universal Teichmüller theory is treated at length in Chapters 11.5 through 11.8 and is quarantined. Mochizuki's papers appeared in *PRIMS* in 2021; Scholze and Stix published a specific objection to the proof of Corollary 3.12; the author disputes it; there is no consensus in the number-theory community. Nothing in this book depends on whether that theory is correct, and the dependency quarantine is checked rather than promised. What is taken from it is structural: reasoning across set-ups with no common ambient universe, the value/structure split, implementation-invariant specification, and the correspondence between three named indeterminacies and the gauge, basepoint and cover-refinement ambiguities that this book carries independently and proves things about.

Five paths through the book

The section above gives three coarse routes and the volume structure. This section is the fuller guide: the book is long, it is not intended to be read linearly, and five paths through the technical parts are useful. Each one names what it is for.

None of the five paths respects the volume boundaries, and none is meant to. The volumes order the argument; the paths order a reading, and a reading that follows an interest will cross volumes freely. The sceptic's path is almost entirely within Volume III and Volume V; the mathematician's begins in Volume IV and returns to Volume I; the engineer's opens with a chapter of Volume I, lives in Volume II and finishes in Volume III; the generative path is Volume IV read first and the rest read as its consequences. A reader who wants the volumes in order should simply read forward, which is the sixth path and the one the division was built for.

The sceptic's path. Begin with the rigour ledger and the twelve disclaimers in Part X, then the corrections ledger in the same part, then Part IX on falsification, then Part XIV, which states the replacement assessment and reaches the sentence displayed in the abstract. Those chapters state plainly what is proved, what is merely measurable, what is design and what is metaphor, and they list every overclaim that the book retracts. If the programme is going to fail your inspection it will fail there, quickly.

The mathematician's path. Part II develops sites, sheaves, descent, cohomology, topoi, internal logic and the structure theory on which the sharper results depend — the Tits alternative, uniform exponential growth, and the holonomy dichotomy. Part II is not self-contained: its relative formulations, its value/structure typing and its invariance declarations come from Chapters 11.1, 11.2 and 11.3, and a reader who wants the generating principles before their consequences should read those three first. Then Part III for the architecture that instantiates the mathematics, and Appendix A1 for the collected proofs. The result we would most like checked is the holonomy dichotomy: that the non-abelian capacity β of an orthogonal cellular sheaf lies in $\{0\} \cup [c(d), \infty)$, so that β is a binary certificate rather than a continuous score. Its *certification* is a separate matter and is currently **UNDETERMINED** for the deployed procedure.

The engineer's path. Part III specifies the architecture with its open contracts, Part IV the mixture-of-experts construction, Part VII the execution model, kernels, distributed training and serving, and Part VIII the API and the executable reference core. Read Chapter 1.3 first: it carries the module compliance matrix, which states for every module which contract cells are discharged and which are **OPEN** or **UNTRAINABLE**, and reading a module without it invites the assumption that an unstated backward map exists. That matrix is a rendering of registry state and is at present transcribed by hand rather than generated, because the claim registry has not yet exported stable identifiers; it therefore records compliance without yet enforcing it, and the reader should treat its discharged cells as claims to be checked against the named authority rather than as certificates. Chapter 3.8 declares the reference configuration, formulae and closed arithmetic cells; complete totals, ratios, atomisation and regenerated-view verification remain **OPEN**. Chapter 8.3 fixes the typed contracts and is the

single claim authority; both precede any implementation decision. The single cheapest thing to implement is the compositionality defect of Part I, which requires no retraining.

The researcher's path. Part I for the diagnosis and the method, Part V for the learning theory — including the collapse theorem, which is the most practically important result here — Part VI for reinforcement learning and self-evolution, and Part IX for the experimental programme. Chapter 10.2 situates the work against state-space models, diffusion language models, test-time training, group-equivariant position encodings, categorical deep learning and modern mixture-of-experts, and says for each whether it is orthogonal, composable, competing, or the correct baseline.

The generative path. For a reader who wants the reasoning that produced the architecture rather than the architecture itself: Chapter 1.3 for the transplantation discipline and the dependency map, then Chapters 11.1, 11.2 and 11.3 for the relative viewpoint, the functor of points, species and the value/structure split, then Chapters 11.5 through 11.8 for what inter-universal Teichmüller theory is and is not, then Chapters 11.9 and 11.11 for the indeterminacy calculus that binds every reported number in the book. Parts II through X can then be read as consequences, and Parts XII and XIII as the lineage-by-lineage execution of the same discipline.

Table FM.4 — Part synopsis, the volume each part belongs to, and the order in which parts depend on one another.

Part	Volume	Content	Depends on
I	I	Diagnosis, thesis, claim ledger, transplantation method and the module compliance matrix	—
II	I	Sites, sheaves, cohomology, descent, Kan extensions and coends, holonomy, the six operations	XI for its relative and typed formulations
III	II	The nine layers L0–L8, the Alien Engine and T-Unit, and the one reference configuration	II, XI
IV	II	Topos-MoE: the expert as an open sub-topos, coverage routing, gluing, the gerbe, systems	II, III, XI
V	III	Objective design, the collapse theorem, pretraining, continual learning, site construction	II, III
VI	III	Policy optimisation, the Čech–Hodge identity for advantage, DGSPO, self-evolution, limits	II, III, V
VII	III	Execution model, kernels, distributed training, inference and serving	III, IV
VIII	III	API specification, executable reference core, typed claim record and contracts	III, IV, VII
IX	III	Validation instruments, metrics and the experimental protocol	III–VIII
X	III	The rigour ledger, the twelve disclaimers, positioning, risks and roadmap	all
XI	IV	The Grothendieck programme, the functor of points, motives, IUT, the indeterminacy calculus	II (it adjudicates for II–X)
XII	V	The seven lineages: transferable cores, their limits and the synthesis	II, XI
XIII	V	Lineage algorithms, the one forward map, exact degeneration, overhead, separation, machine-checking contracts with open cells, and the replacement ledger	II–VIII, XI, XII
XIV	V	The replacement assessment and the conclusion	all

Part	Volume	Content	Depends on
A0–A16	—	*Apparatus*: notation, proofs, glossary, references, worked example and code, algorithm compendium, protocol, long derivations, attribution register, expanded proofs, negative results	all

Conventions

Chapter, theorem, algorithm, table and figure identifiers are produced by one numbering scheme, and every cross-reference is a link. Notation is collected in Appendix A0 and terms in Appendix A2. Figures are vector artwork, numbered within their chapter, each with a short noun-phrase title, printed beneath the image and in the List of Figures. Tables carry a bold noun-phrase caption above them. Citations are numbered and collected in the bibliography of Appendix A16.

British English is used throughout. Every non-trivial claim carries its grade in the text, and no overview, appendix, figure or API entry may state a claim more strongly than the chapter that establishes it.

What this book is not

It is not an empirical paper. It contains no experimental results, because we have run no experiments. Numbers are definitions, complexity bounds, worked arithmetic examples, clearly labelled illustrative computations, or partial generated counts tied to Chapter 3.8's closed cells; publication-level regenerated-view verification is **OPEN**. Where a plot would ordinarily show data it either shows a schematic with the axes labelled and the word *schematic* in the caption, or it does not appear. We consider the temptation to garnish a theoretical work with plausible-looking curves to be the most corrosive practice in the field, and we have not indulged it.

It is not a claim of priority over the many strong lines of work it builds on. Cellular sheaf theory, neural sheaf diffusion, categorical deep learning and group-equivariant position encoding all exist, are cited, and in several cases have already tried, and reported on, ideas close to ours. The most important such report is negative — that sheaf neural networks frequently learn the identity sheaf and thereby collapse to ordinary graph networks — and a substantial part of Part V exists to explain *why* that happens and what to do about it, rather than to hope it will not happen here.

It is not a description of a system that has been built. There is no trained model, no measured latency, no throughput result and no quality comparison anywhere in these pages. The executable material is an algebraic reference core and a conformance harness, not an end-to-end implementation, and Chapter 8.2 says precisely what it does not implement.

It is not, finally, a promise that the programme works. It specifies discharged cells while marking **OPEN** cells and contracts, so presently testable failures can be stated without presenting the whole system as buildable.

%%BODYSTART%%

Part I — Diagnosis and Method

Chapter 1.1 — The five structural deficits

Large language models are, by any reasonable historical standard, astonishing. They have absorbed a substantial fraction of written human knowledge, they manipulate natural language with a fluency that would have been considered science fiction a decade ago, and they perform competently on tasks their designers never anticipated. Any diagnosis that begins by understating this is not worth reading, and we do not intend to offer one.

Nevertheless there is a persistent, characteristic pattern of failure. Models assert things that are false with exactly the confidence they use to assert things that are true. They handle a two-step composition of two facts they know individually and fail on the four-step composition. They can be fine-tuned on a narrow domain and emerge fluent in that domain and subtly damaged elsewhere. They cannot say, in any principled way, *I do not have enough information to answer this*; when they abstain it is because abstention has been trained in as a behaviour, not because a computation returned an obstruction. And they have no internal notion of the difference between a claim that is supported by the context and a claim that merely fits the context's shape.

The standard explanations for this pattern are scale, data, and search. Each is partly right and none is structural. This chapter sets out a structural reading, states exactly what that reading does and does not establish, and lists the conditions under which it would be refuted.

1.1.1 The three standard explanations, and where each stops

Scale. The empirical scaling relations are among the most robust findings in the field, and they describe how a loss falls as parameters, data and compute rise together. What they describe is the quality of interpolation under the training distribution. They say nothing about whether a computation is *obliged* to be consistent with itself, because consistency is not what the loss measures: a model that contradicts itself three paragraphs apart is penalised only to the extent that the training distribution contains the contradiction's continuation less often than the consistent one. Scaling therefore improves the frequency of consistency without ever creating a quantity whose non-vanishing would mark inconsistency. The distinction is between a behaviour that is rewarded and a constraint that is enforced, and no amount of the former becomes the latter.

Data. Curation, deduplication, synthetic augmentation and instruction tuning all demonstrably change what a model does. They act on the distribution, which is the correct place to act for failures whose cause is distributional. Two of the five failures below are not obviously of that kind. The asymmetric erosion of constraints under fine-tuning persists when the fine-tuning data is unrelated to the constraints, which is difficult to attribute to what that data contains; and the absence of a principled abstention is a property of the output type — a distribution over tokens has no coordinate in which "the context does not determine this" can be expressed — rather than of the data the distribution was fitted to.

Search. Chain-of-thought prompting, self-consistency sampling, verifier-guided search, process reward models and test-time compute are all effective, and this book treats them as the correct baselines rather than as rivals; Chapter 10.2 says for each whether it is orthogonal,

composable or competing. The limit is the same in every case: search improves the *selection* among candidate answers using a scoring function, and the scoring function is either another learned model with the same deficits or an external verifier that exists only for domains where verification is cheap. Where an external verifier exists, search is powerful and this programme has little to add. Where it does not, search relocates the problem into the scorer.

None of the three is wrong. What they have in common is that they operate on the distribution or on the search over outputs, and leave untouched the question of what the computation is structurally obliged to satisfy. That question is what the rest of this chapter is about.

1.1.2 What this chapter claims, and at what grade

The chapter's conclusions are, in order of strength: a small number of theorems about what vanishes on a trivial index structure; a small number of well-posed measurable quantities; and a conjecture that the five failure modes below share one origin. The conjecture is the interesting part and it is the weakest part, and the two must not be confused. No intervention experiment has been run. Re-describing an architecture in the vocabulary of sites does not establish that five empirically distinct failures are caused by the feature that the description makes salient, and nothing in this chapter should be read as if it did.

Table 1.1.1 — The five deficits, the surviving claim in each case, and its grade and status.

Deficit	Surviving claim	Grade	Status
Locality without gluing	On a discrete index category with the trivial topology every presheaf is a sheaf, so the sheaf condition constrains nothing	**T**	The identification of a transformer's context with that structure is a diagnostic convention, grade **D**
Composition without compositionality	$\mathrm{CD}(A) = \lVert A^2 - A \rVert_F / \lVert A \rVert_F$ is well defined for a row-stochastic A and computable from any checkpoint	**M**	Its reading as a compositionality measure is **D**; its invariance class must be declared before it is reported
Confidence without justification	A non-vanishing obstruction class certifies that no global section exists over the given cover	**T**	The converse fails; the deployed instrument returns **UNDETERMINED** without independently supplied overlap data
Adaptation without preservation	The inverse image of a geometric morphism preserves exactly $\{\top, \wedge, =, \exists, \bigvee\}$; $\Rightarrow, \neg, \forall$ require openness	**T**	That fine-tuning induces such a morphism is **D** and currently unsupported; the transfer instrument returns `RELOCATION_UNAVAILABLE` when uncertified
Capacity without structure	For an orthogonal cellular sheaf, $\beta \in \{0\} \cup [c(d), \infty)$, so β is a binary certificate rather than a score	**T**	Certification of which side of the gap a given model lies on is **UNDETERMINED** for the deployed procedure
The five share one origin	—	**D** (conjecture)	No intervention evidence exists; the falsification condition is stated at the end of this chapter

MOTIVATION · GRADE D

Five deficits as falsifiable structural hypotheses

The rows are proposed interventions, not consequences proved from a "trivial site". Cohomology supplies only a narrow one-way non-gluability statement.

OBSERVED DEFICIT	D-GRADED STRUCTURAL HYPOTHESIS	ADMISSIBLE CLAIM AND TEST
Hallucination	Explicit overlap evidence and descent checks may expose some incompatible local claims.	A supplied non-exact *cocycle* can certify that those data do not glue. Hallucination $\not\Rightarrow H^1 \neq 0$, and $\delta^0 x$ never detects it.
Over-confidence	Evidence-indexed truth values may separate support scope from scalar confidence.	Compare calibrated confidence with a declared support object; no causal advantage is yet established.
Long-context decay	Typed locality and sparse incidence may reduce indiscriminate mixing.	Measure quality against matched attention baselines; oversmoothing results require the stated sheaf and operator.
Forgetting	Separating geometric from non-geometric probes may predict asymmetric retention.	Run matched probe experiments; no adapter is assumed to realise a site morphism.
Multi-agent coherence	Higher-overlap checks may reveal failures missed by pairwise agreement.	A non-zero checked class concerns the supplied coherence datum, not truth of every agent answer.

COMMON-CAUSE HYPOTHESIS One intervention affecting several rows would support the programme; it would not prove that discrete indexing caused all five deficits.

On a trivial site — one object, only its identity, the trivial topology — descent and every higher-obstruction diagnostic are vacuous, so no row is a consequence of that setting. The checkpoint statistic $\|A^2 - A\|_F / \|A\|_F$ is computable but is not free evidence for compositionality or for a coend, and it needs an invariance declaration or an explicit budget.

Figure 1.1.1. Five structural design hypotheses.

1.1.3 Locality without gluing

Consider what a transformer actually does with a context of n tokens. It embeds each token, and then, at every layer, it forms for each position i a convex combination of value vectors,

$$h_i' = \sum_{j=1}^{n} \alpha_{ij} v_j, \qquad \alpha_{ij} \geq 0, \quad \sum_j \alpha_{ij} = 1.$$

This is a powerful primitive and we have no wish to disparage it. But observe what structure it presupposes. The index set $\{1, \dots, n\}$ carries no relations. Token j is not *part of* token i; token j does not *overlap* token i; there is no notion of the information at j being a *restriction* of information held on a larger region containing j. Each position is an isolated point, and attention is a rule for mixing values across a set of isolated points.

In sheaf-theoretic language: the context is being treated as a **discrete category**, and the topology under which the computation is read is the **trivial** one, in which the only covering sieve of an object is the maximal sieve. Under the trivial topology every presheaf is a sheaf. The sheaf condition — that compatible local data glue uniquely to global data — is satisfied *vacuously*, because there is no non-trivial notion of "local" for it to constrain.

The consequence is not that the model cannot represent global structure. It plainly can; a sufficiently large transformer can represent essentially anything on a bounded context. The consequence is that the model is under **no structural obligation** to be globally consistent. Consistency is available to it, and it will learn as much consistency as the training distribution

rewards, but nothing in the computation forces it. When the model produces two statements that contradict each other three paragraphs apart, no internal quantity has gone wrong. There is no residual to be non-zero.

A precise statement, and the caveat that travels with it. "The transformer's context is a discrete site with the trivial topology" is a claim about the *presentation* we are choosing, not a theorem that no non-trivial structure can ever be present. Positional encodings, causal masks, and learned attention patterns all impose relations among positions. What they are not known to do is impose relations satisfying the coverage axioms — stability under base change and locality — and it is those axioms, not the mere presence of relations, that make descent meaningful. A topos moreover has many inequivalent sites, so no argument of this kind can *classify* a computation. We therefore grade the identification **D**, and Part X records under its first disclaimer that the Transformer is not mathematically classified as topos computation.

Two further caveats belong here rather than four hundred pages later. First, the alternative structure this book proposes is *imposed*, not discovered: the evidence-atom locale of Chapter 3.2 is a construction, and every invariant computed on it inherits whatever arbitrariness that construction contains. Second, a natural candidate for the alternative fails and the failure is instructive. If contexts are taken to be linear subspaces of a representation space, the proposed join coverage is not stable under base change, because the subspace lattice is not distributive; the counterexample is given in Part II. That result does not show that no Grothendieck topology exists on a category of subspaces — every small category admits some — and Chapter 3.2 states the corrected claim. Contexts must instead be **down-sets of evidence atoms**, for which the three coverage axioms are verified in place.

Consider how little room the discrete case leaves. On a discrete category the only sieves on an object X are the empty sieve and the maximal sieve $\{\mathrm{id}_X\}$, so a Grothendieck topology amounts to a choice, independently for each object, of whether the empty sieve covers: there are exactly $2^{|\mathrm{Ob}|}$ of them. If the empty sieve covers X, the sheaf condition forces $\mathcal{F}(X)$ to be terminal, so that object carries no information at all. The trivial topology is therefore the only *non-degenerate* choice available, and the vacuity of the sheaf condition on a discrete index category does not come from a bad choice of topology: it is the whole of what a discrete index category can support. Part II states the classification rather than only its consequence. The escape is not a better topology on the same category — it is a different category, one whose objects contain one another.

The replacement object is named here, in outline, so that the diagnosis is not purely negative. An **evidence atom** is a minimal unit of committed evidence: a retrieved span, a tool result, a premise supplied by the user, a previously asserted and certified proposition. Atoms are partially ordered by entailment, and a **context** is a down-set of that order — the set of atoms available once one has committed to the atoms it contains. Down-sets of a partial order form a distributive lattice, indeed a frame, and covering is join: a family $\{U_i\}$ covers U when $\bigvee_i U_i = U$. Distributivity is what makes the covering families stable under base change, and stability under base change is what the subspace lattice fails. The three coverage axioms are then verified in place rather than assumed, and Chapter 2.1 does that once; Chapter 3.2 gives the construction that turns a query and its retrieved pool into such an order, together with the conditions under which that construction abstains. The point for this chapter is only that the object exists, that it is not the token index, and that it is built rather than found.

This is the first deficit. It is the one from which the language of the other four is borrowed, which is not the same as its being their cause.

1.1.4 Composition without compositionality

The second deficit is the one that can be measured today, on a model that already exists, with no retraining.

If information lived on overlapping regions with restriction maps between them, then composing two restrictions would be a restriction: $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ whenever $W \subseteq V \subseteq U$. Functoriality of restriction is what makes local-to-global reasoning possible at all. It is what allows one to assemble a conclusion from a chain of intermediate steps and know that the chain means something.

Attention has no such law. Applying the attention operator twice does not correspond to attending over a coarser structure; it mixes the mixture. Concretely, if A is the row-stochastic attention matrix, there is no reason for A^2 to bear any particular relation to A, whereas a mechanism with genuine restriction structure and idempotent aggregation over a fixed cover would satisfy $A^2 = A$ exactly. This suggests an essentially free diagnostic. Define the **compositionality defect** of an attention head as

$$\mathrm{CD}(A) \;=\; \frac{\|A^2 - A\|_F}{\|A\|_F}.$$

Rigour grade: M (measurable). $\mathrm{CD}(A)$ is well defined for any row-stochastic matrix and computable in $O(n^3)$ time, or $O(n^2 r)$ for a rank-r factored attention, from any existing checkpoint with no retraining. Its *interpretation* as a compositionality measure is grade **D**: we conjecture, and Part IX pre-registers the experiment, that heads with low CD carry compositional structure and heads with high CD do not. That conjecture may be false. The quantity is nonetheless defined and measurable, and to our knowledge it has not been reported.

Three readings of this quantity must be refused, and each is withdrawn. It is not a distance from the attention operator to the nearest idempotent: it is an idempotence *residual* evaluated at A itself, and the two agree only under hypotheses that are not available here. It is not evidence for or against any coend identification; the corrected coend statement of Part II concerns the unweighted codiagonal quotient of the indiscrete identity diagram, and a generic weighted attention row does not factor through that quotient, so no numerical value of CD bears on it. And it is not a gauge-free number: attention matrices are defined relative to a tokenisation, a positional convention and a head ordering, so CD must be reported with an invariance declaration or an explicit budget, exactly as every other reported quantity in this book must be. Chapter 9.1 fixes the protocol, the controls and the transformation class; Chapter 9.2 records the invariance class in the metric registry.

A fourth caution follows from arithmetic alone, and it determines what the experiment must control for. Small exact instances show that CD vanishes at both ends of the informativeness scale. For the 2×2 identity, $A^2 = A$ and $\mathrm{CD} = 0$. For the uniform row-stochastic matrix with every entry $\frac{1}{2}$, again $A^2 = A$ and $\mathrm{CD} = 0$; the same holds for the uniform $n \times n$ matrix for every n. So the head that attends only to itself and the head that attends to everything equally are both exactly idempotent, and both are exactly uninformative. A defect of zero is therefore not by itself evidence of compositional structure. At the other extreme, a permutation matrix realising a 2-cycle gives $\|A^2 - A\|_F = 2$ and $\|A\|_F = \sqrt{2}$, hence $\mathrm{CD} = \sqrt{2}$, and the 3-cycle gives the same value $\sqrt{2}$; a strict routing of information is maximally non-idempotent. A causal-uniform head on three positions, whose row i is uniform over $j \le i$, gives $\mathrm{CD} = \sqrt{5}/6 \approx 0.373$. These are exact evaluations of the definition on stated matrices, not measurements.

The consequence is that the pre-registered experiment of Part IX must contrast CD against informativeness controls — attention entropy, effective support size, and the two degenerate baselines above — and must report the joint distribution rather than the marginal. A finding

that low-CD heads are compositional is interesting only if it survives the observation that maximally diffuse heads are also low-CD. This is stated here because it is the kind of confound that is easy to discover after the plots have been made.

We flag the quantity early because it is the cheapest falsification gate in the book. If one computes CD across the heads of a strong open-weights model and finds no structure — no relationship to head function, no change across training, no correlation with compositional task performance — then a specific and load-bearing intuition behind this programme is wrong, at a cost of a few GPU-minutes.

1.1.5 Confidence without justification

The third deficit concerns what a language model's output distribution actually means.

A softmax over the vocabulary produces numbers in $[0, 1]$ summing to one, and it is natural to read them as probabilities; under a proper scoring rule they are indeed calibrated probabilities *of the next token under the training distribution*. What they are not — and this is the crux — is a measure of whether the assertion is **supported by the context**.

The two come apart precisely when the context is insufficient. Asked a question whose answer does not follow from anything in its context, a model does not return a low-confidence distribution over answers; it returns a confident distribution over *plausible-sounding* answers, because plausible-sounding answers are exactly what the training distribution contains. The output is a fluent completion of a pattern, and the probability attached to it is the probability of that completion, not the probability that it is true.

What is missing is a **second, structural channel**: a quantity that is not a likelihood, that is computed from the geometry of the evidence rather than from the language-model head, and that is non-zero exactly when the local pieces of evidence cannot be assembled into a coherent global answer. In sheaf terms this is an obstruction. Given local sections s_i on a cover $\{U_i\}$ that agree pairwise, do they glue? On a discrete site the answer is always yes, trivially. On a non-trivial site the answer is governed by cohomology, and when the answer is no, one has a certificate.

Care is needed here. It is tempting to write "hallucination = $H^1 \neq 0$". This is false, and it is false in a specific way that matters. The quantity most easily computed, $\|\delta x\|$, is the norm of a **coboundary**, and a coboundary represents *zero* in H^1 by definition. The correct statement is a one-way implication: a non-vanishing obstruction *class* certifies that no global section exists over the given cover, hence that no globally coherent answer is available there. It does not follow that every hallucination has $H^1 \neq 0$; a model can produce a false statement that is perfectly consistent with everything in its context. Part X records this as the sixth of its twelve disclaimers.

A second and sharper error lies downstream of the first, and it is stated in full because its repair determines what the obstruction head can be. Suppose a design defines the disagreement on an overlap e between charts i and j as $z_e = \rho_{i\to e} v_i - \rho_{j\to e} v_j$, built from per-vertex values v. That is exactly $\delta^0 v$ for the zero-cochain v, *whatever* the v_i are and whether or not the projections are shared. A pairwise difference of vertex data is a coboundary by construction; its harmonic part vanishes in exact arithmetic and its distance to the image of δ^0 is zero. Everything such a head reports is floating-point noise. The construction is withdrawn, not patched.

What replaces it is narrower and sound. A non-vacuous obstruction requires 1-cochain data that is *not* manufactured from vertex data: either agreements given directly on overlaps, or a genuine holonomy around a cycle. The surviving instrument therefore consumes an independently supplied overlap datum with recorded provenance, checks the cocycle condition before reporting any class, and reports the harmonic magnitude separately from the

class. When the required datum is absent or its provenance cannot be certified, the instrument returns **UNDETERMINED**. It does not return zero, and it does not fall back to the withdrawn vertex-derived expression. Chapter 3.5 gives the definitions and the gate; a production source of certified overlap data does not yet exist, so the deployed head abstains.

The value of the structural channel is therefore not that it detects all falsehood. It is that when it fires, it fires for a reason that can be exhibited, and abstention becomes a computation rather than a trained reflex.

That difference is operational and worth stating precisely, because it is easy to mistake for a matter of degree. A trained refusal is a behaviour: the model has learned that certain input shapes are followed by a decline, and the decline is produced by the same head, with the same type, as any other continuation. It generalises the way behaviours generalise — unevenly, and it can be argued away by an input the training distribution did not contain. A structural abstention is a different kind of object. It is emitted by a channel that has no access to the language head, it names the cover and the overlap datum on which it was computed, and it carries a certificate that can be checked by a third party who has the same evidence and none of the weights. The two are also falsified differently: a trained refusal is wrong when the model should have answered, whereas a structural abstention is wrong only if the exhibited certificate is invalid, which is a checkable condition rather than a judgement.

The book does not claim that structural abstention is available today. The deployed head abstains, in the strong sense that it returns **UNDETERMINED** for want of a certified overlap datum, and an instrument that always abstains is not yet an instrument. What the chapter claims is narrower: that abstention *of this type* is well posed, that its output type is a record rather than a probability, and that building it requires overlap-native evidence rather than a better-calibrated softmax. Chapter 3.5 specifies the record; Part IX specifies the gate by which the abstention rate would be judged useful; safety alone does not pass it.

1.1.6 Adaptation without preservation

The fourth deficit is catastrophic forgetting, and it has a sharper form than is usually recognised.

Fine-tune a model on a narrow domain and two things happen. The first is expected: it becomes better at the domain. The second is the problem: it becomes worse elsewhere, and — this is the sharp form — it becomes worse *asymmetrically*. Factual recall degrades gracefully. Constraints, refusals, and universally quantified commitments degrade sharply. A model that reliably declines a certain class of request before fine-tuning may comply after fine-tuning on data that has nothing to do with that class.

Standard explanations invoke the geometry of the loss landscape, interference in shared parameters, or the relative sparsity of the constraint-bearing data. These are real effects. But there is a structural account available, and it is more specific.

Transfer between domains, in the topos-theoretic picture, is a **geometric morphism**. A geometric morphism $f : \mathcal{F} \to \mathcal{E}$ has an inverse image f^* that preserves finite limits and all colimits. The classical fact is that such a functor preserves exactly the **geometric fragment** of the internal logic: the connectives $\top$, $\wedge$, $=$, $\exists$ and arbitrary disjunction $\bigvee$. It does *not* in general preserve implication $\Rightarrow$, negation $\neg$, or universal quantification $\forall$. Those survive if and only if the morphism is **open**.

Now observe what the two lists correspond to. Facts — "X is the capital of Y", "there exists a Z such that …" — are existential and conjunctive; they are geometric. Constraints — "for all requests of type T, refuse", "if P then not Q" — are universally quantified and involve

implication and negation; they are *not* geometric. The structural prediction is therefore that transfer should preserve facts and erode constraints, and this is what is observed.

Rigour grade. That f^* preserves the geometric fragment, and that $\Rightarrow, \neg, \forall$ require openness, is standard topos theory, grade **T**. That fine-tuning a neural network *is* a geometric morphism is grade **D**, and it is currently unsupported: it is not, in any literal sense, and Part II explains why the naive version is vacuous in **Vect**, where every linear functor is exact and left-exactness therefore forbids nothing. The non-vacuous content requires morphisms of *sites*. The resulting prediction about the asymmetry of forgetting is grade **D**, and it is testable; Part IX gives the experiment.

The gap between the theorem and the prediction is wider than earlier formulations allowed, and the document now marks it operationally rather than rhetorically. Ordinary weight-space fine-tuning, low-rank adaptation and weight merging do not exhibit a site inclusion, an inverse image, or a commuting square between the source and target sites; nothing in the training procedure constructs one. Chapter 5.3 therefore requires an independently certified semantic map before any transfer statement may be made in these terms, and returns `RELOCATION_UNAVAILABLE` when the map does not pass its monotonicity, cover and typed commuting-square tests. Safety erosion under fine-tuning is a real and reproducible phenomenon; the geometric mechanism offered here is a hypothesis about *why* it takes the asymmetric form it does, and it can be wrong while the phenomenon remains real.

1.1.7 Capacity without structure

The fifth deficit is the most speculative and we mark it as such.

Everything a transformer does to a representation between residual streams is, to first order, a sequence of linear maps and pointwise non-linearities. The linear part is, in the relevant sense, **abelian**: matrices compose, and although matrix multiplication is not commutative, the group generated by a model's effective transport maps along different paths through a context is, by construction, extremely close to trivial. Transport a representation around a loop in the context and you get back essentially what you started with.

This is a statement about **holonomy**. In a sheaf with orthogonal restriction maps, the holonomy group $\mathrm{Hol}(\mathcal{F}, v_0) \leq O(d)$ is the group generated by transport around all loops based at v_0. If it is trivial, the sheaf is globally trivialisable — there is a single consistent global frame, and every loop is a no-op. If it is non-trivial, there exist path-dependent computations: the answer depends on the route through the evidence, which is what one wants for reasoning that must respect the structure of an argument as well as its content.

The speculative step is to claim that this matters for capability. We do not have a theorem that it does, and we say so. What we do have is a theorem about the geometry of the quantity itself. Writing β for the exponential growth rate of the holonomy group, Part II proves

$$\beta \in \{0\} \cup [\, c(d), \infty)$$

for a constant $c(d) > 0$ depending only on the stalk dimension. There is a **gap**. The argument runs through the Tits alternative — a finitely generated linear group either contains a free subgroup of rank two or is virtually solvable — together with the fact that a compact connected solvable group is a torus, and the uniform exponential growth theorem of Eskin, Mozes and Oh.

The consequence is architecturally significant. β is a **certificate**, not a score to be maximised. A model either has non-abelian transport structure or it does not, and there is no continuum in between. That changes what one should measure, what one should regularise, and what one should report.

Two caveats that must travel with the theorem. First, any naive numerical estimate of β by ε-counting of group elements will return $\beta_\varepsilon \to 0$, always, for trivial metric-entropy reasons. The dichotomy is a statement about the abstract group, so β must be certified *algebraically* and never estimated by enumeration; a programme reporting β from a counting procedure would be reporting noise. Second, certification by a ping-pong argument on the sphere is impossible. Elements of $O(d)$ are isometries, so their powers are recurrent: no positive-margin inclusion $g^n(B) \subset A$ can hold for all $n \neq 0$ between disjoint spherical caps, and there is no hyperbolic-type estimate on the real unit sphere. No implementation of it is admitted. Certification must either pass to a suitable local field with certified non-archimedean inequalities for algebraic generators, or return **UNDETERMINED**. The certificate is binary *plus undetermined*, never a score.

A related monotonicity claim holds only in a weak form. Enlarging the structure group from H_1 to $H_2 \geq H_1$ gives only $(\mathbb{R}^d)^{H_2} \subseteq (\mathbb{R}^d)^{H_1}$, and the inclusion may be an equality; the correct statement is that the invariant space *cannot increase*, with strict decrease exactly when the larger group removes a vector fixed by the smaller. Classification of regimes must be by the dimension of the fixed representation, never by whether a subgroup is proper: a proper closed subgroup may have no non-zero invariant vector at all. The architectural trade-off between capacity and consistency that these facts are used to motivate is therefore grade **D**.

1.1.8 The common origin, stated as a conjecture

The five deficits admit one description. Locality without gluing is the trivial topology. Composition without compositionality is the absence of functorial restriction. Confidence without justification is the vanishing of the obstruction group. Adaptation without preservation is the failure of a transfer functor to be open. Capacity without structure is the triviality of holonomy. In each case the same feature of the description is doing the work: **the site is trivial**, and every invariant that would have carried information about global structure is, on a trivial site, identically zero.

Rigour grade: D (conjecture). That the five failure modes share one origin is a hypothesis, not a result, and it is graded and treated as such throughout. Each of the five clauses in the previous paragraph is a correct statement about what vanishes on a trivial index structure; none of them is an identification argument connecting that vanishing to the observed failure. No intervention experiment has been run, so the chapter's own common-cause test has no result. The defensible form of the thesis is: *we hypothesise that imposing explicit site and descent structure will jointly affect these five failures*, and each causal prediction is separated, graded, and requires intervention evidence before the word "because" may be used.

That is the diagnosis. It has the shape one wants in a diagnosis — it is unified, it is specific, and it is wrong in a checkable way if it is wrong. What it is not is established.

The programme in one sentence. Replace the trivial site with a genuine one; then descent, cohomology, obstruction, transfer and holonomy become non-trivial, computable and architecturally usable, and each module is required to exhibit the explicit hypotheses under which it specialises one-way to its standard counterpart. The specialisations are proved individually and under mutually incompatible hypotheses; they are not composed into a global containment statement, and no performance guarantee of any kind follows from them.

1.1.9 The observables this chapter licenses

A diagnosis that generates no measurement is a position, not a diagnosis. Each of the five deficits licenses at least one observable, and the table below states, for each, where it is defined, what its invariance status is, and what would count as a refutation. The invariance column is not decoration: the binding rule of this book is that no quantity may be reported anywhere without either a proof of exact invariance under the declared transformation class or a stated

budget, and a quantity whose invariance class is unknown may not be plotted, gated on, or compared across models.

Table 1.1.2 — Observables licensed by each deficit, with their invariance status and gate.

Deficit	Observable	Defined in	Invariance status	Refuting outcome
Locality without gluing	Coverage defect of a constructed cover against its atom set	Chapters 3.2, 5.4	Exact under relabelling of atoms; budgeted under change of atomiser	A site that cannot be built within budget, returning `SITE_CLOSURE_UNAVAILABLE` at corpus scale
Composition without compositionality	CD(A), jointly with attention entropy and effective support	Part I, Chapters 2.9, 9.1	Exact under permutation of positions; budgeted under tokenisation change	No relationship to head function, to training time, or to compositional task performance
Confidence without justification	Harmonic magnitude of an independently supplied overlap cochain, and the class when the cocycle gate passes	Chapter 3.5	Gauge- and basepoint-declared; class reported only after closure is checked	Abstention that does not concentrate on underdetermined queries
Adaptation without preservation	Fragment-wise retention: geometric versus non-geometric assertions before and after adaptation	Chapters 5.3, 9.2	Budgeted; requires a certified semantic map or returns `RELOCATION_UNAVAILABLE`	Symmetric erosion of both fragments, or asymmetric erosion under an adaptation that provably is not a site morphism
Capacity without structure	The certified side of the holonomy dichotomy, plus the naturality residual R_{nat}	Chapters 2.11, 3.4, 3.6	R_{nat} is measured, never imposed; the certificate is exact or **UNDETERMINED**	Capability differences uncorrelated with the certified side of the gap

Two entries in that table deserve emphasis because they are places where the book measures its own failure rather than its own success. The naturality residual is one: a coordinatewise nucleus commutes with restriction only when every restriction map is non-negative row-monomial, which the anti-collapse constraints of Part V force to be a partial permutation, so no sheaf network with learned orthogonal transports and a pointwise activation is a morphism of sheaves. "The model computes with a sheaf" is therefore at best approximately true, and the approximation is quantified and reported rather than assumed. The other is the abstention column: an instrument that returns **UNDETERMINED** more often than it returns a class is not thereby broken, but it is also not yet useful, and the frequency is itself a reported quantity.

1.1.10 Which declaration repairs which deficit, and which part is not repaired

A diagnosis that names five absences and then changes the subject is a complaint. This section discharges the obligation the diagnosis creates: for each deficit, the numbered declaration later in this book that addresses it, the sense in which it does so, and — set out at the same length, because it is the part a reader is entitled to and rarely receives — the residue that no declaration in this book addresses at all.

Definition 1.1.1 (D) — repair relation. A numbered declaration R *repairs* a deficit F when three conditions hold: R constructs or characterises an object whose absence is what F names; R is proved, or its proof obligation is explicitly recorded as open with the failing hypothesis identified; and at least

one observable licensed by F in Table 1.1.2 is computable from the data R constructs. A declaration that satisfies the first two conditions and not the third *addresses* F without repairing it, and is recorded as such.

The third condition is the one that does work. It refuses the move by which a structural absence is answered with a structure that generates no measurement, and it is why several rows below read *addresses* rather than *repairs*.

Table 1.1.3 — Repair map: deficit, repairing declaration, and the residue.

Deficit	Repairing declarations	Sense of repair	Residue: what is not repaired
Locality without gluing	Theorem 2.1.6; Theorem 3.2.15; Theorem 3.2.18; Algorithm 3.2.4 with Theorem 3.2.19	Repairs. A covering relation exists, is proved to satisfy the three axioms, saturates in finitely many steps, and the greedy cover attains the submodular-cover bound relative to its bank	The site is imposed rather than discovered; Theorem 3.2.20 shows deciding sufficiency is intractable in general, so absolute sufficiency returns **UNDETERMINED** and the minimal-sufficient-cover claim is not certified
Composition without compositionality	Lemma 2.9.5; Lemma 2.9.6; Theorem 2.9.7; Theorem 3.3.19	Addresses only. The coend calculation is proved and the "truncated coend" reading is withdrawn; a functorial restriction is constructed at L1	No certificate follows: CD is descriptive telemetry at grade **M**, and Theorem 2.9.7 shows a generic weighted row does not factor through the codiagonal quotient, so the construction that would make composition functorial is not the attention actually computed
Confidence without justification	Definition 3.5.6; Theorem 3.5.31; the cocycle gate of Chapter 3.5	Addresses only. An obstruction group is available and non-vanishing is defined; the logical fragment that transports is characterised	No certified source of an `IndependentEdgeDatum` exists, so L5 returns **UNDETERMINED** by design; an implementation built from vertex differences measures a coboundary and is withdrawn; a manufactured difference can never certify a class
Adaptation without preservation	Theorem 3.5.31; Theorem 3.6.41; Definition 5.3.4; Theorem 5.3.3	Addresses only. Transfer is typed as a site morphism and the preserved fragment is characterised exactly	The asymmetry prediction is grade **D**; the transfer layer has no gradient and is marked **UNTRAINABLE** in Table 1.3.5; without a certified semantic map the instrument returns `RELOCATION_UNAVAILABLE`; the claim that left-exactness prevents forgetting is *rejected* as vacuous in **Vect**
Capacity without structure	Theorem 2.11.12; Theorem 2.11.17; Theorem 2.11.26; Theorem 3.6.9	Repairs the mathematics, not the measurement. The fixed-space theorem, the exact capacity–consistency trade-off and the growth dichotomy are proved	Proposition 2.11.31 exhibits explicit failure witnesses for the ping-pong certificate on the real sphere, so for a given trained model the side of the gap is **UNDETERMINED**; Corollary 2.11.13 shows no diagnostic factoring through the fixed space alone can decide it; the capability wager remains grade **D**

Proposition 1.1.2 (T) — the repair map does not compose into a repair of the common-origin conjecture. Suppose every row of Table 1.1.3 were upgraded to *repairs* in the sense of Definition 1.1.1. It would not follow that the common-origin conjecture of the previous section is established.

Proof. The conjecture asserts that the five deficits share one cause, namely triviality of the site. Definition 1.1.1 quantifies over declarations that construct an object and license an observable; it says nothing about causation, and a construction repairing five symptoms is consistent with the five symptoms having five independent causes each of which the construction happens to touch. Formally, let $F_1, \dots, F_5$ be the deficits and let R be the conjunction of the repairing declarations. From $R \to \neg F_i$ for each i one cannot infer that the F_i have a common cause, since the implication is compatible with any assignment of causes; the inference requires intervention evidence of the kind the chapter's own common-cause test demands and which has not been produced. □

The proposition is stated because the table invites exactly the wrong reading. A repair map that looks complete would make the conjecture look established, and it is not: the conjecture is refuted by an intervention that improves one deficit and leaves the others untouched, and it is supported by an intervention that moves all five together. No such intervention has been run.

Proposition 1.1.3 (T) — three of the five residues are blocked on the same missing object. The residue entries for the second, third and fifth deficits each reduce to the absence of a *certified independent quantity*: a weighting that is not manufactured from the values it is used to compare, an edge datum that is not a coboundary, and a growth certificate that is not obtained by ε-counting.

Proof. For the second deficit, Theorem 2.9.7 states that a weighted contraction factors through the codiagonal quotient exactly when its coefficients are equal; the coefficient system that manufactures a weighted sum is non-canonical and is graded **D**, so the quantity that would certify compositional structure is constructed from the same data it certifies. For the third, Definition 3.5.6 requires an `IndependentEdgeDatum`, and any datum formed as a difference of per-vertex values is a coboundary, hence represents zero in the first cohomology; independence from the vertex values is precisely what is unavailable. For the fifth, Proposition 2.11.35 shows that ε-counting saturates, so the growth quantity computed from tolerance-thresholded words carries no information about the group, and the algebraic certificate that would replace it is blocked by Proposition 2.11.31. In each case the missing item has the same form: a quantity whose value is not determined by the object it is supposed to adjudicate. □

That is a sharper statement of the programme's central practical difficulty than the deficit list itself provides, and it is the reason the observables table above has an abstention column. Three of the five instruments are worse than imprecise: they are blocked on the construction of an independent input, and no amount of engineering on the instrument side will supply one.

The repair map is a reading of declarations, not a generated artefact. Table 1.1.3 was compiled by reading the named statements. Nothing generates it from the claim registry, no cross-reference checker verifies that the cited numbers say what the row asserts, and the residue column is an absence claim of the kind that decays silently when later chapters add material. It may be cited as a route into the book and may not be cited as evidence that a deficit is repaired.

1.1.11 What this diagnosis is not competing with

Several established lines of work address symptoms in this list, and confusing the present claim with theirs would make it both less useful and easier to dismiss. Four boundaries are worth drawing, and none of them is a claim of superiority.

Calibration. A large literature makes a model's confidence match its accuracy, by temperature scaling, by ensembling, by conformal prediction, or by training on abstention labels. That work succeeds at what it sets out to do, and conformal methods in particular deliver distribution-free coverage guarantees that nothing here matches. The difference is in what is being calibrated. A calibrated probability is a statement about the frequency of correctness in a reference population; the structural channel described above is a statement about whether the *specific* local evidence in front of the model can be assembled at all. The two are complementary and answer different questions, and a well-calibrated model can still be confidently wrong on an individual query whose evidence does not glue.

Oversmoothing in graph networks. The observation that repeated neighbourhood averaging drives representations towards a low-dimensional invariant subspace is closely related to the first and fifth deficits, and Part II makes the connection explicit rather than avoiding it: the constant sheaf is precisely the case in which the Laplacian kernel is large and transport is trivial. The relationship is one of specialisation, not rivalry. What sheaf structure adds is a non-trivial kernel to smooth *towards*, and the published negative result — that such structure is often not learned — is the reason Part V exists.

The forgetting literature. Elastic weight consolidation, replay, parameter isolation and adapter methods all target the fourth deficit and several of them work. The claim here is not that they do not; it is that they do not predict the *asymmetry*, and that a mechanism which predicts the asymmetry is worth having even if the mitigations remain the same. If a retention-preserving method turned out to preserve constraints and facts equally, the geometric account would be wrong and those methods would be unaffected.

Retrieval and neuro-symbolic hybrids. Retrieval supplies evidence; symbolic layers supply inference rules with guarantees. Both are, in the vocabulary of this book, ways of enlarging or constraining the evidence available at a context, and neither constructs a covering relation among contexts or an obstruction to gluing. Retrieval in particular is treated throughout as an input to L0 rather than as an alternative to it, and Chapter 5.2 makes a retrieval trigger a gated, controlled decision rather than a default.

In summary, this book proposes a different *object* to compute — a site with covers, restrictions and invariants — and argues that several familiar failures become expressible in it. Whether that object earns its cost against the methods above is exactly what Part IX is for, and Chapter 10.2 states for each named alternative whether it is orthogonal, composable, competing, or the correct baseline. Nothing in Part I settles that question.

1.1.12 The negative results the diagnosis must respect

A diagnosis is only as good as the counter-evidence it has already absorbed. Six negative results constrain everything in the remaining eleven parts, and they are collected here rather than distributed as footnotes, because a reader who does not know them cannot judge whether the constructive programme is serious.

Sheaf structure collapses to the identity when it is rewarded directly. The published negative result on sheaf neural networks is that they frequently learn the identity sheaf and degenerate to ordinary graph networks. Part V shows that this is not bad luck. A gluing or sheaf-consistency term in the loss is a homogeneous quadratic in the restriction maps and has a data-independent minimiser at $\rho \equiv 0$: the trivial restriction minimises it regardless of the data. The prohibition that follows — such a term may never be a primary training objective — is a theorem, not a preference, and it is the point on which designs of this kind most often go wrong.

The countermeasures are partial. Two of the four admissible fixes are not interchangeable: heterogeneous stalks make the *identity* attractor inexpressible but leave the *zero* attractor intact, and the Stiefel constraint removes the zero attractor; they must be combined. Even combined, they exclude only zero and literal-identity parameter collapse. Functional transmission collapse, rank loss and task bypass remain possible, and at the declared ranks of the reference configuration the joint frame condition that would close activation-side collapse is infeasible. Any statement that anti-collapse is complete is **WITHDRAWN**.

The degree-two obstruction theory is vacuous by default. On a graph, $C^2 = 0$ and hence $H^2 \equiv 0$ identically. Under standard top-k routing the expert nerve is a full simplex, hence contractible, hence again $H^2 \equiv 0$. Any reported gerbe class or degree-two diagnostic on such

a nerve is identically zero, and reporting it is meaningless. The minimal non-vacuous configuration is four experts arranged as $\partial\Delta^3$, and it must be engineered deliberately. Engineering a structure into existence in order to measure it is a move that deserves suspicion, and we say so where it is done.

A manufactured difference cannot certify a class. The result recorded above under the third deficit generalises: any quantity built as a difference of per-vertex values is a coboundary and therefore represents zero. Every instrument in this book that reports a cohomology class must first prove that its input is not manufactured, and must return **UNDETERMINED** otherwise.

There is no ping-pong certificate on the real sphere. The impossibility recorded under the fifth deficit removes the most direct concrete certification procedure for the holonomy dichotomy. The theorem survives; the ability to decide, for a given trained model, which side of the gap it lies on does not.

The structural metrics are attackable at their source. Every invariant in this book is computed against a site, and a site is a design object. A system that is rewarded for low obstruction can reduce the obstruction by trivialising the site rather than by becoming coherent. The defence — committing the site externally before a rollout is scored against it — works only where an exogenous measure is available, and no such measure exists for contexts without precedent. This is an unsolved problem and is stated as one in Part VI, not deferred to future work.

1.1.13 What the diagnosis costs

A structural diagnosis is only useful if the structure can be computed at a price someone would pay, and the price is stated here rather than in Part VII, because it bounds how the rest of the book should be read.

Table 3.8.9 reports checked arithmetic for the single reference configuration, but it contains partial subtotals only. Complete architecture totals and architecture-to-comparator ratios are **OPEN**, and atomisation cost is **OPEN**. Consequently no 8,192-token sequence headline total or multiplier may be quoted from that table.

The comparator rows fix shape, not capability, and do not constitute a matched-total-parameter, matched-quality baseline with a named checkpoint, recipe, hardware, precision, batch, optimiser and serving contract; no such baseline manifest exists in this book. Any sentence that turns the partial subtotals into a statement about efficiency or value for compute is an error against the cost discipline of Part VII.

Three limits are more serious than the arithmetic. The coverage-based router that motivates the whole mixture-of-experts construction has no published backward map with a bias budget, and is marked **UNTRAINABLE**; only the top-k fallback is executable, and the fallback is not the object the theory is about. Learned site construction, the abstention head and the self-evolution proposal step carry the same mark. And the descent solve has exactly one normative training program, differentiated by reverse mode through its realised arithmetic and masks; exact and tolerance-stopped solves are certificate-only, because differentiating an exact converged inverse while executing a truncated iteration is a different gradient, not a better one.

The reader should hold these facts alongside the diagnosis. The claim is not that a cheap fix has been found. It is that a specific structural absence can be named, that naming it generates a programme, and that the programme is expensive and incompletely trainable as specified.

1.1.14 What would make this diagnosis wrong

The ways in which the diagnosis could fail are set out explicitly, because a diagnosis that cannot fail is not a diagnosis. Five conditions would refute it, and each has a corresponding experiment in Part IX with a pre-registered threshold.

It would be wrong if the five deficits had five unrelated causes, each better addressed by a targeted mechanism — retrieval for the third, curriculum for the second, parameter isolation for the fourth. The unification would then be an aesthetic preference rather than an explanation. The evidence that would show this is failure of the interventions to correlate: fixing the site should improve all five, and if it improves one and leaves the others untouched, the common-origin conjecture is refuted while the individual mechanisms may survive.

It would be wrong if the structure is already present. Transformers are not required to leave their index structure trivial; they might, through positional encoding and learned attention, be constructing something with genuine covering structure that we have simply failed to look for. The compositionality defect is the first place to look. If CD is already near zero in strong models, the second deficit does not exist and the first is doubtful.

It would be wrong if the structure is present but cannot be learned. This is the most likely failure mode and the one we take most seriously, because published evidence for it already exists, and because the trainability status recorded above shows that several components of the proposed architecture do not yet have a gradient contract at all. A structure that must be imposed by hand at every scale is a research instrument, not an architecture.

It would be wrong if the invariants can be driven to zero by trivialising the site. Then the metrics measure the design rather than the model, and the programme is self-referential. The external-commitment protocol is a partial mitigation, and Part VI states plainly that it is not a solution.

And it would be wrong, finally, if the whole apparatus is real but the effect is small. A correct theory of a negligible phenomenon is still a correct theory and still not worth building an architecture around. This is the risk we can say least about in advance, and it is why Part IX insists on pre-registered effect sizes rather than on statistical significance.

The rest of Part I makes the diagnosis precise enough to be worth arguing with. Chapter 1.2 states the central thesis, fixes the sentence by which the whole system is to be assessed, and enumerates every load-bearing claim with its grade, including the rows marked *rejected* and *withdrawn*. Chapter 1.3 sets out the method of transplantation — how one moves from a mathematical structure to a computational module without deceiving oneself — the compliance matrix that records which modules have actually met that standard, and the dependency map showing where the Grothendieck programme of Part XI is used before Part III begins.

1.1.15 Deliberate demotions

"Five deficits" is a choice of granularity, not a partition proved to be exhaustive. Nothing here shows that the list is complete or that its members are independent. Two of them — locality without gluing and composition without compositionality — could be read as one deficit at a coarser grain, and a reader who prefers four or six is not contradicted by any statement in this chapter. The list earns its place by generating distinct observables, which is a weaker property than being a partition.

Where a repair is attempted is recorded in Table 1.1.3, and three of its five rows say *addresses* rather than *repairs*. The distinction is Definition 1.1.1's third clause, and it is doing real work: for the second, third and fourth deficits the constructed object exists and the observable it would license is either telemetry at grade **M** or an instrument that abstains.

Presenting this chapter as a diagnosis with a matching cure would misstate the position by three rows out of five.

The repair map is not common-cause evidence. Proposition 1.1.2 bars that reading of Table 1.1.3; Proposition 1.1.3 identifies only a shared residue form. The three blocked instruments each want an independently constructed quantity. That identifies the obstruction's shape but does not establish that one construction would unblock all three. The three candidate constructions — a canonical coefficient system, a certified edge datum, an algebraic growth certificate — are unrelated pieces of mathematics with no shared proof strategy identified here.

The partial cost subtotals quoted above bound nothing about value. Table 3.8.9 does not supply a complete architecture total: complete totals and ratios remain **OPEN**, atomisation cost remains **OPEN**, and there is no matched-quality baseline anywhere in this book. Any sentence converting the subtotals into an efficiency claim is an error against the cost discipline of Part VII.

Chapter 1.2 — The central thesis and the claims it rests on

Chapter 1.1 offered a diagnosis. This chapter states what follows from it, in a form precise enough to be disagreed with, and enumerates every load-bearing claim in the book together with the grade it carries.

1.2.1 The thesis

Central thesis. The characteristic failures of contemporary language models — incoherence across a context, brittle multi-step composition, confidence uncorrelated with evidential support, asymmetric catastrophic forgetting, and the absence of path-dependent structure — are conjectured to share an origin in computing over a *trivial site*. Replacing the trivial site with a genuine Grothendieck site is proposed to make descent, cohomology, obstruction, transfer and holonomy simultaneously non-trivial, computable, and architecturally usable. The resulting system is assessed by exactly one permitted sentence, fixed by §Q2 of the synthesis and repeated verbatim wherever the question is raised:

> **GM-Ω∞ is an unvalidated structural augmentation and research architecture for Transformer/MoE systems; it has several proved local specialisations but no proved global subsumption or replacement result.**

The vocabulary this document may not use as an achieved result. No abstract, table, figure, claim record, ledger row or conclusion in this specification may describe GM-Ω∞ as *containing*, *strictly subsuming*, *replacing* or supplying an *exact performance floor* for the Transformer. Those are falsifiable research targets, and their falsification conditions are stated in Chapter 10.2 and Appendix A7. The obstacle is not modesty but arithmetic: the advertised end-to-end degeneration is not a pre-norm Transformer block — it omits both normalisations, evaluates the MLP on the wrong state and replaces two ordered residual sublayers by parallel terms — while the purported floor is elsewhere obtained by retaining a complete Transformer branch, and the normative model still contains grouped-query Transformer attention. Individually proved *local specialisations* survive, each with its own full hypothesis list; they do not compose, because their hypotheses use incompatible sites, routers, objectives, shape conventions, residual order and solver semantics. Chapter 14.1 discharges the same obligation at length and reaches the same sentence.

The thesis has three parts and they fail independently.

The **diagnostic** part says the deficits share an origin. It is falsified if interventions targeting the site improve one deficit and leave the others untouched.

The **constructive** part says a non-trivial site can be built, learned and computed with, at tolerable cost. It is falsified by cost — if the descent solve dominates the forward pass, or if site construction does not scale — or by trainability, which is the more likely failure and the one for which negative evidence already exists.

The **specialisation** part says that individual modules reduce, one-way and under explicitly stated hypotheses, to their standard counterparts. This is the part for which proofs exist, and it is also the part most easily inflated into a global containment claim that no proof supports. We state it carefully below.

1.2.2 What a local specialisation does and does not mean

Every module in Parts III through VIII carries a *degeneration condition*: an explicit set of hypotheses under which the module provably reduces one-way to its standard counterpart, on the same objective and the same forward map. Set the comparison graph complete, the stalk dimensions equal and the restrictions to identities, and the displayed L2 numerical kernel specialises to softmax attention; take the cover to be a top-k selection under the five hypotheses of Theorem 4.4.1 and the Topos-MoE forward map is that of a standard sparse mixture of experts. Each such statement is a proposition about one module under its own hypotheses.

Local specialisations do not compose, and there is no floor. These results establish that a standard *module* lies inside the corresponding module's hypothesis class. They do **not** establish a global function-class containment, because the hypotheses differ from module to module: they demand incompatible sites, routers, objectives, shape conventions, residual order, normalisation placement and solver semantics, and no theorem in this document assembles them into one forward map. They also do not establish that training will find the standard solution, that the larger class is as easy to optimise, or that the additional parameters are harmless. A hypothesis class containing a good solution is a necessary and entirely insufficient condition for finding one. We state this once, here, at full strength, and we do not soften it later. Any sentence elsewhere in this document that appears to promise "at least as good as a transformer", or a "performance floor" of any kind, is an error against §Q2 and should be read against this paragraph and reported.

The distinction matters because the history of architecture research is substantially a history of strictly-more-general models that trained worse. Generalising a construction is easy; generalising it without destroying its optimisation landscape is the whole difficulty, and Part V is about a case where we can prove the landscape is destroyed unless a specific countermeasure is taken.

1.2.3 The claim ledger

What follows is the executive list of principal claims and known adjudicated failures. The versioned ledger, with precise statements and locations, is in Part X; neither list claims completeness while corpus propagation remains under review.

Table 1.2.1 — the principal claims and their grades.

#	Claim	Grade	Where established
1	The transformer's default context carries the trivial topology, on which every presheaf is a sheaf	**D**	Part I, Part II
2	The indiscrete identity coend has the unweighted codiagonal quotient; generic weighted attention does not factor through it	**T**	Part III, App. A1
3	The compositionality defect $\|A^2 - A\|_F / \|A\|_F$ is well defined and computable from any checkpoint	**M**	Part I, Part III
4	Low compositionality defect indicates compositional structure	**D**	Part IX (experiment)
5	ReLU is the closed nucleus at 0 on $(\overline{\mathbb{R}}, \min, \max)$; via sublocale it corresponds to a sub-topos	**T**	Part III, App. A1
6	LayerNorm and max-pooling are **not** closure operators	**T**	Part III, App. A1
7	GELU, SiLU and sigmoid are not nuclei	**T**	Part III

#	Claim	Grade	Where established
8	The context site is a genuine site when contexts are down-sets of evidence atoms	**T**	Part III, App. A1
9	It is **not** a site when contexts are subspaces — the subspace lattice is non-distributive	**T**	App. A1
10	$H^0 \cong (\mathbb{R}^d)^{\mathrm{Hol}}$ for an orthogonal cellular sheaf	**T**	Part III, App. A1
11	Holonomy dichotomy: $\beta \in \{0\} \cup [c(d), \infty)$, so β is a binary certificate	**T**	Part III, App. A1
12	β must be certified algebraically; ε-counting always yields $\beta_\varepsilon \to 0$	**T**	Part III
13	β governs capability	**D**	Part IX (wager, with protocol)
14	A specified non-zero relative connecting class obstructs extension of the registered partial section	**T** abstractly; implementation **withdrawn**	Part III, Part IX
15	Hallucination *equals* $H^1 \neq 0$	**rejected**	Part X, corrections ledger
16	Geometric morphisms preserve exactly the geometric fragment $\{\top, \wedge, =, \exists, \bigvee\}$	**T**	Part II
17	Left-exactness "prevents forgetting"	**rejected** — vacuous in **Vect**	Part X
18	Fine-tuning should retain facts and erode refusals	**D**	Part V, Part IX
19	An expert is an open sub-topos $\mathcal{E}/U_i$; $\Sigma_U \dashv U^* \dashv \Pi_U$ is dispatch/combine	**D** (construction), **T** (adjunction)	Part IV
20	Nucleus sub-toposes and sub-sites cannot express `combine` — they lack Σ_U	**T**	Part IV
21	The L0/L4 routing greedy maximises a monotone submodular coverage objective, so greedy attains the $(1 + \ln n)$ submodular-cover bound *relative to its candidate bank*; absolute sufficiency and injectivity of the amalgamation map are not verified, so the routine is not certified as a *minimal sufficient cover* and returns UNDETERMINED when the absolute threshold is unchecked (§Q5.1)	**T** for the relative bound; **UNDETERMINED** for sufficiency	Part IV
22	At matched active-expert budget $k \geq 2$, the repaired top-k redundancy gap is independent of k	**T**	Part IV, App. A1
23	The stated determinant gerbe cochain is always exact	**rejected**	Part IV, Part X
24	$H^2 \equiv 0$ on a graph and on a full-simplex nerve; the gerbe programme is vacuous by default	**T**	Part II, Part IV
25	The gluing loss has a data-independent global minimiser at $\rho \equiv 0$	**T**	Part V, App. A1
26	Heterogeneous stalks make the identity sheaf inexpressible	**T**	Part V, App. A1
27	For $\sigma > 0$, GRPO's advantage is $\Delta_0 R / \|\delta^0 R\|_2 = \Delta_0 R / (G\sigma)$, σ the population standard deviation, for the Čech–Hodge 0-Laplacian of the group nerve; at $\sigma = 0$ both sides are $0/0$ and the registered zero-variance action applies (Thm 6.2.4)	**T**	Part VI, App. A1
28	The advantage is "an H^1 class"	**rejected** — $\delta^0 R$ is a coboundary; the home is C^0/H^0	Part VI, Part X

#	Claim	Grade	Where established
29	A token-level importance ratio is generally intractable for a parallel revisable decoder: the terminal token conditionals exist but are not recoverable from one sampled trajectory without marginalising the latent revision history, while the *augmented round-action* trajectory likelihood is exactly evaluable when every action component, including the route, is scored (Def. 62.15b, Def. 65.1)	**T**	Part VI
30	Restricting self-edits to geometric morphisms constrains the logic that transports	**T**	Part VI
31	That restriction is an alignment guarantee	**rejected** — it constrains logic, not behaviour	Part VI, Part X
32	Each module is intended to specialise one-way to a registered baseline under fixed hypotheses and objective	**D / UNDETERMINED** until each conformance proof passes	Parts III–X
33	Therefore performance is at least that of the standard model	**rejected** — no global subsumption or replacement result is proved (§Q2); local specialisations do not compose	Part I, Part X
34	Tannakian reconstruction, the section conjecture, motives and IUT transfer to this setting	**X**	Part X
35	Cover refinement reproduces the GRPO / Dr. GRPO / RLOO taxonomy as a theorem, and degree weighting is the unique refinement-stable estimator	**withdrawn** (§P5) — what survives is a classification inside one restrictive exchangeable estimator family and exact invariance under mass-preserving vertex splitting	Part VI, Part X

Of the **35** principal rows, **six** are marked *rejected* (15, 17, 23, 28, 31 and 33) and **one** is marked *withdrawn* at claim level (35). Row 14 is different: its abstract theorem is retained at grade T while only the attempted implementation is withdrawn, so it is not counted among those seven claim dispositions. We consider this explicit negative ledger to be among the more valuable contributions of the revision, and we have not buried it: the records appear here, in the corrections ledger of Part X and at the point of use.

1.2.4 The five contributions we would defend

If the programme is judged on its strongest points rather than its most ambitious, these are the five we would put forward.

First, the corrected coend limit. A first formulation claimed attention was a *truncated* coend recovered by taking $\mathcal{C}$ discrete. Over a discrete category the coend is a direct sum; over the indiscrete identity diagram its canonical quotient is the unweighted codiagonal. A generic weighted attention row does not factor through that quotient. Thus attention remains a barycentric numerical kernel unless a separate functor and universal property are supplied. The idempotence defect remains descriptive telemetry, not a coend certificate.

Second, the holonomy dichotomy. That the non-abelian capacity of an orthogonal cellular sheaf satisfies $\beta \in \{0\} \cup [c(d), \infty)$ is, as far as we know, new in this setting, and it changes the epistemics of the quantity: β is a certificate, not a score. The proof composes standard results — the Tits alternative, the structure of compact connected solvable groups, uniform

exponential growth — but the composition is not, we think, obvious, and its architectural consequence is immediate.

Third, the collapse theorem and its remedies. That the sheaf/gluing loss has a data-independent minimiser at $\rho \equiv 0$ explains a published negative result rather than acknowledging it, and it converts an empirical disappointment into a design constraint with four provable countermeasures. Of these, heterogeneous stalks is the one we would build on, because it does not only disincentivise the trivial solution — it makes it *inexpressible*.

Fourth, the Čech–Hodge identity for group-relative advantage. That GRPO's advantage is exactly $\Delta_0 R/\|\delta^0 R\|_2$ for the 0-Laplacian of the sampling group's nerve, wherever $\sigma > 0$, is an exact identity about an algorithm that is already widely deployed. It cannot by itself improve GRPO — an exact reformulation improves nothing — and, following §P5, we no longer claim that it *forces* the generalisation. What it does is *suggest* one: a family of graph baselines indexed by a nerve, of which the complete-graph member is GRPO and the i.i.d. leave-one-out member is RLOO. Whether any other member of that family reduces variance at matched cost is an open empirical question, tested by claim C11 of `A7` and unanswered here.

Fifth, the expert-as-open-sub-topos construction. Of the three candidate formalisations of "expert as a piece of a topos", exactly one supports the operation MoE actually needs. Covering must be a frame join $\bigvee_i U_i = 1$; `combine` must exist; and only the open sub-topos supplies the left adjoint Σ_U that `combine` is. Nucleus sub-toposes and sub-sites fail, and they fail for a reason one can state. That is the kind of argument that makes a categorical framing worth having: it eliminates alternatives rather than decorating a choice already made.

1.2.5 The five things we are least sure of

Symmetry requires the other list.

The site must be imposed. We do not know how to *discover* the evidence-atom locale from a corpus; we construct it. Everything downstream inherits whatever arbitrariness that construction contains, and a critic who says the invariants measure our design choices rather than the model's structure has a real point that we cannot fully answer.

The gerbe programme is vacuous unless deliberately rescued. On a graph, and under standard top-k routing, H^2 vanishes identically. We give the minimal non-vacuous configuration, but one should be clear that the degree-two theory does not arise naturally here; it has to be engineered into existence, and engineering a structure into existence in order to measure it is a move that deserves suspicion.

Trainability. The collapse theorem tells us one specific way training fails and how to prevent it. It does not tell us that nothing else goes wrong, and the empirical record of sheaf methods in deep learning is, so far, not encouraging.

The capability wager. That β governs capability is grade **D** and we have no proof. It is the most interesting claim in the document and the least supported.

Reward hacking at global scale. Descent verification provably catches *local* inconsistency. Against a globally coherent falsehood it offers nothing, and we have no structural defence to propose. Part VI says so at length rather than gesturing at future work.

1.2.6 The claim ledger read a second time: licence grade and falsification condition

The table above answers *how strong is the evidence*. It does not answer the two questions a reader actually needs answered before using a row: *what sentence am I permitted to write on the*

strength of it, and *what observation would take it away*. This section supplies both, and it does so with the licence vocabulary fixed in Definition 13.15.4 rather than inventing a second scale.

Definition 1.2.1 (D) — licence grade of a claim row. A claim row of this chapter carries the licence **S0** when the only sentence licensed is an assertion about a formal statement: that the statement has been written down and, where a proof is printed, that the printed proof establishes it. It carries **S1** when a declared finite fixture has been discharged by kernel check in the sense of Definition 13.15.27 and the sentence is therefore an assertion about that fixture and no larger object. It carries **S2** when a length-uniform or configuration-uniform check has been performed and the sentence quantifies over the declared family. The licence is assigned by the strongest check *actually executed*, never by the strength of the claim.

Definition 1.2.2 (D) — falsification condition of a claim row. The falsification condition of a row is a predicate over observable quantities, stated before any measurement, whose truth obliges the row to be regraded or withdrawn. A condition is admissible only if it names the quantity, the comparator, the direction and the threshold symbol; "the effect is not observed" is not a falsification condition, because it fixes neither the quantity nor the threshold.

Proposition 1.2.3 (T) — no row of this chapter exceeds S0. Every claim row of Table 1.2.2 below carries **S0**. No row carries **S1** or **S2**.

Proof. By Definition 1.2.1 the licence is the strongest executed check. **S2** requires a length-uniform or configuration-uniform check over a declared family. This document reports no executed experiment of any kind: every empirical gate in Chapter 10.2 and Appendix A6 is signed but unrun, and the conformance harness has not been executed against a frozen manifest, which is recorded in the `Gate` column of Table 1.3.4 as **OPEN** throughout. **S1** requires, by Definition 13.15.4, status `AXIOM_AUDITED` across a fixture's dependency closure together with a satisfiability witness for its hypotheses. The worked instance of Appendix A3 and the arithmetic of its source program are *stated* fixtures in the sense of Definition 13.15.27 and have been discharged by no kernel, the release environment carrying no Lean toolchain; no artefact of this release therefore reaches `AXIOM_AUDITED`, as the census of Chapter 10.1 records. No row exceeds **S0**. □

Proposition 1.2.3 is the most consequential statement in this chapter and it is a statement about the document rather than about the architecture. A reader who wants a single sentence for the present epistemic position may take this one: *the mathematics is written and partly proved, one finite instance has been computed, and nothing has been trained or measured.*

Table 1.2.2 — Licence grade, falsification condition and numbered carrier, by claim row. Row numbers are those of the principal-claims table above. The *carrier* column names the numbered declaration that would have to be edited if the row changed.

#	Grade	Licence	Falsification condition	Carrier	Endpoint
1	D	S0	Exhibition of a covering family on the default context index of a standard decoder for which some presheaf fails gluing	Definition 2.1.5, Theorem 2.1.6	none; definitional
2	T	S0	A weighted attention row with unequal coefficients shown to factor through the codiagonal quotient	Theorem 2.9.7	A3 fixture
3	M	S0	CD not computable from a checkpoint within the declared budget, or not invariant under position permutation	Chapter 9.1 instrument table	E-series telemetry
4	D	S0	CD uncorrelated with head function, training time and compositional task performance at the registered threshold	Chapter 9.2	H-family, exploratory

#	Grade	Licence	Falsification condition	Carrier	Endpoint
5	T	S0	A point of $(\overline{\mathbb{R}}, \min, \max)$ at which the ReLU closure conditions fail	Theorem 3.4.7	none; proved
6	T	S0	An affine LayerNorm exhibited as idempotent and order-preserving on the declared domain	A1, affine characterisation	none; proved
7	T	S0	A nucleus law verified for GELU, SiLU or sigmoid on the declared lattice	A1, activation counterexamples	none; proved
8	T	S0	A down-set covering family shown to fail pullback stability	Theorem 2.1.6	none; proved
9	T	S0	A distributive subspace lattice on the declared instance	Proposition 2.1.7, A3	A3 fixture
10	T	S0	A global section of an orthogonal cellular sheaf outside the holonomy-invariant subspace	Theorem 3.6.9, A1 Haar projection	A3 fixture
11	T	S0	A transport group with word growth strictly between bounded and $c(d)$	Chapter 2.11, dichotomy	none; proved
12	T	S0	An ε-counting procedure with a non-degenerate limit on the declared family	Chapter 2.11, counting trap	A3 fixture
13	D	S0	Capability differences uncorrelated with the certified side of the gap, at the registered interaction threshold	Chapter 3.6 wager	H10 with E11
14	T abstract; implementation withdrawn	S0	The registered partial section extended while the relative connecting class is non-zero	Chapter 3.5	E09 gate
15	rejected	S0	Not applicable; the row records a refutation	Part X ledger	none
16	T	S0	A geometric morphism whose inverse image fails to preserve a geometric formula	Chapter 2.3	none; proved
17	rejected	S0	Not applicable; vacuous in **Vect**	Part X ledger	none
18	D	S0	Symmetric erosion of geometric and non-geometric fragments under an adaptation that is a site morphism	Chapter 5.3	fragment-retention endpoint
19	D construction, T adjunction	S0	A `combine` operation realised without a left adjoint to the dispatch functor	Chapter 4.1	none; proved for the adjunction
20	T	S0	A nucleus sub-topos or sub-site shown to carry the required left adjoint	Chapter 4.1	none; proved

#	Grade	Licence	Falsification condition	Carrier	Endpoint
21	T relative; UNDETERMINED absolute	S0	Greedy output beating the submodular-cover bound relative to its bank, or an amalgamation map shown injective without the absolute check	Algorithm 3.2.4	E02, E03
22	T	S0	A matched-budget redundancy gap that varies with k on the declared family	A1, redundancy gap	A3 fixture
23	rejected	S0	Not applicable; the cochain is exact	Part X ledger	none
24	T	S0	A non-zero degree-two class on a graph or on a full-simplex nerve	Chapter 4.3	A3 fixture
25	T	S0	A gluing loss whose minimiser depends on the data at fixed architecture	Chapter 5.1 collapse theorem	E05, E06
26	T	S0	An identity sheaf expressed with heterogeneous stalk dimensions	A1, heterogeneous stalks	A3 fixture
27	T	S0	The Dirichlet identity failing at $\sigma > 0$ on the declared group nerve	Theorem 6.2.4	A3 fixture
28	rejected	S0	Not applicable; $\delta^0 R$ is a coboundary	Part X ledger	none
29	T	S0	A tractable token-level ratio exhibited for the parallel revisable decoder without marginalising the revision history	Definition 6.3.16, Definition 6.7.1	F2 stage
30	T	S0	A non-geometric assertion transported by a geometric morphism	Chapter 6.4	none; proved
31	rejected	S0	Not applicable; logic is constrained, behaviour is not	Part X ledger	none
32	D / UNDETERMINED	S0	Any conformance proof failing on its own stated hypotheses	Parts III–X specialisation statements	conformance harness
33	rejected	S0	Not applicable; no global result is proved	Part X ledger	none
34	X	S0	Not applicable; illustration only	Part X	none
35	withdrawn	S0	Not applicable; what survives is the exchangeable-family classification	Part VI, Part X	E13 with E14 gate

Three regularities in that table are worth naming because they are the ones a summary would lose.

The rejected rows have no falsification condition, and this is correct rather than convenient. A refuted claim is not a hypothesis awaiting a test; it is a recorded error with a proof or a counterexample attached. Writing a falsification condition for row 15 would suggest that some future measurement could restore "hallucination equals non-vanishing first cohomology," and

no measurement can, because the objection is that a manufactured difference is a coboundary and therefore represents zero. Reviving such a row requires a *new statement* about a differently constructed quantity, which enters the ledger as a new row with its own grade.

The rows touched by the one finite instance are licensed no higher than the rest. The instance is the worked example of Appendix A3, and the rows it touches are those whose content is arithmetic on a small complex: the codiagonal factorisation, the redundancy gap, the degree-two vanishing pattern, the Dirichlet identity, the heterogeneous-stalk inexpressibility, the holonomy invariant subspace, and the counting trap. Because that fixture is stated rather than discharged, it rules out the possibility that the statements are vacuous on every instance, which is more than nothing, and it licenses nothing beyond **S0**, which is less than a reader is most likely to want.

The endpoint column is sparse. Of thirty-five rows, eight name an endpoint in the pre-registered statistical plan; the remainder are proved, refuted, definitional, or graded **D** with no registered measurement. That ratio is itself a claim about the document: the programme is at present far more a piece of mathematics with a design attached than a piece of empirical science.

1.2.7 The three parts of the thesis, and their logical independence

The thesis was said above to have three parts that fail independently. Independence is a technical claim and is stated precisely here, because a reader who assumes the parts stand or fall together will misread every negative result in the document.

Proposition 1.2.4 (T) — pairwise logical independence of the three thesis parts. Let P_d be the diagnostic part, P_c the constructive part and P_s the specialisation part, as stated above. No two of the three imply the third, and no one of them implies any other.

Proof. It suffices to exhibit, for each ordered pair, a consistent state of affairs satisfying the antecedent and refuting the consequent. P_s without P_c: the specialisation statements are theorems about hypothesis classes and are already proved, while the constructive part could fail for cost — if site construction does not scale, every specialisation statement remains true and the architecture remains unbuildable. P_c without P_s: a site could be constructible and trainable while the module reductions failed, since a reduction is an identity on a forward map and a construction may be realised by a forward map that is not an extension of the standard one. P_c and P_s without P_d: both could hold while the five deficits had five unrelated causes, in which case the architecture works and the diagnosis that motivated it is wrong. P_d without P_c: the deficits could share a structural origin that no tractable construction repairs, which is the ordinary situation of a correct diagnosis without a cure. P_d without P_s and P_s without P_d follow from the two preceding constructions by deleting the irrelevant conjunct, since neither argument used it. □

The independence is not symmetry of evidence. Proposition 1.2.4 says the parts do not entail one another. It does not say they are equally supported. P_s has printed proofs and is the only part with any; P_d is a conjecture with a stated common-cause test that has not been run; P_c is contradicted in part by the trainability record of Table 1.3.4, in which four modules have no usable gradient. A reader who converts independence into "three equally open questions" has made the document sound better than it is.

Proposition 1.2.5 (T) — a rejected row cannot be revived by evidence alone. Let r be a row of the principal-claims table marked *rejected* on the ground that a printed argument refutes it. Then no empirical outcome licenses restoring r to a positive grade; the only admissible repair is a new row stating a different proposition.

Proof. Each rejection cited here is a proof or a counterexample: row 15 rests on the coboundary argument, row 17 on vacuity of left-exactness in **Vect**, row 23 on exactness of the stated cochain, row 28 on $\delta^0 R$ being a coboundary, row 31 on the fact that the geometric-morphism restriction constrains a logical fragment rather than behaviour, and row 33 on the absence of any composition of the local specialisations. In each case the negation of the row is a member of the target setting's proved

statements in the sense of Definition 1.3.2. An empirical observation cannot refute a proved statement without refuting its proof, and refuting a proof is a mathematical act, not a measurement. Hence a positive measurement bearing on r is evidence about some other proposition, which by definition enters as a new row. □

Proposition 1.2.5 governs release-row transitions. Even if abstention later correlates with a structural residual, row 15 stays rejected: support would attach to a *particular independently supplied cochain* and its harmonic magnitude, while row 15's quantity remains a classless coboundary.

1.2.9 Deliberate demotions

The licence assignment in Table 1.2.2 is an audit and is not generated. The `Licence` column was assigned by reading which checks this document reports as executed. There is no export that computes it from a registry. No row rises above **S0**, because promotion to **S1** would require an artefact at status `AXIOM_AUDITED` and this release has none. The `Endpoint` column still records where the worked instance touches a row; that record is an aid to later work and is not a licence.

The falsification conditions are stated at the level of the quantity, not of the threshold. Definition 1.2.2 demands a threshold symbol, and the table supplies one only where the statistical plan already registers an endpoint. The remaining rows name the quantity, the comparator and the direction and leave the threshold to the signing step, which means they are not yet admissible conditions by the definition's own standard. This is a deficiency of the present issue and is recorded rather than papered over by inventing numbers.

The licence census will decay. The proof of Proposition 1.2.3 depends on the current facts that no experiment has been run and that no artefact has been kernel-checked. The proposition is therefore true of this book and false of any issue in which a confirmatory run completes, and it must be re-derived rather than carried forward. A statement about the document's own state is the one kind of statement that cannot be inherited.

Independence is proved for the thesis parts as stated here, not for their informal readings. Proposition 1.2.4 quantifies over the three sentences printed in this chapter. Readers commonly substitute stronger readings — for instance, taking the constructive part to include trainability at scale — and under such substitutions the independence argument would have to be redone. Where a later chapter uses a stronger reading it must prove its own independence claim or cite one.

Chapter 1.3 — The method of transplantation

A great deal of unserious work has been produced by taking a theorem from pure mathematics, observing that some object in machine learning superficially resembles an object in the theorem, and concluding that the theorem applies. The resulting papers are recognisable: they contain correct mathematics and correct machine learning connected by a sentence that is neither.

This chapter sets out the discipline imposed to avoid that. It states the failure mode precisely, gives the three methodological moves — the relative viewpoint, the rising sea, and representability — that generate the architecture rather than decorating it, states the obligations every architectural module must discharge, and then publishes the **compliance matrix** recording, one row per module, which of those obligations are actually discharged and which are **OPEN** or **UNTRAINABLE**. It closes with the dependency map that shows where the fused Grothendieck programme is used, so that a reader reaches Part III already knowing which of its constructions would break if Part XI were deleted.

1.3.1 The failure mode, stated precisely

Let T be a theorem: *if X satisfies hypotheses $H_1, \dots, H_n$, then X has property P.* Let Y be a computational object. The transplantation is legitimate exactly when one has checked $H_1, \dots, H_n$ for Y. It is illegitimate — and it is the usual practice — when one has checked that Y *resembles* X and inferred P.

The reason this is so tempting is that the hypotheses of deep theorems are often the parts that look like technical bookkeeping. Tannakian reconstruction recovers a group from a category of representations, and the statement is beautiful; its hypotheses include rigidity, exactness, and the existence of a fibre functor to vector spaces over a field. A neural network's representation spaces satisfy none of these in any but a formal sense. One may still *design* with Tannakian reconstruction in mind — asking, for instance, what symmetry group a family of learned representations implicitly determines — and that is a perfectly respectable thing to do. What one may not do is claim the group is recovered.

Hence the grading scheme of the front matter. **T** requires the hypotheses to be checked. **D** is the accurate home for design guided by a theorem whose hypotheses fail. **X** is for illustration. The scheme is not modesty; it is a mechanism for keeping the document's own reasoning sound, because a **D** claim cannot be used as a premise in a **T** argument, and enforcing that rule mechanically catches errors that prose does not.

The rule that does the work. Metaphor-grade material may motivate a design but may never discharge a proof obligation. If a proof in this document depends on a step graded **D** or **X**, the proof is invalid and the result must be regraded. The rule is enforced by the dependency validator of Chapter 8.3, which rejects a `uses` edge from a record at grade **T** into any record at grade **M**, **D** or **X**, and rejects an edge at any grade into a record whose status is withdrawn or superseded. The rule was applied to earlier formulations, and it is why several rows of the claim ledger of Chapter 1.2 are marked *rejected* or *withdrawn*.

1.3.2 Six status words, and what each licenses

The grading scheme answers *how strong is this claim*. It does not answer *has this obligation been met*, and conflating the two is how a document acquires an architecture that reads as specified and cannot be built. Seven words are therefore used with fixed meanings throughout, four for grades and four for statuses, with **UNDETERMINED** doing double duty as the accurate output of an instrument and as the status of a question the specification cannot presently answer.

Table 1.3.1 — The status vocabulary and what each word licenses.

Word	Kind	What it licenses	What it forbids
T	grade	Use as a premise anywhere, subject to its stated hypotheses	Use outside those hypotheses
M	grade	Reporting the quantity, with its invariance class or budget	Any interpretation not separately established
D	grade	Motivating a construction; guiding a design choice	Appearing as a premise in a proof
X	grade	Illustration and exposition	Every argumentative use
OPEN	status	Recording the obligation and naming who must discharge it	Treating the module as specified; any downstream number that depends on the cell
UNTRAINABLE	status	Shadow, audit or frozen-external use of the component	Any empirical claim about learned behaviour of that component
UNDETERMINED	status	Abstention as a legitimate return value; continuing with the rest of the computation	Substituting zero, or falling back to a withdrawn procedure
WITHDRAWN	status	Citing the claim as history, with its refutation	Using it as a premise, plotting it, or gating on it

Two consequences of that table are worth making explicit because they are the ones most often violated in practice. First, a module whose forward pass is specified and whose backward map is not is **not specified**: trainability is an obligation of the contract, not an implementation detail, and this outranks mathematical elegance, since an architecture that cannot be trained cannot replace anything whatever its categorical merits. Second, **UNDETERMINED** is an output, not a failure: an instrument that abstains because its precondition is unmet is behaving correctly, and the error is to replace that abstention with a number.

1.3.3 The three moves

Three ideas from Grothendieck's practice generate most of what follows. They are methodological rather than technical, and they are the reason the architecture has the shape it does. Each is supplied formally by a named chapter of Part XI; this section states the computational reading and the constraint it imposes.

1.3.3.1 The relative viewpoint

Grothendieck's first move was to stop studying objects and start studying *morphisms*. An object X becomes a morphism $X \to S$ over a base S; properties of X become properties of the morphism; and constructions become base change. The payoff is that everything becomes parametrised, and parametrised constructions compose.

The computational reading: do not ask what a representation *is*; ask what it is a representation *over*. A token embedding is not a point in $\mathbb{R}^d$; it is a section of a sheaf over a region of a

context site. This is not a rewording, because it immediately makes sense of operations that are otherwise unmotivated — restriction to a subregion, comparison on an overlap, transport along a path — and it makes nonsense of operations that ignore the base, which is a useful thing for a design principle to do.

Concretely, the relative viewpoint is what forces stalks to have *heterogeneous* dimension. If $\mathcal{F}(U)$ is the space of sections over U, there is no reason for $\dim \mathcal{F}(U)$ to be independent of U; a larger region should generally support more information. Once dimensions differ, the identity map between stalks does not exist, and — as Part V shows — that single consequence makes collapse to the identity sheaf inexpressible rather than undesirable. It does not make collapse impossible: the zero map remains available and is excluded separately, by a constraint rather than by a penalty. A methodological principle that produces a concrete architectural constraint against one known empirical failure, while leaving a second failure to be handled by other means, is about as much as one can ask of a methodological principle, and overstating it would be the first mistake this chapter exists to prevent.

The formal apparatus is Chapter 11.1: the relative form of a construction, the fibration of the section category over the site category, and the triage that decides when base change is free.

1.3.3.2 The rising sea

The second move is the one Grothendieck described as letting the sea rise until the problem dissolves: rather than attacking a hard problem directly, build up the general theory around it until the problem becomes a special case of something easy.

The computational reading is a layering discipline. Do not build a mechanism that solves the observed failure; build the structure in which the failure has a name, and let the check be the computation of an invariant rather than a heuristic. Concretely: rather than adding a consistency-checking module that detects contradictions, construct the computation so that inconsistency is a non-vanishing class, and then diagnosing it is a computation.

This move has a characteristic failure, and it has already occurred once in this programme. The sea rises, the problem does not dissolve, and one is left with an elaborate structure and the original difficulty — or worse, with an invariant that is identically zero for structural reasons and therefore measures nothing. That is exactly what happened to the first obstruction head, whose disagreement term was a coboundary by construction, and to the first gerbe construction, whose class was always exact. Both are withdrawn. The guard is not optimism but the non-vacuity obligation: any invariant this document reports must be accompanied by a proof that it is not identically zero on the configuration where it is being reported, and by the falsification gate that names the measurement which would show the structure is buying nothing.

1.3.3.3 Representability

The third move is to define an object by the functor it represents. Rather than constructing a thing and then proving it has a property, one specifies the universal property and asks whether it is representable; if it is, the object exists and is unique up to unique isomorphism.

The computational reading is that a module should be specified by what it must satisfy, not by how it is implemented, and that the specification should be tight enough to determine the implementation up to isomorphism. When it is, one gets a derivation instead of a design choice. The dispatch–combine pair of Topos-MoE is the clearest instance: once one requires that experts form a covering family and that `combine` be left adjoint to `dispatch`, the two rival formalisations are eliminated, because a nucleus sub-topos and a sub-site do not

supply the left adjoint Σ_U that `combine` would have to be. That elimination is a genuine argument and is graded **T**.

What follows from it is narrower than earlier formulations claimed. The production gather/scatter implementation is *motivated* by the adjunction; it is not that adjunction, and it does not become so until the categorical data and the tensor semantics are constructed and shown to agree. The implementation is graded **D**, the adjunction **T**, and the bridge between them is **OPEN**. Chapter 11.2 supplies the functor-of-points apparatus that makes the distinction precise, and Chapter 4.1 marks the boundary where it is used.

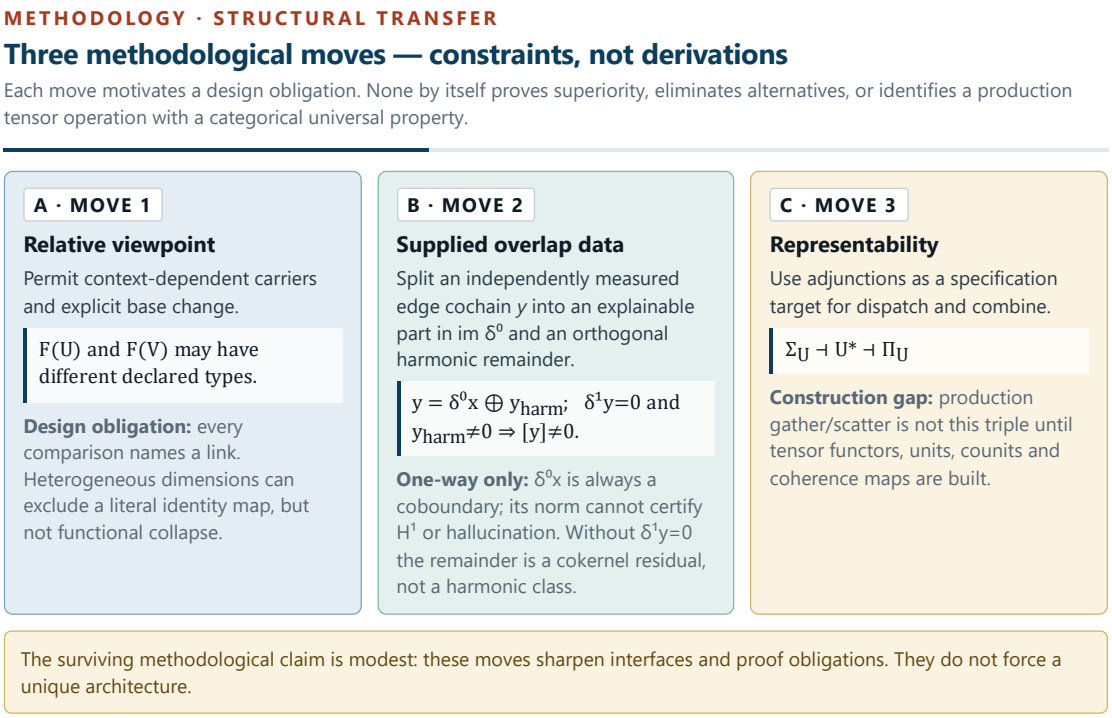

Figure 1.3.1. Three methodological transfers.

1.3.4 The module contract

Every architectural module in Parts III through VIII is *required* to discharge the following nine obligations. The requirement is normative; whether it has been met is a separate, recorded fact, and the matrix in the next section states which cells are discharged. A universal assertion that every module already satisfies the contract would be false, and this chapter does not make one.

1. Definition (`forward_type`). Formal, with types: what the module consumes, what it produces, what data it carries, and on which axes. No appeal to intuition at this stage.

2. Mathematical status. The grade — **T**, **M**, **D** or **X** — with a sentence justifying it. If **T**, the hypotheses being checked. If **D**, the theorem that inspired it and the hypothesis that fails.

3. Theorem. The precise statement of what is claimed, with all hypotheses.

4. Proof. Given in full, or given as a sketch with an explicit statement of what the sketch omits and where the full argument appears.

5. Algorithm and backward map (`backward_definition`). Numbered steps with real tensor shapes, and the gradient that flows through them: whether it is exact, implicit-function-theorem-based, or a reverse pass through a truncated program, and if it is a

surrogate, the bias budget. A module with a forward pass and no backward map is marked **UNTRAINABLE**, not silently accepted.

6. Complexity and hardware (`cost, communication`). Time, memory and communication against the single reference configuration, with constants where they matter, and the arithmetic intensity where the operation is a candidate for a custom kernel.

7. Specialisation hypotheses (`specialisation_hypotheses`). The exact hypotheses under which the module reduces one-way to its standard counterpart, on the same objective and the same forward map, *proved*.

8. Observables (`invariance_or_budget`). What telemetry the module emits, at what sampling rate and overhead, and — binding, without exception — the invariance class of every reported quantity or its explicit budget.

9. Conformance test and falsification gate (`conformance_test, empirical_gate`). An executable test that an implementation either passes or fails, and the measurement whose outcome would require the module to be abandoned, with its threshold stated in advance.

Why the ninth element. A specification in which nothing can fail is a specification in which nothing is claimed. Pre-committing to a falsification threshold before the measurement exists is the only reliable defence against the retrospective reinterpretation of a disappointing result as a partial success. The gates are collected in Part IX; each is stated with its baseline, its statistic, and the effect size that counts as confirmation and as refutation.

1.3.5 The compliance matrix

The matrix below is the register that replaces the universal assertion. It has one row per module and one cell per obligation, and a cell that is not discharged by a named authority in the current sources reads **OPEN**. A module may be called *specified* only when every required cell is discharged and every not-applicable cell carries a validated reason.

1.3.5.1 Rendering contract

The matrix is a **rendering of registry state, not an independent claim authority**. Its schema is the `ModuleCompliance` record of Chapter 8.3, whose fields are exactly the obligation names introduced above, together with `module_id` and `evidence_ids`; its statuses are that chapter's `ComplianceStatus` values; and its dependency admissibility is decided by that chapter's executable `check_dependencies`. Nothing in this chapter overrides a registry status, defines a new grade, or resolves a conflict between two normative definitions; where this chapter and the registry disagree the registry governs and the disagreement is a defect to be reported.

Column	Resolved from	Rule

Table 1.3.2 — Rendering contract: where each column of the matrix is resolved from.

Column	Resolved from	Rule
`module_id`	The registry key	Stable across issues; never a source filename
Normative statement	The single named definition, theorem or algorithm that defines the module	One per module; every other occurrence cites it
Operator ID	The named executable, where one is deployed	Absent where no executable is deployed; never inferred from prose
Tensor ABI	The declared axes of the module's inputs and outputs	Structural state and value state are typed separately; the only bridge is a registered pooling map and its transpose
Backward authority	The `BackwardWitness` row of Table 10.1.9, or the owning chapter's own backward table where it has one, such as Table 3.7.3	Cited, never restated in different terms
Nine obligation cells	`ModuleCompliance` fields	`DISCHARGED`, `OPEN`, or `NOT_APPLICABLE` with a reason record
Evidence	`evidence_ids`	A claim identifier or a named statement; a chapter number alone is not evidence

Generation status: BLOCKED. The intended production of this matrix is mechanical: the registry exports its `ModuleCompliance` records against the frozen manifest, and this table is generated from that export so that a status can change here only by changing the registry. That export does not yet exist. Chapter 8.3 defines the schema and the validator but has not yet published stable identifiers for every principal claim and normative executable, so no generated artifact is available to render. The table below is therefore a **hand-transcribed provisional rendering**: it was compiled by reading the named normative statements and the central trainability register, and every cell not discharged by such a source is printed **OPEN** rather than estimated. It may not be cited as evidence that an obligation is discharged, and it must be regenerated and re-verified before release. Until it is generated, the correct reading of this section is that module compliance is *recorded but not yet mechanically enforced*.

1.3.5.2 Module authorities

Table 1.3.3 — Module authority index: normative statement, deployed operator, and tensor ABI.

Module	Normative statement	Operator ID	Tensor ABI
`L0-SITE`	Algorithm 3.2.4, relative max-coverage greedy	none deployed	site, nerve and cover; sequence-level structure, no token axis
`L1-SHEAF`	Proposition 3.3.3 and Algorithm 3.3.1; parameters typed by Definition 5.1.29	none named	restrictions $\rho \in \mathbb{R}^{d_e \times d_i}$, $\rho\rho^\top = I_{d_e}$; edge-indexed
`L2-ATTN`	Algorithm 3.3.4; specialisation at Theorem 3.3.31	none named	token-indexed values; overlap transport on nerve edges
`L3-NUCLEUS`	Theorem 3.4.7	none named	coordinatewise on each stalk
`L4-SOLVE`	Definition 5.1.35 and Algorithm 3.4.2	`L4-PCG6-RM-v1`	pooled `[sequence, chart, d]` state; registered pool and transpose are the only bridge
`L5-EDGE`	Definition 3.5.6 with an `IndependentEdgeDatum`	none deployed	edge-indexed 1-cochain with provenance record
`L6-LOGIC`	Theorem 3.5.31	none named	support sieves over the site

Module	Normative statement	Operator ID	Tensor ABI
L7-HOL	Theorem 3.6.9 with the ternary outcome	none deployed	reads restrictions and nerve cycles; no parameters
L8-TRANSFER	Theorem 3.6.41 and Definition 5.3.4	none named	site morphism π and induced base change
MOE-ROUTE	Algorithm 4.2.1, executable token router with a detached coverage shadow	TOPK-LINEAR-v1	token-level linear scores; coverage audit is sequence-level and detached
MOE-COMBINE	Definition 4.1.11 and Algorithm 8.3.3	none named	two-channel expert output; shapes fixed by Table 8.3.1
MOE-GERBE	Definition 4.3.12, links failing to compose	none deployed	sparsified $\geq$ 2-dimensional nerve; vacuous below $\partial\Delta^3$
T-UNIT	Definition 3.7.1	tunit.forward.v1	typed record over a base; token-local and sequence-level
ALIEN-ENGINE	Definitions 3.7.11 to 3.7.16	alien.transport.v1	comparison link, coricity residual, indeterminacy vector
COUNTER-EXPERT	Algorithm 4.1.1, the declared counter_support gate	none named	separate declared gate; not a Gödel pseudocomplement
DGSPO	Definition 6.3.16 with Definition 6.7.1	none named	joint route-and-written-action likelihood per round
SELF-EVO-GATE	Algorithm 6.4.1 with the normative gate conditions of Section 63-GC	none deployed	sandboxed proposal, ledgered side effects, global budget

1.3.5.3 Obligation status

Table 1.3.4 — Compliance matrix: one row per module, one cell per obligation.

Module	Fwd	Bwd	Obj	Cost	Comm	Inv	Spec	Conf	Gate
L0-SITE	OK	UNTRAINABLE	OPEN	OPEN	OPEN	OPEN	OK	OPEN	OPEN
L1-SHEAF	OK	OK	OK	OPEN	OPEN	OPEN	OK	OPEN	OPEN
L2-ATTN	OK	OK	OK	OK	OPEN	OPEN	OK	OPEN	OPEN
L3-NUCLEUS	OK	OK	OK	OK	n/a	OPEN	OK	OPEN	OPEN
L4-SOLVE	OK	OK	OK	OK	OK	OK	OK	OPEN	OPEN
L5-EDGE	UNDETERMINED	n/a	n/a	OPEN	OPEN	OK	OK	OPEN	OPEN
L6-LOGIC	OK	UNTRAINABLE	OPEN	OPEN	OPEN	OPEN	OK	OPEN	OPEN
L7-HOL	UNDETERMINED	n/a	n/a	OPEN	n/a	OK	OK	OPEN	OPEN
L8-TRANSFER	OK	UNTRAINABLE	OPEN	OPEN	OPEN	OPEN	OK	OPEN	OPEN
MOE-ROUTE	OK	OK for top-*k*; UNTRAINABLE for coverage	OK	OK	OPEN	OPEN	OK	OPEN	OPEN
MOE-COMBINE	OK	OK	OK	OPEN	OPEN	OPEN	OK	OPEN	OPEN
MOE-GERBE	UNDETERMINED	n/a	n/a	OPEN	OPEN	OPEN	OK	OPEN	OPEN
T-UNIT	OK	OK; read-out channels carry no gradient by construction	OK	OK	OK	OK	OK	OK	OPEN
ALIEN-ENGINE	OK	PARTIAL; two rows OPEN	OK	OPEN	OK	OK	OK	OK	OPEN

Module	Fwd	Bwd	Obj	Cost	Comm	Inv	Spec	Conf	Gate
COUNTER-EXPERT	OK	OK	OK	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN
DGSPO	OK	OK	OK	OPEN	OPEN	OK	OPEN	OPEN	OPEN
SELF-EVO-GATE	OK	UNTRAINABLE	n/a	OPEN	OPEN	OPEN	n/a	OPEN	OPEN

Note. Columns are the `ModuleCompliance` fields in the order of the contract above: forward type, backward definition, objective, cost, communication, invariance or budget, specialisation hypotheses, conformance test, empirical gate. `OK` abbreviates `DISCHARGED` by the authority named in Table 1.3.3, in Table 10.1.9, or in the owning chapter's own declaration; `n/a` abbreviates `NOT_APPLICABLE` and requires a reason record. `OPEN` means *not discharged by a named authority in the current sources*; it is not a claim that the obligation cannot be met. The `T-UNIT` and `ALIEN-ENGINE` rows are transcribed from Table 3.7.9, which is their owning declaration. Every `Conf` and `Gate` cell that is not marked `OK` is `OPEN` because no conformance harness has been executed against a frozen manifest and no experimental threshold has been signed; the two `OK` conformance cells record a specified test, not an executed one.

1.3.5.4 Gradient status and blocking evidence

Table 1.3.5 — Backward authority and the item blocking each undischarged gradient.

Module	Backward authority	Resolved gradient status	Blocking item
L0-SITE	Table 10.1.9, context-site row	absent	No differentiable construction; admissible only as a frozen external compiler
L1-SHEAF	Table 10.1.9, restrictions row	manifold gradient, polar or Cayley retraction	Retraction cost not charged; QR excluded
L2-ATTN	Table 10.1.9, descent-attention row	dense exact	Saved-state, backward-memory and communication contract
L3-NUCLEUS	Table 10.1.9, nucleus row	subgradient with a declared tie rule	none
L4-SOLVE	Table 10.1.9, training-solve row	exact reverse through the six realised steps and masks	none for the executed program; distance to the exact shifted solve is recorded as a budget
L5-EDGE	Table 10.1.9, obstruction row	no gradient by design	No certified `IndependentEdgeDatum` source exists
L6-LOGIC	Table 10.1.9, logic-head row	score-function estimator	No baseline, variance or sample-complexity statement
L7-HOL	Table 10.1.9, holonomy row	no parameters	Certification returns **UNDETERMINED**; the real-spherical procedure is withdrawn
L8-TRANSFER	Table 10.1.9, site-morphism row	gradient stopped through the site morphism	Cost and certified semantic map
MOE-ROUTE	Table 10.1.9, coverage-router row	exact through selected gate values for top-k; surrogate for coverage	Coverage path has no custom VJP and no bias budget
MOE-COMBINE	Chapter 8.3 backward contract	dense exact	Cost and communication under the corrected two-channel path
MOE-GERBE	audit only	none	Non-vacuity requires an engineered $\geq$ 2-dimensional nerve
T-UNIT	Table 3.7.3	declared absent on the truth, witness and behaviour channels	none; the absence is by construction and is declared, not undischarged

Module	Backward authority	Resolved gradient status	Blocking item
ALIEN-ENGINE	Table 3.7.3	Danskin subgradient; reverse of realised capped iterations	Two rows OPEN; discharged only when the manifest fixes the iteration cap
COUNTER-EXPERT	Table 10.1.9, counter-head row	dense exact	Counter-head arithmetic omitted from the published cost table
DGSPO	Table 10.1.9, update row	exact reverse of the clipped surrogate, advantages detached	Rollout, verifier, audit and rejected-compute accounting
SELF-EVO-GATE	Table 10.1.9, proposer row	absent	No proposer-cost bound; sandbox and shadow use only

Read together, the three tables say something the prose of an architecture chapter cannot say compactly. One layer — the descent solve — has every structural obligation discharged and fails only the two release cells that no module can pass before a frozen manifest exists. Four modules have no usable gradient at all: the context site, the logic head, the transfer layer and the self-evolution proposer. A fifth, the coverage router, has one only for the ordinary top-k substitute, which is not the object from which the mixture-of-experts construction takes its meaning. Three modules abstain by design in their current configuration, returning **UNDETERMINED** rather than a number. And the empirical-gate column is empty throughout, which is the strongest single statement in this chapter: nothing here has yet been tested against a signed threshold.

1.3.6 The regression obligation

The specialisation obligation deserves separate treatment because it is easy to satisfy trivially and easy to overinterpret.

Satisfied trivially: any module with enough parameters can be made to *approximate* its predecessor, and a claim of the form "with suitable weights this reduces to attention" is nearly vacuous. What is required is stronger — an *exact* reduction under stated structural hypotheses, proved as an identity on the same objective and the same forward map, not an approximation argument, and not a proof that quietly changes the objective or the router half way through. Changing the algorithm being recovered is a change of algorithm, not a degeneration.

Overinterpreted, and this is the more damaging error: an exact reduction establishes containment of one *hypothesis class* by another and nothing about optimisation. It does not show that training will find the standard solution, that the larger class is as easy to optimise, or that the extra parameters are harmless. Nor do the individual reductions compose. Each is proved under its own hypotheses, and those hypotheses are mutually incompatible — they demand different sites, routers, objectives, shape conventions, residual orders and solver semantics — so there is no forward map on which they hold simultaneously, and the end-to-end statement is **UNDETERMINED**. No performance floor of any kind follows, and the vocabulary of containment, subsumption and replacement is not available to this document as an achieved result. Replacement remains a falsifiable research target with a stated discharge condition, and Part XI works through what would be required.

There is a third thing the specialisation condition does, which is less obvious and quite practical: it defines the *ablation*. If a module reduces to the standard component under stated hypotheses, then those hypotheses are the control condition for every experiment involving the module, and it is a control that shares the codebase, the initialisation and the data pipeline. Ablations built this way are more trustworthy than ablations built against a

separately implemented baseline. They are not, however, a substitute for a matched external baseline, because a control that shares an implementation also shares its bugs.

1.3.7 Where the generative principles is used

The final obligation of this chapter is to make the document's dependency on Part XI visible before Part III begins. The fusion commissioned here is not an added Part but a re-derivation: Part XII states the generative principles and Parts II through X are written to embody them, and the test is that a reader who deletes Part XI should be left with a broken document rather than a shorter one. A principle that changes no definition, theorem, algorithm or contract has failed that test and is cut or demoted to analogy; length does not rescue it.

Table 1.3.6 — Fusion principles and the module obligations they change.

Principle	Supplier	Obligation it changes	Consequence if the principle is deleted
M1 relative point of view	Chapter 11.1	`forward_type`: every layer signature carries a base argument and declares whether it is cartesian	Base change, site versioning and expert-local execution become three unrelated mechanisms; the transfer definition of Chapter 5.3 loses its object
M2 value versus structure	Chapter 11.3	`forward_type` and `backward_definition`: structure-carrying data selects an optimiser and may not receive a value-path gradient	The router's route variable is typed as a value, and the gradient passes through a decision as though it were a number
M3 explicit links	Chapter 11.3	`objective` and `specialisation_hypotheses`: every comparison of two locally defined objects names its link	`combine` compares sections across experts with no declared identification; the gerbe obstruction loses its definition as failure of links to compose
M4 indeterminacy budgets	Chapters 11.2, 11.9, 11.11	`invariance_or_budget`: no quantity is reported without proved invariance or a stated budget	Gauge-dependent quantities are plotted and compared across models; several gates become unfalsifiable
M5 species and mutations	Chapter 11.2	`conformance_test`: conformance is invariance under a declared class of implementation change	Bit-comparability becomes a property of one build; the conformance harness has no object to test
M6 dévissage	Chapter 11.1	`specialisation_hypotheses`: each reduction names exactly what structure it strips	The specialisation table becomes a list of coincidences with no accounting of what was removed

Two disciplines govern that material and are applied without exception. Every item is placed in one tier: **Tier G**, transferred mathematics whose hypotheses are verified here and which is used as a theorem; **Tier S**, a pattern of reasoning imported as a design principle, which must state in the same paragraph that no theorem is being borrowed; **Tier A**, analogy, which may motivate and may never justify. And every construction of this kind must be restated in ordinary mathematical language with the imported vocabulary removed: if it survives and is still non-trivial, it is kept and graded on its own merits; if it evaporates, it was decoration and is cut rather than softened.

1.3.8 What is done with the deep theorems that cannot be used

Several results that motivated this programme cannot be transplanted, and it is worth being explicit about what is done with them instead, since deleting them would misrepresent how the architecture was arrived at.

The section conjecture motivates the obstruction head: the idea that the existence of a global object is controlled by a group-theoretic condition, and that its non-existence is detectable. What is taken is the *shape* of the idea — that failure to glue should be certified by an invariant rather than diagnosed by a heuristic — implemented with Čech cohomology, which can actually be computed. The conjecture proves nothing here, is cited as motivation, and is graded **X**. It is also worth recording that the shape alone was not enough: the first implementation of that idea measured a coboundary and had to be withdrawn.

Tannakian reconstruction motivates the holonomy layer: the idea that a symmetry group is recoverable from the category of its representations. What survives is the design question — what group do the learned transport maps generate? — and that question has an answer that can be computed, the holonomy group. The reconstruction theorem is not used; the group is defined directly. Graded **D**.

Inter-universal Teichmüller theory motivates the transfer layer and the treatment of experts, through the idea of relating structures that live in different universes by controlled comparison rather than by identification. Its status must be stated accurately and never softened: the papers were published in *PRIMS* in 2021, Scholze and Stix published a specific objection to the proof of Corollary 3.12, the author disputes it, and there is no consensus in the number-theory community. Nothing whatever is transplanted, and nothing in this specification depends on whether that theory is correct. What is taken is an operational skeleton — that comparison across incompatible contexts should be mediated by an explicit correspondence with recorded distortion — together with the observation that three named indeterminacies correspond to the gauge, basepoint and cover-refinement ambiguities this document carries independently. The skeleton is Tier S and grade **D**; the theory itself is Tier A and grade **X**; and the phrases "IUT proves" and "by the theta-link" are not available to any chapter.

Motivic cohomology motivates the layered invariant structure: the idea of a universal cohomology theory through which others factor. It is used as an organising picture for why H^0, H^1 and H^2 carry different kinds of information. Nothing is transplanted. Graded **X**.

This list is included because the alternative — silently removing the inspirations once the mathematics is written up — would make the document look more rigorous and be less accurate about its own genesis. The reader is entitled to know which parts of the architecture were arrived at by proof and which by analogy, and the grading scheme is precisely the mechanism for telling them apart.

1.3.9 The transplantation type check

The rule stated at the head of this chapter — check the hypotheses — is correct and, taken alone, unusable. It presupposes that one already knows which object of the target setting is being offered as the source theorem's X, and in practice that identification is exactly where the error lives. A hypothesis cannot be checked before the thing it is a hypothesis about has been named, and the sentence "the context of a language model is a site" names nothing until someone says which category, which covering families, and which pullback stability. The discipline below therefore runs in two stages: first a *type check*, which decides whether the identification is even expressible, and only then a hypothesis check.

The type check is the reason this chapter can say something sharper than "be careful." It has a definite output on every attempted transfer, and on several of the transfers that motivated this programme its output is a named sort that has no image in this book at all. That is a more useful report than "the analogy is loose," because it says which object would have to be constructed before the question of hypotheses could be reached.

Definition 1.3.1 (T, definitional) — source setting. A *source setting* is a triple $\mathsf{S} = (\Sigma_{\mathsf{S}}, H_{\mathsf{S}}, C_{\mathsf{S}})$ in which Σ_{S} is a finite list of sort names with their intended interpretations, H_{S} is a finite conjunction of hypotheses whose free sort names all lie in Σ_{S}, and C_{S} is a conclusion whose free sort names all lie in Σ_{S}. The setting is *closed* when every notion appearing in H_{S} or C_{S} is either a sort name of Σ_{S} or is defined from those sort names by the source's own definitions.

Definition 1.3.2 (T, definitional) — target setting. The *target setting* is the pair $\mathsf{B} = (\Sigma_{\mathsf{B}}, A_{\mathsf{B}})$ in which Σ_{B} lists the sorts this document actually constructs — the context site and its covering families, the sheaf of sections with its restriction maps, the expert nerve, the cochain complexes in degrees zero to two, the transport groupoid, the site morphism used by the transfer layer, the round-action trajectory of the decoder, and the typed runtime records of Chapter 8.3 — and A_{B} is the set of statements about those sorts that this document proves at grade **T**.

Definition 1.3.3 (T, definitional) — transplantation typing. A *transplantation typing* from S to B is a partial function $\tau : \Sigma_{\mathsf{S}} \rightharpoonup \Sigma_{\mathsf{B}}$ together with, for each $s \in \operatorname{dom}\tau$, a *coercion witness*: a statement of A_{B} asserting that $\tau(s)$ belongs to the class that S requires of s. The sort names of $\Sigma_{\mathsf{S}} \setminus \operatorname{dom}\tau$ are the *untyped residue* of the transplantation, written $\operatorname{res}(\tau)$.

A coercion witness is a proof obligation, not a declaration of intent. "Let the context lattice play the role of the site" is not a witness; Theorem 2.1.6, which establishes that join coverage on the down-set frame satisfies the three Grothendieck axioms, is one. The asymmetry is the whole point: a typing whose witnesses are all present is cheap to audit, and a typing whose witnesses are absent announces itself immediately as an assertion rather than an argument.

Definition 1.3.4 (T, definitional) — discharge status. Relative to a typing τ, each hypothesis $h \in H_{\mathsf{S}}$ receives exactly one status. It is **DISCHARGED** when the translate $\tau(h)$ is a member of A_{B}. It is **FAILED** when the negation of $\tau(h)$ is a member of A_{B}. It is **UNTYPED** when some sort name occurring in h lies in $\operatorname{res}(\tau)$, so that $\tau(h)$ is not a well-formed statement of the target setting. It is **UNDETERMINED** in every remaining case: $\tau(h)$ is well formed and neither it nor its negation is proved here.

Proposition 1.3.5 (T) — the transfer trichotomy. Let τ be a typing of S into B and suppose C_{S} contains no sort name of $\operatorname{res}(\tau)$.

1. If every $h \in H_{\mathsf{S}}$ is **DISCHARGED**, then $\tau(C_{\mathsf{S}})$ holds in the target setting relative to the source's own proof of C_{S}, and the resulting declaration is capped at grade **M** by the no-T-from-import rule unless this document also prints a proof of C_{S} from H_{S}.
2. If some $h \in H_{\mathsf{S}}$ is **FAILED**, then $\tau(C_{\mathsf{S}})$ has no support from S at any grade; what may still transfer is the *method* of the source proof, at grade **D**, carrying no conclusion with it.
3. Otherwise at least one hypothesis is **UNTYPED** or **UNDETERMINED** and none is **FAILED**, and the transplantation status is **UNDETERMINED**, with the offending hypothesis and, where applicable, the offending untyped sort naming the work order that would resolve it.

Proof. Clause 1 is the substitution property of the source's proof: a derivation of C_{S} from H_{S} in the source signature is a derivation of $\tau(C_{\mathsf{S}})$ from $\tau(H_{\mathsf{S}})$ in the target signature, because τ is a translation of signatures and the derivation uses no axiom outside H_{S} and the source definitions, each of which is transported by τ with a coercion witness. The grade cap is Axiom A9.11 applied to the transported derivation: removing the source's own proof leaves the target statement unproved here. Clause 2 is the observation that a derivation of a conclusion from a hypothesis conveys nothing about the conclusion once the hypothesis is refuted; formally, $H \to C$ together with $\neg H$ leaves C undecided, and A_{B} contains $\neg\tau(h)$ by the definition of **FAILED**. Clause 3 is the complement of the first two, and the work order is well defined because Definition 1.3.4 assigns statuses to a finite list. □

Proposition 1.3.6 (T) — a method transfers where a theorem does not, and arrives without a grade. Let S have a **FAILED** hypothesis under τ and let P be the proof of C_{S}, regarded as a finite sequence of construction and inference steps. Then the image sequence $\tau(P)$, formed by applying τ

stepwise and deleting every step that uses the failed hypothesis, is a well-formed construction in the target setting whenever every retained step is typed; but its terminal statement is in general not $\tau(C_S)$, and it carries no grade until this document proves it.

Proof. Each retained step is either a definition, an instance of a target axiom, or an inference whose premises are earlier retained steps; by induction the sequence is well formed. That the terminal statement need not be $\tau(C_S)$ is immediate, since the deleted steps were the ones consuming the failed hypothesis and the source proof was, by assumption, a proof that used it. Grading is by the definitions of the front matter: a construction is graded by the proof this document supplies for it, and $\tau(P)$ supplies none. □

Proposition 1.3.6 is the formal content of the slogan that this chapter transfers *methods rather than theorems*. It is worth reading twice, because it is weaker than the slogan sounds. What transfers is a sequence of moves that is guaranteed only to be well formed; it is not guaranteed to end anywhere interesting, and the interesting endpoint, when there is one, is a new statement with new proof obligations. Every grade-**D** entry in the tables above is in that position.

Table 1.3.7 — Type check of the seven principal transplantations. *Sort assigned* names the element of Σ_B chosen by τ; *residue* names the source sort with no image.

Source setting	Sort assigned by τ	Coercion witness	Untyped residue	Worst hypothesis status	Verdict
Grothendieck topology on a small category	context site: down-set frame with join coverage	Theorem 2.1.6	none	DISCHARGED	clause 1; grade capped at M for the imported morphism-of-sites step
Cellular sheaf cohomology in degrees 0 and 1	sheaf of sections on the expert nerve	Proposition 3.3.3, Theorem 3.6.9	none	DISCHARGED	clause 1 for H^0; clause 3 for the certificate side of the dichotomy
Almost mathematics over a valuation ring	almost-isomorphism calculus on restriction maps	the idempotent-ideal construction of Chapter 2.7	the valuation ring and its rank-one non-discrete valuation	UNTYPED	clause 3
Weight filtration on a mixed object	weight-graded layer of Chapter 3.9	the declared grading of the layer	the algebraic variety and its Frobenius action	UNTYPED	clause 3
Tannakian reconstruction	holonomy group of the transport groupoid	Theorem 3.6.4	the neutral fibre functor to finite-dimensional vector spaces over a field	UNTYPED	clause 3; group defined directly instead
Anabelian section conjecture	obstruction cochain of Chapter 3.5	none supplied	the arithmetic fundamental group and its outer Galois action	UNTYPED	clause 3, and the first implementation additionally FAILED
Inter-universal comparison of distinct universes	gauge, basepoint and refinement indeterminacies	Definitions 11.9.2 to 11.9.4	the theatre, the theta-link, and everything they are built from	UNTYPED	clause 3; nothing is imported

Four observations follow from that table and are the reason it is printed rather than summarised.

The successful rows are the shallow ones. The two transplantations that reach clause 1 are the ones whose source hypotheses are about finite combinatorial data: a category with pullbacks and a coverage, and a finite cell complex with linear restriction maps. Nothing about a valuation, a Frobenius, a fibre functor or a fundamental group appears in either. This is not a coincidence and it is not modesty; it is the mechanical consequence of Definition 1.3.3, since a coercion witness must be a proved statement of this document and this document proves statements about finite combinatorial data.

The failures are failures of sorts, not of hypotheses. In five of the seven rows the block occurs before any hypothesis is examined, because a sort name of the source has no image in Σ_B at all. One cannot ask whether the fibre functor is exact when there is no fibre functor; the accurate report is **UNTYPED**, and the work order is "construct a fibre functor, or stop invoking Tannakian reconstruction as more than a design analogy."

An untyped residue is not automatically fatal to the design. Rows three and four are the cases where the residue is small and the surviving structure is still doing work: the almost-isomorphism calculus is defined here directly on restriction maps and its own statements are proved here, and the weight-graded layer is a declared grading whose properties are established without reference to any variety. In both, what has happened is exactly Proposition 1.3.6 — the method is retained, the theorem is not, and the resulting statements are graded on the proofs printed here.

One row is worse than untyped. The section-conjecture row carried, in a first formulation, an implementation whose reported quantity was a difference of per-vertex values. That is a coboundary by construction, hence cohomologically zero, so the translated hypothesis "the reported cochain is not a coboundary" is refuted in the target setting and the row is **FAILED** in the sense of Definition 1.3.4. It is the only row in the table where the type check would have been passed by a careless auditor and the hypothesis check still catches the defect, which is a useful reminder that the type check is a filter and not a substitute.

1.3.10 The evaporation test

The type check decides whether an import is expressible. It does not decide whether the imported vocabulary is *doing* anything, and that is a separate and more embarrassing failure mode: a construction that is perfectly well typed, whose hypotheses are all discharged, and which would survive the deletion of every word borrowed from the source. Such a construction is decoration. The test below is the mechanical form of the discipline stated in the fusion section above.

Definition 1.3.7 (T, definitional) — the evaporation test. Let K be a construction of this document and let V be the set of imported terms occurring in its statement. Write $K \setminus V$ for the *unfolded restatement*: every occurrence of a term of V is replaced by the definition this document gives for it, and every sentence whose only content is an appeal to a source is deleted. The construction K *passes the evaporation test* when $K \setminus V$ satisfies three conditions:

1. **Well-typedness.** $K \setminus V$ is a well-formed statement about sorts of Σ_B.
2. **Non-vacuity.** There is at least one configuration reachable in the reference configuration of Chapter 3.8 on which the conclusion of $K \setminus V$ is not satisfied identically for structural reasons.
3. **Consequence.** $K \setminus V$ still implies at least one numbered declaration for which K is cited elsewhere in this document.

Definition 1.3.8 (T, definitional) — evaporation verdicts. K is **LOAD-BEARING** if it passes all three clauses; **DECORATIVE** if it satisfies clause 1 and fails clause 3, so that deleting the import removes nothing that any declaration uses; and **VACUOUS** if it fails clause 2, so that the quantity it licenses is identically zero or identically true on the configuration where it is reported. A construction failing clause 1 is not graded: it is not yet a statement.

Algorithm 1.3.1 (M) — executing the evaporation test on one construction.
Input. A construction K, its imported term set V, and the list of declarations citing K.

1. Unfold every term of V using the definition given in this document; if any term has no such definition, halt and report **UNTYPED**, naming the term.
2. Delete every sentence of K whose predicate is an appeal to source authority rather than to a sort of Σ_B.
3. Type-check the residue against Σ_B; on failure halt and report *not a statement*, naming the offending sort.
4. Instantiate the residue on the reference configuration and on the minimal non-degenerate configuration recorded for its degree — the graph for degree one, the boundary of the three-simplex for degree two — and evaluate whether the conclusion is forced.
5. If the conclusion is forced on every such configuration, report **VACUOUS** with the forcing reason.
6. Otherwise, for each declaration citing K, check whether the residue still implies it; report **LOAD-BEARING** if at least one implication survives and **DECORATIVE** if none does.

Output. One verdict of Definition 1.3.8, together with the naming data required by whichever branch was taken.

Proposition 1.3.9 (T) — vacuity implies evaporation failure, and the converse fails. If an invariant reported by K is identically zero on every configuration in which K licenses its report, then K fails clause 2 of Definition 1.3.7. The converse fails: a construction may pass clause 2 and still be **DECORATIVE**.

Proof. The first statement is immediate from Definition 1.3.7, clause 2, since the conclusion is then satisfied for structural reasons on every admissible configuration. For the converse, take K to be any true, non-vacuous statement about the transport maps that no declaration cites — for example, that the product of the transports around a cycle is orthogonal. It passes clauses 1 and 2 and fails clause 3, because deleting it removes no premise of any declaration. □

The proposition is worth stating because vacuity is the failure that gets caught and decorativeness is the failure that does not. A vacuous invariant announces itself the first time someone plots it and sees a flat line at zero. A decorative import produces respectable-looking mathematics that no later argument consumes, and the only way to detect it is to ask, for every import, which numbered declaration would lose a premise if the import were deleted. That question is clause 3, and it is the clause that costs work.

Table 1.3.8 — Evaporation verdicts on the imports of this document.

Construction	Imported vocabulary V	Unfolds to	Verdict	Declaration that would lose a premise
Context site	site, coverage, sieve, sheaf	down-set frame, join-generated covering families, order-theoretic sieve, gluing condition	LOAD-BEARING	Theorem 2.1.6, Lemma 3.2.2
Descent solve	descent, sheaf Laplacian	shifted linear solve on an edge-weighted complex with orthogonal transports	LOAD-BEARING	Definition 5.1.35, the L4 backward contract
Holonomy layer	holonomy, monodromy, Tannakian	group generated by cycle transports; word growth of that group	LOAD-BEARING for the group, DECORATIVE for the Tannakian vocabulary	Theorem 3.6.9 keeps the group; no declaration consumes reconstruction
Obstruction head, first issue	non-abelian cohomology, section conjecture	difference of per-vertex values	VACUOUS; withdrawn	none survives

Construction	Imported vocabulary V	Unfolds to	Verdict	Declaration that would lose a premise
Gerbe diagnostic, first issue	gerbe, determinant class	a cochain proved exact	VACUOUS; withdrawn	none survives
Degree-two capacity, present issue	gerbe	scalar second cohomology of an engineered nerve	LOAD-BEARING only on $\partial\Delta^3$; VACUOUS on a graph or full simplex	the non-vacuity clause of Chapter 4.3
Almost-descent engine	almost mathematics, almost isomorphism	maps invertible modulo a declared idempotent ideal of tolerances	LOAD-BEARING	the tolerance-composition statements of Chapter 3.10
Weight-graded layer	weight filtration, purity	declared grading with a monotonicity constraint across layers	LOAD-BEARING for the grading, DECORATIVE for purity	Chapter 3.9's grading statements; nothing consumes purity
Motivic justification unit	motive, universality	a typed record with a factorisation obligation	DECORATIVE until the factorisation obligation is discharged	none at present
Indeterminacy calculus	theta-link, multiradiality	three declared ambiguity groups acting on reported quantities	LOAD-BEARING for the group actions, DECORATIVE for the source vocabulary	the invariance-or-budget rule; nothing consumes multiradiality

Two entries in that table are deliberately unflattering. The motivic justification unit is marked **DECORATIVE** at present, not because the record is ill formed but because no numbered declaration currently consumes its factorisation obligation, and by clause 3 that is exactly what decorative means. The Tannakian and purity vocabularies are marked decorative while the constructions they sit beside are marked load-bearing, which is the correct outcome of separating a construction from the words used to introduce it: the holonomy group is real and its growth dichotomy is proved, and none of that depends on reconstruction; the layer grading is real and its monotonicity constraint is used, and none of that depends on purity.

What a decorative verdict does and does not require. A **DECORATIVE** verdict requires the vocabulary to be demoted to Tier A and the sentence introducing it to say that no theorem is borrowed. It does *not* require deleting the construction beside it, which may be load-bearing on its own proofs, and it does not license retaining the vocabulary on the grounds that it is "merely expository." The test for the retained sentence is Proposition 1.3.9's clause 3, applied to the sentence rather than the construction: if no declaration loses a premise when the sentence is deleted, the sentence is expository and must be marked as such where it stands.

Proposition 1.3.10 (T) — the residue is non-empty and its emptiness is not a target. The union of the residues $\mathrm{res}(\tau)$ over the seven typings of Table 1.3.7 contains at least the valuation ring, the Frobenius-equipped variety, the neutral fibre functor, the arithmetic fundamental group, and the theatre. No construction in this document is proposed as an image for any of them.

Proof. Each listed sort is required by its source setting and is absent from Σ_{B} as fixed by Definition 1.3.2, which enumerates the constructed sorts; absence from a finite enumeration is decidable by inspection. That none is proposed as an image is a statement about this document's declarations and is likewise checked by inspection of the transplantation table. □

The second half of that proposition is the methodological point and is easy to lose. A programme that measures its own success by shrinking the residue will manufacture images: it will define something, call it a fibre functor, and proceed. The residue is a report, and the correct response to a large residue is to state which theorems are consequently unavailable, which is what Table 1.3.7 does and what the grade-**X** entries of the next section record.

1.3.11 The canonical locus, the two names, and the named variants

The instrument defined above is used in four later Parts, and it is introduced there four more times under two different names. That is a defect of this document, and it is repaired here rather than in the later chapters, because the earliest statement is the one a reader meets first and the later ones are the ones that drift.

The two names denote one instrument, and the identity is demonstrable rather than asserted. Section L3 of Chapter 11.1 introduces the procedure under the heading *the evaporation test*; Chapter 11.7 and the lineage kernels of Part XII cite that same section as *the §L3 restatement test*; and Chapter 11.7's own statement of it says, of a construction that fails, that it has evaporated and must be cut. One section, two names, used of each other in the same sentence. The names are not loose synonyms: *restatement* names the operation, the formation of the unfolded residue $K \setminus V$ of Definition 1.3.7, and *evaporation* names the failure mode, the case in which that residue carries nothing. A procedure named for its input and the same procedure named for its negative outcome are one procedure.

What is not identical is the criterion. The four later forms differ from Definition 1.3.7 in ways that are substantive rather than editorial: one is strictly weaker, one is strictly stronger, one is incomparable with it, and one is a record schema that prescribes what an audit entry must contain while delegating the decision. Replacing them with a bare cross-reference would silently change the strength of every argument that invokes them. They are therefore adopted here as named variants, and each later occurrence cites the variant it actually uses.

Definition 1.3.11 (D, normative) — the canonical locus. Definition 1.3.7, Definition 1.3.8 and Algorithm 1.3.1 are the sole definition of the evaporation test in this document. Every other occurrence of the evaporation test or of the restatement test, in any Part or appendix, is one of the following three things: an *application*, which runs the test on a named import and reports a verdict of Definition 1.3.8 while defining nothing; a *named variant*, one of Definitions 1.3.12 to 1.3.14, cited by number; or a *record schema*, which prescribes what an audit entry must contain and delegates the decision to an application or a variant. An occurrence that is none of the three is a defect of this document and is to be reported rather than obeyed.

Definition 1.3.12 (D) — the binary variant, Part XI form. Let K and V be as in Definition 1.3.7, with V restricted to the vocabulary of algebraic geometry, topos theory and inter-universal Teichmüller theory. The *binary variant* asks clause 1 of Definition 1.3.7 alone and returns *keep* when the residue is a well-posed definition or a provable statement and *delete* when the residue is empty. It admits no verdict between the two. In particular it does not separate **DECORATIVE** from **LOAD-BEARING**, because it never reaches clause 3.

Definition 1.3.13 (D) — the design-constraint variant, lineage-kernel form. Clauses 1 and 2 of Definition 1.3.7 are retained unchanged, and clause 3 is replaced by the requirement that the residue force, forbid or change at least one definition, theorem, algorithm, release gate, datum or report of this document. A construction passes when all three modified clauses hold, and passes *in part* when the residue splits into a component satisfying all three and a component failing clause 3, in which case the surviving component is retained under its own identifier and the remainder is cut and reported.

Definition 1.3.14 (D) — the strict-necessity variant, admission form. Clause 1 of Definition 1.3.7 is strengthened to require that the residue carry a falsifiable hypothesis and a falsifiable conclusion; clause 3 is strengthened to require both that some later result use a premise supplied by the residue and that deleting that premise render the later proof invalid or strictly weaker; and a fourth clause is added, that the import's release grade and its bibliography status be printed where the import is used. An import which satisfies the unstrengthened clause 3 but not the strengthened one is decorative and is cut.

Proposition 1.3.15 (T) — the variants are ordered, and one pair is incomparable. Write E for the verdict **LOAD-BEARING** under Definition 1.3.7, and B, G, N for the passing verdicts of Definitions 1.3.12, 1.3.13 and 1.3.14 respectively. Then N implies E and E implies B; neither implication reverses; and E and G are incomparable, neither implying the other.

Proof. That *N* implies *E*: Definition 1.3.14 retains clause 2 unchanged, strengthens clause 1 by adding a falsifiability requirement to a well-typedness requirement, and strengthens clause 3, since a later proof which becomes invalid or strictly weaker when the premise is deleted is in particular a numbered declaration which the residue implies. That *E* implies *B*: clause 1 of Definition 1.3.7 is the passing condition of Definition 1.3.12, and the vocabulary erased by Definition 1.3.12 is contained in the imported-term set *V* of Definition 1.3.7, so a residue that is well-formed after unfolding all of *V* is well-formed after unfolding a subset of it.

That *E* does not follow from *B*: the Tannakian entry of the holonomy row of Table 1.3.8 has a well-posed residue, hence *keep* under Definition 1.3.12, and is consumed by no declaration, hence **DECORATIVE** under Definition 1.3.7. That *N* does not follow from *E*: clause 3 of Definition 1.3.7 is an implication test and the strengthened clause is a necessity test, and a redundant premise implies its consumer while being unnecessary to it, so a residue whose consumer is derivable from the ambient definitions without it satisfies the former and fails the latter.

Incomparability. *G* without *E*: an entry of the forbidden list at §F3 of Chapter 11.1 has a well-posed and non-vacuous residue which forbids a construction, satisfying clause 3 of Definition 1.3.13, and is cited by no numbered declaration, failing clause 3 of Definition 1.3.7. *E* without *G*: the redundant premise of the previous paragraph implies the declaration citing it, satisfying clause 3 of Definition 1.3.7, and forces, forbids and changes nothing, since deleting it leaves every definition, theorem, algorithm, gate, datum and report of the document unaltered. □

The ordering matters for reading the later Parts. A construction reported as surviving the test in Part XI has cleared the weakest of the four criteria, and the report is silent on whether anything consumes it; a construction admitted under Theorem 2.17.39 has cleared the strongest. Two verdicts printed with the same word are therefore not comparable evidence, and the table below is the translation a reader needs.

Table 1.3.9 — Where the test is stated, under which name, in which form, and with which verdicts.

Locus	Name used there	Form in force	Verdict alphabet	Relation to Definition 1.3.7
Chapter 1.3, Definitions 1.3.7–1.3.8, Algorithm 1.3.1	evaporation test	three clauses, four verdicts	LOAD-BEARING, DECORATIVE, VACUOUS, ungraded	canonical
Chapter 11.1, §L3	evaporation test	Definition 1.3.12	keep, delete	strictly weaker; clause 3 never reached
Chapter 11.7, opening discipline	restatement test	Definition 1.3.13	pass, cut, pass in part	incomparable; consumption replaced by design constraint
Chapter 11.7, Rule 11.7.I	restatement test	record schema	RETAIN, CUT, PARTIAL, UNRESOLVED	delegates the decision; adds no criterion
Chapter 2.17, Theorem 2.17.39	restatement-and-consumer admission test	Definition 1.3.14	admitted, cut	strictly stronger; adds necessity and printed grade
Appendix A15, Definitions A15.1–A15.2	evaporation test	a fifth form, stated there	KEEP, CUT, DEMOTE, SPLIT, REGRADE→M	see the warning below

Definition A15.1 is outside the authority of this chapter and is not reconciled by Definition 1.3.11. It differs from Definition 1.3.7 in two respects that a reader of the negative-results appendix must know. Its non-vacuity clause asks whether the residue is an instance of a standard elementary fact already available to the reader, which is a judgement about the reader, whereas clause 2 of Definition 1.3.7

asks whether the conclusion is satisfied for structural reasons on a named configuration, which is a check on the configuration. Its consumption clause is the necessity test of Definition 1.3.14 rather than the implication test of Definition 1.3.7, so its `CUT` verdicts are issued under a stricter criterion than its own preamble claims, that preamble attributing the instrument to Chapter 11.1, whose form is the weakest of the five. The discrepancy is recorded here and is not silently repaired, because repairing it would change the strength of verdicts already printed in that appendix.

1.3.12 The document's own falsifiability

Finally, a note on what would make this document as a whole a failure rather than a partial success, since the module-level gates do not aggregate automatically.

The programme fails if the compositionality defect turns out to be structureless, because the diagnosis of the second deficit is then wrong and no measurable consequence follows from the coend correction. It fails if the collapse remedies do not work — if models with heterogeneous stalks and Stiefel-constrained restrictions still degenerate functionally — because the constructive part is then unrealisable regardless of its elegance, and the anti-collapse argument is already only partial. It fails if the invariants can be driven to zero by trivialising the site and no defence is found, because the metrics are then self-referential. It fails if the components now marked **UNTRAINABLE** cannot be given gradients at a stated cost, because an architecture whose defining mechanism must be replaced by a conventional substitute in order to train is a conventional architecture with extra telemetry. It fails if the descent solve and the routing cannot be made cheap enough, because an architecture that is right and slow loses to one that is wrong and fast. And it fails, most quietly, if everything works and the effect is small.

Each of these has a corresponding experiment in Part IX with a pre-registered threshold. It would be better for the document to be shown wrong quickly than to be left standing because nobody could tell.

1.3.13 Deliberate demotions

The type check and the evaporation test make this chapter look more decisive than it is. Four places where they are weaker than they appear are recorded here rather than left for a reader to find.

The type check is a filter on expressibility and decides nothing about mathematics. A typing whose coercion witnesses are all present and whose hypotheses are all **DISCHARGED** still yields only clause 1 of Proposition 1.3.5, which is a transported conclusion capped at grade **M**. It is therefore possible for a transplantation to pass every mechanical check in this chapter and remain a statement this document has not proved. The mechanism is deliberate — Axiom A9.11 is the cap — but the reading it invites, that a passed type check is a proof, is wrong and is the most likely misuse of this section.

The type-check audit in Table 1.3.7 is hand-executed and is not generated. The same generation block that applies to the compliance matrix applies here: there is no export that produces these rows from the registry, the statuses were assigned by reading the named declarations, and a row is therefore evidence about what was read rather than about what the registry holds. Proposition 1.3.10 governs the residue column: non-emptiness records unavailable source theorems, not a completion target. Its **UNTYPED** verdicts are absence claims that can decay silently as later chapters add constructions.

Clause 3 of the evaporation test is a search over the citing set, and the citing set is not mechanically maintained. Algorithm 1.3.1 step 6 asks whether the unfolded residue still implies a declaration that cites the construction. That question is decidable only relative to a

complete list of citing declarations, and the cross-reference machinery that would produce such a list mechanically is not part of this chapter's authority. Where the list used was compiled by reading, a **DECORATIVE** verdict may understate the import's role and a **LOAD-BEARING** verdict may overstate it; the direction of the error is not known in advance.

No procedure for deciding non-vacuity in general is given by Proposition 1.3.9. Step 4 of the algorithm evaluates the residue on the reference configuration and on the recorded minimal non-degenerate configuration for its degree. That is a finite check on two instances, not a proof that the conclusion is unforced in general, and a construction that is non-vacuous on both while being forced on the configurations actually used in training would pass a test it should fail. The degree-two row of Table 1.3.8 is the case where this is known to bite, and it is marked with the configuration on which the verdict holds rather than being marked good.

No gate enforces the normative claim made by Definition 1.3.11. The claim that this chapter holds the sole definition of the evaporation test was established by reading every occurrence of both names in the book, not by a check that runs at release. A sixth statement of the test added later, or an existing occurrence edited to add a clause, would leave Definition 1.3.11 false and Table 1.3.9 stale without any gate failing. The table is therefore evidence about the book as read, and the reconciliation it performs is only as current as the reading. Definition A15.1 is the case that demonstrates the exposure: it is a formally numbered fifth statement of the test in an appendix this chapter cannot edit, and it was found by search rather than by any release check.

Vincent Liu
WiseMotif.ai

WiseMotif Monograph of Mathematical Intelligence
Series 02 · Mathematics and Artificial Intelligence

GM-Ω∞ · ONE ARCHITECTURE · FIVE VOLUMES

The complete monograph

I–V

Topos-Native Intelligence sets out the meta-reasoning kernel of the Grothendieckian Programme. A context is a site, its evidence a sheaf, and a conclusion a global section that exists only where local evidence glues. On this foundation the work builds nine layers over one base, experts as open sub-topoi, and a learning theory that preserves the structure.

Every claim carries its grade where it is made; the experiments that will decide the theory are specified in advance.

PUBLICATION RECORD

Edition
Research Architecture Edition 1

Version
1.0.0

Language
English

Format
Complete five-volume monograph

FIVE-VOLUME ARCHITECTURE

From diagnosis to the replacement case

I **Diagnosis and the Source Mathematics**
Five structural deficits, the transplantation discipline, and the source mathematics.

II **The Architecture and Its Mixture of Experts**
Nine layers over one base, and a mixture of experts whose gating is a site of local topoi.

III **Learning, Systems, Validation, and Position**
Objectives, reinforcement learning, systems, instruments, and the rigour ledger.

IV **Provenance and Adjudication**
The Grothendieck programme and inter-universal structure, adjudicated import by import.

V **Transferable Cores and the Replacement Case**
Seven lineages reduced to transferable cores, their algorithms, and the replacement case.

A **Apparatus · A0–A16**
Notation, proofs, glossary, worked code, protocol, negative results and bibliography.